



FITTER-EU

DELIVERABLE D2.1

Embedding Social Justice in Twin Transition Research: Concepts, Definitions and Methodological Approach



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Summary

D2.1 presents just transition concepts, provides an operationalised definition and describes the research methodology to be undertaken in FITTER. D2.1 will be updated in M13, and M21, following the completion of T2.3, T2.4 and T3.1.

Table of Contents

1. Introduction.....	5
1.1. Overview of FITTER-EU	5
1.2. Purpose and scope of the Deliverable	5
1.3. Structure of the Deliverable	6
1.4. Abbreviations and Acronyms.....	6
1.5. Key Terms	7
2. Intersectional inequalities: theoretical perspectives on the social contract, justice, and power	10
2.1. The Natural Individual and the Social Contract	10
2.2. Understandings of justice	12
2.2.1. Subjects, Objects and Domains of Justice	12
2.3. Understandings of Power	19
2.3.1. Modalities of Power	19
2.3.2. The Power Cube.....	20
2.3.3. An Intersectional Approach to Power Relations	22
3. Existing structural and spatial inequalities in Europe	25
3.1. Structural inequalities and vulnerabilities in the context of the twin transition	25
4. Key academic debate on twin transition: shifts, impacts and inequalities	33
4.1. Green, digital and twin transition: main trends and relationship with social and structural inequalities in academic debate at the European level	33
4.2. Twin transition and spatial inequalities in the academic debate: key lessons from regional variations	38
5. Twin transition in four FITTER sectors.....	40
5.1. Agriculture	40
5.2. Construction, building and housing sector	41
5.3. Energy.....	44
5.4. Transport and mobility	46

6.	Twin transition in public debate: analysis from the six FITTER case studies	50
1.1.	Government	51
1.1.1.	Twin transition: main government strategies	51
6.1.2.	Just transition	52
6.1.3.	Opposition parties, counterarguments and other voices from the political actors 52	
6.2.	Industry: four FITTER sectors	53
6.3.	Trade Unions	54
6.4.	NGOs and other Civil Society Organisations	56
6.5.	Additional issues highlighted by the case studies in relation to public debate on twin transition.	57
7.	Just transition: FITTER approach	59
7.1.	A Just Transition	59
7.1.1.	Context	59
7.1.2.	Conceptual widening of the Just Transition Approach – A Socio-Ecological Contract 60	
7.2.	Approaches to Policy Design, Implementation and Evaluation	63
7.3.	FITTER Approach	72
7.3.1.	Subject, Domain and Object of Justice.....	73
7.3.2.	Adopting a Grounded Approach to Justice.....	73
8.	Conclusions	82

1. Introduction

1.1. Overview of FITTER-EU

The FITTER-EU project aims to contribute to existing research on the origins, dynamics and determinants of inequalities and enable anticipatory governance to support a fair and inclusive twin transition across Europe. The project innovates through the formulation and development of an ecosystem (i.e., the FITTER ecosystem) that pivotally includes a highly interactive and gamified Digital Platform powered by an Intelligent Predictive Decision Support System. Through a co-creation methodological approach, this ecosystem aims to enable policymakers to predict which social groups may be at risk of being adversely affected by twin transition policies under different scenarios, with the option to simulate the implementation of these policies for the assessment of inequalities and social exclusion risks. The Digital Platform within the ecosystem aims to provide proposed mitigation measures and better practice guides that can be used to curve potential negative effects of policies on identified at-risk groups. In addition, the FITTER ecosystem incorporates wide platform connectivity capability, which helps to support the needs of the engaging stakeholders within the community of the FITTER Cluster Group (CG). The group includes key national and pan-European members such as policymakers, civil society organisations including trade unions and business angels.

At the core of FITTER is the commitment to take an intersectional approach to tool-building that will assist policy development which addresses, or at least does not further exacerbate, societal inequalities across four sectors – energy, transport, building and housing, and food and agriculture.

1.2. Purpose and scope of the Deliverable

This deliverable articulates the just transition concept and related definitions and provides the foundations for the methodological approach to be undertaken in FITTER, thus guiding the data collection process. At the core of this deliverable is a literature review of theories, approaches and indicators of social justice and of equality. Drawing on various disciplines (including philosophy, sociology, law, economy and political science) and schools of thought, the deliverable also examines how the concept is articulated in academic and public debate. To fully explore the multiple aspects of the just transition, the analysis also considers different drivers of inequality in contemporary European Union (EU), with structural and spatial inequality at its focus. The twin transition concept is also operationalised through in-depth scrutinisation of the ‘green’ and ‘digital’ aspects of the socio-economic shifts, with a further examination of their impact on inequalities in four sectors: Energy; Transport; Building & Housing; and Food & Agriculture.

The FITTER project includes six national case studies, conducted by the consortium’s national research teams, and covering the following countries: Germany, Hungary, Ireland, Italy, Portugal and Spain. In T2.1, desk research was conducted by each of these research teams, with the aim to gather information and data on existing structural inequalities at the national level, taking into account regional and sectoral variations, and the implications for the twin transition. In conducting this part of the research, all national research teams utilised standardised enquiry forms and grids, developed by the Work Package (WP) 2 leader. To gather data which allows for both a national overview and comparisons of differences and similarities between the countries, the information requested varied from a general overview of the country’s socio-economic context to more specific mapping of issues related to twin transitions, the four FITTER sectors, structural and spatial inequalities, and previous crises’ impact and response. All six national teams were asked the same set of questions to provide a descriptive, narrative account of the selected topics, and to provide statistical tables related to the specific topics. These were based on the national statistics office databases and on the Eurostat

dataset, with detailed definition of the variables provided by the WP2 leader to allow for cross-country comparisons. As this was an initial mapping, the teams were asked to utilise the Labour Force Survey data which is standardised across EU member states. For this reason, the focus of the mapping was guided by labour market-related data. The statistical mapping task had two aims: (1) to provide data which will allow observing general trends within the countries, and (b) to identify gaps in data which will need to be explored through future iterations of the data gathering process within the project. The overall results of this desk-based research are presented across relevant sections of this deliverable. In addition, public debate was also analysed for each country, with different thematic areas highlighted by each national research team. This activity was structured around different types of stakeholders and the national teams used desk research methods (e.g. the analysis of newspaper content, key word search, and the analysis of the content of specific institutional websites). These findings are presented separately in Section 6.

1.3. Structure of the Deliverable

The deliverable is divided into 7 sections. Section 1 introduces key twin transition concepts. Section 2 outlines key areas of focus for building a theoretical framework – understandings of the social contract; distinct understandings of the subject, object and domain of justice; applied theories of justice; and understandings of power relations. Section 3 presents an overview of existing structural and spatial inequalities in contemporary Europe, with examples from the FITTER case studies also provided. Section 4 presents an overview of the academic debate on the twin transition. With structural vulnerabilities at the core of the analysis, it examines distinct aspects of the ‘green’ and ‘digital’ transition and also provides a summary of some aspects of the academic debate on the just transition. Section 5 considers the impact of the twin transition on the four FITTER sectors, with specific focus on social impacts and inequalities, with country case studies also included. Section 6 focuses on the public debate in the six national case studies, and provides perspectives from various stakeholders, including political parties, industry, trade unions and NGOs. Finally, Section 7 considers the FITTER approach to the just transition and provides the methodological foundations for the data gathering process to be conducted in subsequent stages of the project.

1.4. Abbreviations and Acronyms

CG	FITTER Cluster Group
CCIRs	Coal and Carbon Intensive Regions
EGD	European Green Deal
ESCO	European Skills, Competences, Qualifications and Occupations
EED	Energy Efficiency Directive
EPSR	European Pillar of Social Rights
EU	European Union
ISCO	International Classification of Occupations
GHG	Green House Gas
CGIL	General Confederation of Italian Workers
GDP	Gross Domestic Product
ILO	International Labour Organisation
ITUC	International Trade Union Confederation
LGBTQI	Lesbian, gay, bisexual, transgender and queer people
NEETs	Not in Employment, Education or Training
NGEU	Next Generation EU
NGOs	Non-Governmental Organisations
NRRPs	National Recovery and Resilience Plans

OECD	The Organisation for Economic Co-operation and Development
PV	Photovoltaics
RRF	Recovery and Resilience Facility
STEM	Science, Technology, Engineering, and Mathematics
UN	United Nations
UNRISD	United Nations Research Institute for Social Development
WP	Work Package

1.5. Key Terms

The **twin transition** is expected to have a considerable impact on the population - an impact that will be differentiated by interrelated dimensions such as skills, modalities of participation in the labour market, and financial resources. This impact will also vary depending on existing structural inequalities on the grounds of gender, race and ethnicity, age, ability, or socio-economic background, amongst others. The impact of the twin transition on inequalities remains quite uncharted in the literature. Furthermore, as the concept of the ‘twin transition’, combining the green with the digital, is still relatively new, these impacts have been mainly discussed separately, although there is significant overlapping. For example, the shift towards green jobs will require the re-skilling of certain categories of workers – with a certain level of digital skills also needed. This is important in terms of individual capacities, but also in terms of the availability and accessibility of digital tools. The issue further relates to location-specific labour markets: for example, while the reduction of certain jobs may go hand in hand with the creation of new ones, the two processes may not necessarily happen in the same location, as carbon-intense industries do not always share the same spaces as the new technology hubs.

There is ample evidence showing that systemic shocks, such as the financial crisis of 2008 or the COVID-19 pandemic, exacerbated existing patterns of inequalities, and some authors already highlight that the twin transition may amplify existing social inequalities, particularly those related to gender or race (Dodato et al., 2023). Given this, there is an urgency to put in place twin transition policies that effectively ward-off societies from these negative impacts and that act to enhance, rather than curtail, the pursuit for social justice. In doing this, a holistic approach is needed (Galgóczi, 2023). As inequality structures are quite complex, an intersectional perspective can provide a nuanced understanding of appropriate responses to the abovementioned challenges (Kuran et al., 2020). Here, the principle of ‘leaving no one behind’ is often adopted, with application to either vulnerable groups (Sachs et al, 2019) or places and regions which develop unequally (EXIT, 2023), with the overall ‘just transition’ concept used as guiding principle for the twin transition. However, the ‘just transition’ concept is often deployed without a detailed definition or explanation of how to make the transition ‘just’.

In providing a comprehensive overview of the FITTER approach to research on justice in relation to the twin transition, the following key terms will be used throughout the document. These definitions are based on the literature review conducted for the purpose of the WP2, with some specific terms aligned with the Eurostat glossary.

The Green transition can be defined as a response to climate change, which has been characterised as a ‘statistically significant variation in the mean state of the climate or in its variability, persisting for an extended period’ (IUCN, 2011) and can be caused by natural processes or human activity (Eurofound, 2023b). This response implies the transition to a low carbon or zero carbon economy. The shift will have an impact on jobs and labour markets, broadly defined as the ‘world of work’ (Mandelli, 2022). While it will most likely result in job creation (ILO, 2018), this will not necessarily happen in

regions and localities which are affected by carbon-intensive job losses (Atteridge and Strambo, 2020). Similarly, while a transition to a low-carbon economy can create ‘multiple economic opportunities’ with co-benefits such as better air quality and energy security (Karlsson et al., 2020) these opportunities will not be distributed evenly between and within countries (Gambhir et al., 2018; Green and Gambhir, 2019). The following terms are also linked to the theme of green transition and (in)equality (based on Eurostat glossary and the EU Parliament terminology):

- **Energy poverty** refers to a situation which occurs when a household must reduce its energy consumption to a degree that negatively impacts the inhabitants’ health and wellbeing. There are three main causes: (a) a high proportion of household expenditure spent on energy); (b) low income and (c) low energy performance of buildings and appliances.
- **Transport poverty** refers to a lack of adequate service transports necessary to access general services and work, or the inability to pay for these transport services. The following can lead to transport poverty: (a) no transport availability (mobility poverty); (b) no accessibility (e.g. disabled people); (c) low affordability; (d) too much time spent travelling (time poverty); (e) inadequate transport conditions (dangerous or unsafe).

The Digital transition can be defined as a transition from mechanical and electronic to digital electronics – a shift that started in the 1970s and included the adoption of personal computers and mobile phones and, more recently, the growth in usage of new technologies, including AI, robotics, IoT (Fouquet and Hippe, 2022). These shifts are also referred to as the ‘Fourth Industrial Revolution’. The digital transition is set to transform the way in which the EU economy will work (Maucorps et al., 2023), particularly considering its acceleration by the COVID-19 pandemic (Revoltella, 2020). While digitalisation is changing the world of work (Brynjolfsson and McAfee, 2011), the shift can have negative repercussions for income inequalities (European Commission, 2019) and may have adverse consequences for employability of groups which are considered structurally vulnerable (Lyons et al., 2019). The following are terms also related to the theme of digital transition, inclusion and inequalities (based on Eurostat Glossary):

- **E-inclusion** refers to the situation where everyone in society can participate in the information society. This requires affordable access to technologies, the accessibility and usability of ICT tools and services, and the ability and skills of all individuals to use these tools.
- **Digital divide** refers to the distinction between those who have access to the internet or other digital technologies and are able to make use of online services, and those who are excluded from these services. The digital divide can be classified according to criteria that describe the difference in participation according to gender, age, education, income, social groups or geographic location.
- **Digital literacy** lays out five digital competence areas and a total of 21 digital competencies. The digital competence areas include information and data literacy, communication and collaboration, digital content creation, safety, and problem-solving.

The Twin transition has been defined as a process which intertwines the green and the digital transitions with a goal to consider them equally important for the future of societies (Kounduri et al., 2023: 3). It highlights the potential synergies between the green and digital transition, for example, the role of digital technologies in fighting climate change and mitigating environmental damage (Diodato et al., 2023). However, some criticism within the debate notes that digitalisation may have negative impact on jobs (Charles et al., 2022) while others also question whether the ‘digital’ will

necessarily lead to more sustainable economy (Best et al., 2020). Nevertheless, the motto ‘for greener, more digital and resilient future’ (Next Generation EU) relates to post COVID-19 recovery which is focused on strengthening resilience to climate, energy, health and digital threats (Celeste and Dominioni, 2023). In sum, the twin transition refers ‘not only to two concurrent transformational trends (the green and digital transitions)’ but rather to a process that unites the two and possibly accelerates changes towards the level of transformation that is needed (Muench et al. 2022).

2. Intersectional inequalities: theoretical perspectives on the social contract, justice, and power

FITTER has committed to taking an intersectional approach to the analysis of inequalities across the four sectors and to the development of intersectional mitigation plans to address potential hazards and negative impacts of the twin transition.

Taking an intersectional approach to sectoral analysis and policy development requires a good understanding of, first, why norms, structures and practices divided populations into groups in line with their identity formations (by gender, race, sexual orientation, ability etc.). In understanding the ‘why’ of these norms, structures and practices, we can better understand how processes of categorisation led to dynamics of oppression and dynamics of privilege, and how we can best address those in order to enable liveable lives for all. In this section, we consider concepts of the social contract, understandings of justice, and perspectives on power as key building blocks for a definition of a just transition.

2.1. The Natural Individual and the Social Contract

The construct of the ‘social contract’, understood as a pact between governments and peoples, is central to our understanding of the structure of social and political life. At the centre of the Western construct of the social contract is an understanding of the individual as autonomous, unencumbered, and someone who ‘opts in’ to the social contract of their own volition, thereby agreeing to fulfil certain responsibilities, and able to avail of their rights. In reality, though, we are born into specific societies and thus we do not ‘opt in’ as autonomous individuals, yet the primacy of the individual as autonomous and unencumbered in Western political thought nevertheless shapes our ideals and orientations e.g. pursuit of happiness, individualism. This is not the only understanding of the relation of self to society. Social contracts have been structured differently depending on different understandings of the relation of the self to society, and the rights and obligations that that entails. For example, the African philosophy of Ubuntu – ‘I am because we are’ – establishes the relation of self to society and nature as fundamentally interdependent. This leads to an understanding of one’s own wellbeing as interdependent with the wellbeing of others and an orientation to the common good.

In theory, ‘the individual’ to which rights and responsibilities are assigned by law is gender neutral. However, for the construct of the social contract to function, the individual cannot be gender neutral for two reasons. First, the law needs us as individuals to have a substantive continuous core upon which our rights and responsibilities can be based. This substantive core is characterised by the capacity to speak, to reason, and to exercise moral judgement. Having, being or living an intelligible gender provides us with a stable, continuous and coherent self. Such a concept of the self is required for the substantive core to remain operative within the liberal political imaginary (Poovey, 1992). Second, the construct of the social contract, which is centred on the needs of an unencumbered, autonomous individual necessitates the ‘backgrounding’ of the relations of care (breast/bottle-feeding, toilet training, language development to name but a few) and sustenance (clean air, water, stable food supply etc.) that we are all in fact dependent upon. In order to ‘background’ those relations of care and sustenance, differentiations between people based on ascribed characteristics such as gender, race, ability and sexuality were established, and those defined as subsets of the human were excluded from the public sphere.

Binary thinking, modelled on the masculine/feminine binary, is a primary form of thinking in Western thought. It is not just that living an intelligible gender provides us with the appearance of a substantive

and coherent core that is at issue here; it is also the way in which that is organised in line with masculine positions – understood as the norm, unmarked, universal – which define themselves *over and against* feminine positions – understood as particular, marked, and a subset of the human within particular contexts. The form of the masculine/feminine binary has historically been utilised to signify a whole range of hierarchical dichotomies which positioned that associated with the masculine as superior, and that associated with the feminine as inferior e.g. rational/irrational, strong/weak, active/passive, culture/nature, mind/body, white/not white, heterosexual/not heterosexual. We all move between occupying unmarked (masculine) or marked (feminine) positions in line with societal dynamics of oppression and privilege. Take the example of ‘Karol’ whose whiteness, passport (EU) and style of dress (Western) means that they have navigated any airport they have ever been in with ease. These experiences are not a function of who they are, but instead a function of the histories of the ‘natural individual’ and the dynamics of oppression and privilege that work to uphold it.

Liberal political theory established a public/private divide with the ‘private’ domain placed outside the power of the State. The Western social contract is thus based upon a sexual contract, within which women are subordinate and which secures the basis of the political realm on the patriarchal family (Pateman, 1998). With the realisation that power relations are just as operative in civil society as they are within the state, this distinction later evolved to a tripartite division between the state, civil society, and the personal. In liberal theory, however, the personal did not specifically refer to family life, and indeed the domestic realm has generally not been theorised in relation to the pursuit of freedom or to structural constraints on freedom (Kymlicka, 1990). Feminist theorists have argued that this neglect of the domestic is due to the fundamental premises of liberalism regarding subjectivity, the social contract, and non-intervention in civil society and personal relations (Pateman, 1988; Shklar, 1991; Young 1999). These premises work to efface the relations of care and the social and institutional structures which produce the liberal individual and enable the social contract. They also render those charged with reproduction, care work and emotional labour as ‘subordinate’ to the ideal of the rational individual (Squires, 1999: 26-31). Examining the role of colonisation within the construction of the social contract, it is further argued that the social contract is based on a sexual *and* a racial contract. Colonised peoples were subject to genocide, enslavement and segregation and defined as less than fully human. Societies were thus formed through the systematic exclusion of people of colour from contracting and the political sphere (Mills, 1997; Bhambra 2021).

Given that the social contract is built upon, and at the same time seeks to efface, these gendered and racialised histories, Linda Zerilli argues that current political systems are constitutive of, and dependent upon, certain subjects as in some way second class citizens. The Western social contract is not a universal concept and its logic cannot just be straightforwardly expanded to include those who were previously excluded (Zerilli, 2006: 110-111). Indeed, there is growing recognition that the social contract, as it has evolved up to the 21st century, is no longer fit for purpose. Multiple entities such as the United Nations (UN), the International Trade Union Confederation (ITUC), the International Labour Organisation (ILO), the G20 Leaders’ Summit, the McKinsey Global Institute, and various social movements (Fridays for Future, Black Lives Matter, Extinction Rebellion) are calling for an updated social contract - understood as a ‘**socio-ecological contract**’ (UNRISD, 2021; Shafik, 2018; ITUC, 2021; ITUC, 2023; Manyika et al 2020; Galgóczi, 2023).

One of the more advanced conceptualisations of a socio-ecological contract has been developed by the United Nations Research Institute for Social Development (UNRISD) (discussed further in Section 7). The need to ensure human rights for all (including women and migrants); larger freedom for all including security and protection; and the transformation of economies and societies to halt climate change and environmental destruction are core to the UNRISD socio-ecological contract. It provides

explicit recognition of three constitutive yet undervalued or ignored elements of the 20th century social contract associated with welfare capitalism (largely implemented in Global North countries). These are:

1. The low to no value attributed to carework (the sexual contract);
2. The legacies of colonialism and coloniality including the present-day structuring of the global economy (the racial contract); and
3. The availability of the planet's resources as raw materials for the production of goods and as a sink for production's waste.

That is to say that, although the welfare state in Global North countries was an amelioration of conditions of deprivation within nation states, it was funded and enabled by the low to no value attributed to: a) labour and resources in poorer countries (many of which are former colonies and located in the Global South); b) natural resources globally; and c) both paid and unpaid carework across all economies (see Bhambra, 2021 and Fraser, 2016).

2.2. Understandings of justice

The fact that we need to transition to an ecologically sustainable way of life globally is well established (IPCC, 2023). To do so in a 'just' manner necessitates an understanding of different conceptualisations of justice. Distinct conceptualisations lead to distinct configurations of the social-environmental-economic nexus. In this sub-section we first consider the subject, domain and object of justice in liberal, feminist and postcolonial theories of justice. This is followed by the outline of perspectives on applied theories of justice – climate, environmental, energy, and spatial.

2.2.1. Subjects, Objects and Domains of Justice

The primary subject of **liberal theories of justice** is the 'natural individual' i.e. an able-bodied, full-time employed, citizen with no significant caring responsibilities. The objects of justice are generally understood as equal basic liberties and fair distribution of social and economic goods. The domain of justice are public institutions and relations. Liberal theories often either implicitly or explicitly exclude the family as a concern of justice. As recently as 1971 John Rawls held that those in the 'original position' in his theory were 'heads of families' and that the family could be understood as beyond justice.

Ideal theory is prominent in the post-Rawlsian literature, the aim of which is to work out the principles of justice that should be met before we consider a society to be just. Ideal theory often utilises idealisations (e.g. societal members do not engage in actions or choices that are influenced by prejudice) as part of the process of developing principles of justice. Ideal theorists justify the use of idealisations (e.g. the absence of societal prejudices) by arguing that where we have 'strong first principles justifying why these factors can never be justified then we don't need to theorise about the question whether they are just or unjust; in other words, they don't need to be part of the theory' (Robeyns, 2008: 354). Ideal theory can be comprehensive (specifying the conditions to be met before each and every injustice is addressed) or partial. It can be partial in several ways:

- Specifying minimal principles of justice while acknowledging further principles would need to be set once minimal principles are met
- Focused on one domain of justice e.g. gender, health
- Focused on one locality e.g. a nation-state
- Focused on one set of institutions e.g. political institutions and agents (Robeyns, 2008: 345)

Nonideal theory does not focus on principles of justice, instead focusing on developing an evaluative framework. Robeyns (2008: 347) argues that nonideal theory has two main functions. First to enable comparisons between different social states and evaluate which is more just, and second to guide actions that would move us toward the ideals of a just society. Robeyns argues that *ideal theory* (focused on endpoint) and *nonideal theory* (focused on evaluation of ideals and/or social states) are insufficient for *action design and implementation*. As part of action design and implementation we need to take further account of feasibility constraints (e.g. our dependencies on clean air and water, the impact of citizenship status on entitlements) and unintended consequences (e.g. feminist strategies that fail to interrogate race). We also need to ensure the buy-in of all affected communities.

Robeyns (2008) argues that there are two problems with the use of idealisations within ideal theory. First, ideal theory does not generally provide good guidance on what to do with an idealisation when moving to the nonideal level. One solution could be to focus our energies on first enabling the idealisation (e.g. societal members do not engage in actions or choices that are influenced by prejudice), before working on realising other principles of justice (e.g. we all have entitlements to basic goods to enable liveable lives). However, that would be inefficient and arguably not morally defensible. The second problem with idealisations is that some of them are theoretically unjustified. Robeyns gives the example of the assumption that a person 'is not dependent for care upon others, nor constrained in his actions and plans of life by caring duties' (2008: 358) as an example of a bad idealisation. Human interdependencies across the life cycle are inevitable, an inescapable constraint, and this makes 'the just distribution of care (both at the receiving and giving end) an important aspect that a theory of justice, both at the ideal and the non-ideal level, should work out' (2008:358). Robeyns concludes by arguing that ideal theorists need to:

1. Properly define and differentiate between ideal theory, nonideal theory, and action design and implementation
2. Investigate and analyse the relationship between ideal theory and idealisations
3. Specify and debate the ways in which the role of ideal theory is limited
4. Work through cases of flawed idealisations to 'better understand when and why idealizations are bad or unwanted and when and why they are useful' (2008: 362)

A central concern of **feminist theories of justice** is that women must be understood as individuals with full and equal status. Feminist theorists understand that the 'individual' is always already gendered, and indeed the gender binary is an essential part of the construction of the individual. As 'individuals' we come into being through intersecting power dynamics, initially at the level of the nuclear family. The family – culturally established as man/woman/child – is a basic structure of society which institutes unequal power relations. It is thus a subject of justice in feminist theories, with the domestic situated as a domain of justice, and responsibility for unpaid care-work understood as an object of justice (Okin, 1979). As opposed to ideal theory, feminist theories of justice are often grounded theories. As Irish Marion Young notes (1990: 4-5) we must 'begin from historically specific circumstances, because there is nothing but what is, the given, the situated interest in justice, from which to start.' A focus on the injustices in the here and now (forms of domination and oppression), with a primary focus on gendered injustices, is thus the starting point for feminist theory and praxis.

Within feminist theories of justice, redistributive justice is important as it addresses the structures of society that produce exploitation and economic inequalities. However, as Fraser (2007, 2008) argues, recognitive justice and representative justice are equally important. Recognitive justice can be understood as the absence of disrespect, marginalisation, exclusion and cultural domination. Representative justice is equality in political participation and having an equal democratic voice. Feminist theories also understand justice as 'spatially complex' (Mitchell, 2023: 62) with

interconnected relations of domination needing to be addressed at all levels simultaneously (household, local, national, global). Feminist theories of justice stress that we can act in accordance with the dominant norms, law, due process, and institutional regulations – which are all assumed to be just – and still produce unjust outcomes. In these circumstances no one actor is to blame but we all bear some responsibility to identify how injustices are reproduced across norms, institutions and systems and work to transform them (see Young, 2011).

In line with the wider feminist argument that the social contract is built upon, and at the same time seeks to efface, a sexual contract within which women are subordinate and which secures the basis of the political realm on the patriarchal family, feminist theories of justice are concerned with social reproduction i.e. the ways in which pregnancy and birth, unpaid care work (for children, for elders, housework), and affective labour all sustain capitalist societies yet remain undervalued and inhibit the possibilities for justice.

Northern feminists often describe their focus as the ‘balance between family and work’. But struggles over social reproduction encompass much more: community movements for housing, healthcare, food security and an unconditional basic income; struggles for the rights of migrants, domestic workers and public employees; campaigns to unionise service-sector workers in for-profit nursing homes, hospitals and child-care centres; struggles for public services such as day care and elder care, for a shorter working week, for generous paid maternity and parental leave. Taken together, these claims are tantamount to the demand for a massive reorganisation of the relation between production and reproduction: for social arrangements that could enable people of every class, gender, sexuality and colour to combine social-reproductive activities with safe, interesting and well-remunerated work (Fraser, 2016:116)

Black feminist theories have done much work to expose the intersectional nature of dynamics of oppression and dynamics of privilege (discussed further in 2.3). Jagger (2009) notes that critical race theory has expanded the domain of justice to ‘include institutions as well as individuals, the subjects of justice to include groups as well as individuals, and the objects of justice to include recognition as well as redistribution’ (Mitchell, 2023: 65).

Mitchell (2023: 68-69) summarises the concerns of feminist theories of justice as:

- justice substantively concerns the just treatment of individuals, but only insofar as that treatment is just for the group
- the substance of justice inheres not in some ideal, but arises from within, and is defined by struggles against actually-existing injustice.
- the substance of gender justice is that which counteracts gender-based domination and oppression while not enhancing (indeed while seeking to counteract) domination and oppression operating through other group differences
- the substance of justice also therefore inheres in power – in this case the ‘power to define’

Similar to feminist theories of justice, **postcolonial and decolonial theories of justice** focus on examining and addressing existing conditions of injustice. A key throughline is an analysis of how the legacies of colonialism, enslavement, segregation and racism shape contemporary structures, institutions, and norms. There is also a focus on how dynamics of oppression (racism, sexism, colonialism) can be ‘discursively built into dominant, western theories of justice, even of the most radical sort’ (Ohlsson and Mitchell, 2023: 92).

The subject of justice is not the individual but instead the subaltern (Spivak) and the racialised subject. The domain of justice is global and particularly the inequalities between the Global North and the

Global South. The object of justice is the historical and contemporary relations of power that oppress (post)colonised people and lead to injustices, including epistemic injustice (Ohlsson and Mitchell, 2023). The ways in which Western knowledge is continually centred and Global South perspectives ignored, devalued and misinterpreted is a key focus. Postcolonial and decolonial theories of justice not only seek to recover and further develop local epistemes that were devalued or subject to elimination efforts through colonisation. They also seek to demonstrate indigenous and/or pre-colonial understandings of subjectivity, ethics and justice as applicable at the global level (Mohanty 2003; Quijano, 2007; Lugones, 2007; Mignolo, 2009). Postcolonial and decolonial theories of justice stress the importance of rectificatory processes and practices for the harms of colonisation, including reparations, compensation, and apology.

Applied Theories of Justice

Environmental justice combines the concerns of the environmental *protection* movement, which was concerned with the protection of nature, and the environmental *justice* movement, which was primarily concerned with addressing how environmental harms and goods/amenities were distributed across communities. Bryant (1995: 589) defined environmental justice as the ‘cultural norms and values, rules, regulations, behaviours, policies and decisions to support sustainable communities, where people can interact with confidence that their environment is safe, nurturing, and protective.’ Although the scope of environmental justice can be global, it is often focused at a more local level and concerned with the justice of decision-making on land-use outcomes (Wood-Donnelly, 2023: 142).

The means through which environmental hazards and harms are located in or close to poorer communities, and environmental amenities are developed in or close to wealthier communities is a central concern of environmental justice.

Unjust distributions can be assessed through ‘unit based, distance based and exposure/risk based analyses (He et al 2019: 2 cited in Wood-Donnelly, 2023: 145) against sociodemographic indicators.

Maldistribution can partly be the result of procedural injustice, wherein disadvantaged communities lack social power, capacities and influence within decision-making processes on the distribution of environmental harms and amenities. Although in theory more participatory decision-making processes can mitigate against this, if power inequalities are not addressed within those processes they are unlikely to do so.

Environmental justice is also concerned with corrective or retributive forms of justice. Principles of corrective justice that have been promoted include the ‘polluter pays’ and the ‘beneficiary pays’. However, specific liability for environmental harms can be difficult to identify and the principles can be difficult to enforce particularly across international boundaries (Wood-Donnelly, 2023: 147-148).

Although sometimes used interchangeably with the term environmental justice, **climate justice** is broader in scope, relating to ‘concerns about the inequitable outcomes of climate change for differing peoples, communities, contexts and generations’ (Skillington, 2023: 155).

As climate change and climate breakdown has happened over time - through the cumulative effects of resource extraction and pollution across borders - and the benefits and harms of these activities are inequitably distributed, the scope of climate justice is global. A climate justice perspective highlights that climate breakdown is the result of specific norms, beliefs and decisions/actions that are committed to preserving existing social and economic arrangements, and that the organisation of social and economic life sustaining those arrangements can be transformed (Skillington, 2012).

Principles of right, responsibility, equality and fairness are central to climate justice. States in the Global North which have contributed most to GHG emissions, and which have benefited from resource

extraction from the Global South, are seen as morally obliged to pay more toward environmental protection. The most affected communities by climate breakdown (which are disproportionately located in the Global South and least responsible for GHG emissions) are seen as entitled to special rights and compensation (Stern 2006, Shue, 2010, UN 1992, Singer, 2011, Skillington 2016).

States in the Global South call for a 'backward looking' polluter pays perspective regarding who has caused climate change and is responsible for costs (Harris, 2010 cited in Skillington, 2023). States in the Global North argue against the polluter pays principle saying that moral responsibility for climate harms only applies in cases where a free agent had full knowledge of the full effects of their actions. They argue that there was an 'excusable' degree of ignorance of the full effects of climate harms prior to 1990 (Shue, 1997, Skillington, 2023).

As Skillington (2023) notes, the construct of the social contract (through which societies in the Global North are organised), poses problems in terms of assigning responsibility for long-term harms. Climate change is a *collective action* problem that has been produced *over time* and will have effects long into the future. However, the construct of the social contract is based on an exchange model between *individuals in the present* only. The focus on individual rights and responsibilities effaces the collective harms we can cause over time. Rawls (1999) argues that it is unfair to expect present-day individuals to refrain from wealth generating carbon intensive activities, in the interests of future peoples whom we do not know and who cannot repay us. This position aligns with the phenomenon of 'intertemporal discounting' which positions the short-term effects of climate harms as more important than the long-term effects, and the harms we produce as individuals as more important than the harms we produce as members of a collective e.g. generations or citizens of a nation state. Skillington (2023) argues that States – as collective entities that endure over time - need to initiate a deeper framework of climate justice. This would include an intergenerational contract model of just distribution that addresses issues of 'profound irresponsibility, non- accountability and expanding inequalities between generations, regions and, indeed, species' (2023: 160). She argues that a justice as fairness approach must begin with the experiences and needs of all affected communities.

It is only when publics are enabled in their social and civic capabilities to be 'positive climate influencers' in this moment of 'last opportunity' to avoid irreversible climate freefall (IPCCC, 2022), that such aspirations can become a reality. What is critical, however, is that more ambitious deliberative structures and legal applications of universal principles be recognised and enforced in ways that support mutual advantage and a cosmopolitan model of relationality, highlighting interconnections between peoples, regions, climate actions and outcomes, be explored more thoroughly in terms of its practical relevance (Skillington, 2023: 166).

Climate justice is increasingly pursued through a human rights framework although it has been difficult to prove the traceability of climate harms in the courts. The UN has recognised the right of all people to a clean healthy and sustainable environment, and in 2021 a Special Rapporteur on human rights and climate change was appointed by the UN Human rights Council. Also in 2021, the UN Child Rights Committee rules that States bear cross-border responsibilities for the impact of climate harms on children's rights (Skillington, 2023).

There have been tensions between the community-based work of the environmental justice movement which focuses on the local material impacts (e.g. health, jobs, energy infrastructure) and some climate-policy focused NGOs which have worked with market actors and promoted a 'voluntary or market-based approach to meeting ethical demands' (Schlosberg and Collins, 2014: 366). Initially many academic and elite policy practitioners of climate justice focused heavily on prevention and mitigation of climate breakdown. However, 'environmental justice advocates have from the very early days been concerned about the inequitable impacts of climate change on vulnerable communities'

(2014: 368), in terms of the individual (health, income), community (culture, businesses) and political structures (transparency, governance). The environmental justice concern with the local has sometimes been construed as nimbyism and an obstacle to a focus on climate breakdown at the global level. However, there is much overlap between contemporary environmental and climate justice concerns and they have fused in many ways. A key part of this has been a shift to thinking of **adaptation as transformative** in line with a revised eco-social contract. The focus is on building local capacities and development out of poverty and injustice as the basis for adaptation. There is recognition that justice movements need to be both local and international in order to build social systems that are 'based on the protection of sustainable socio-ecological systems, governmental accountability to all those impacted by climate change, and a notion of security built on basic human needs (2014: 370).

Energy justice is concerned with developing mechanisms for more just production, transportation, distribution and use of energy. The concept of fuel poverty, which is defined as spending above 10% of a household's income on energy, is applicable within Global North countries and preceded the concept of energy justice. The concept of energy poverty is broader than that of fuel poverty and more applicable to countries in the Global South. Energy poverty is a lack of energy services which could provide for basic human needs. Responsibility for energy poverty is placed on energy service providers (Sidorstov and McCauley, 2023: 177-178).

Heffron and McCauley (2017) take a *systems approach* to energy justice. They place the energy life cycle at the centre of the framework. This includes 1) Extraction; 2) Production; 3) Operation and supply; 4) Consumption; 5) Waste Management. They advocate a three-step analysis of a given issue:

- Analysis of the issue vis-à-vis distributional, procedural and recognition justice tenets
- Place the issue in the broader energy life cycle taking global interdependencies and issues into account.
- Employ applied principles for guidance on practical action and restorative justice as a condition for action.

Sovacool et al (2014, 2016) take a foundations approach to energy justice (cited in Sidorstov and McCauley, 2023). This is based on four assumptions:

1. Every human being is entitled to the minimum basic goods of life that is still consistent with respect for human dignity
2. The basic goods also include the opportunity to develop the characteristically human capacities needed for a flourishing human life
3. Energy is only an instrumental good – it is not an end in itself
4. Energy is a material prerequisite for many of the basic goods to which people are entitled

Based on these assumptions they develop two principles:

- Prohibitive Principle – Energy systems must be designed and constructed in such a way that they do not unduly interfere with the ability of any person to acquire those basic goods to which he or she is justly entitled.
- Affirmative Principle – If any of the basic goods to which every person is justly entitled can only be secured by means of energy services, then in that case there is also a derivative right to the energy service.

Sidorstov and McCauley (2023: 175) highlight how the prohibitive and affirmative principles foreground the balance that must be struck between the obligation to deliver energy services, and the

conditions under which those services ought to be delivered. The foundations approach thus posits that people have a right to energy service but also specifies the extent of that right.

Sovacool et al (2014) develop a Five Dimensions of Energy Justice Framework which can be used for descriptive, evaluative and/or prescriptive accounts of energy (in)justice.

1. Temporal Dimension – includes injustices arising from climate harms that will impact future peoples
2. Economic Dimension – includes inequitable distribution of energy services, energy poverty, economic impacts of resource depletion
3. Sociopolitical Dimension – includes human rights abuses and erosion of democratic institutions resulting from resource conflicts, resource curse, social marginalisation, corruption
4. Geographic Dimension – includes inequitable spatial distribution of risks and benefits e.g. establishment of sacrifice zones, community displacement, climate refugees
5. Technological Dimension – includes injustices embedded in the design and production of energy technologies.

Procedural injustices (whereby affected communities are entirely excluded or only nominally included in decision making processes) have led to the unjust distribution of power plants and waste facilities, and human rights abuses across affected communities. As in the literature on climate justice, energy justice scholars stress the need for much broader understandings of responsibility, beyond a focus on individual rights and responsibilities. Energy systems will possibly move away from fossil fuels (large urban plants, big employers) to decentralised alternative fuel systems (micro and medium-sized (rural) infrastructures for which the raw materials are often imported). Through local and household-level infrastructures individuals can decide to provide for their own energy needs and those of others. As this shift happens, so too our understandings of rights and responsibilities are likely to shift (Sidorstov and McCauley, 2023).

Spatial justice is concerned with framing the sites and scale of injustices. Its usefulness lies in how it helps identify where injustices are taking place as well as how others beyond the immediate location are impacted. Space can be conceptualised as absolute, relative, and relational at the same time (Przybylinski, 2023: 192).

- Absolute – a specific space that can be mapped e.g. the town centre
- Relative – the proximity of one space to others measured in time or physical distance e.g. the industrial estate is a mile away from the town centre/a 25 minute walk from the town centre
- Relational – the social relations that are active within a given space e.g. the town centre is where we go to socialise, shop and attend town hall meetings. The way in which the town centre is organised produces these social relations.

In order to understand how space is relational we need to understand how scale is constructed and why it matters. The urban, national, and global are socially constructed scales at which there is an interplay of power dynamics. Activities at each scale can be restructured and/or relocated e.g. glocalisation.

David Harvey (1973) first developed the concept of territorial justice arguing that it would require ‘a form of spatial organisation that maximises the prospects of the least fortunate region’ (2009: 110). Harvey criticised liberal theories of justice for not considering both distribution and production, arguing that ‘injustice in general derives from the ways in which capitalist production produces unequal social relations in space’ (Przybylinski, 2023: 193). Harvey added the concept of ‘eco-

generational’ to Young’s (1990) five faces of oppression -exploitation, marginalisation, powerlessness, cultural imperialism and violence. Eco-generational oppression refers to how ‘the necessary ecological consequences of all social projects have impacts on future generations as well as upon distant peoples’ (Harvey, 1992: 600 cited in Przybylinski, 2023: 194).

Marcuse (2009) suggests five propositions that could help identify spatial injustices:

1. It has two forms – 1) the involuntary confinement of any group to a limited space e.g. segregation; and/or 2) the allocation of resources unequally over space
2. It is derivative of broader social injustices
3. Social injustices always have a spatial aspect and social injustices cannot be addressed without also addressing their spatial aspect
4. Spatial remedies are necessary but not sufficient to remedy spatial injustice – let alone social injustices
5. Spatial injustice is dependent on changing, social, political, and economic conditions (Marcuse, 2009 cited in Przybylinski, 2023: 197).

Concepts that help identify how spatial injustice is continually reproduced by capitalism include the Right To The City (RTTC), Accumulation By Dispossession (ABD) and uneven development. Theorists argue that capital accumulation requires the perpetual remaking of geographical space for its continuance. Capitalist development produces uneven development because ‘capital requires a site of investment within which to engage in production, while at the same time, capital must remain mobile to circulate as value and thus remain available for investment elsewhere with higher rates of profit’ (Przybylinski, 2023: 201).

Worker’s rights are the central focus of **just transition** –with a focus on safeguarding livelihoods and employment rights during the shift to a sustainable economy. The focus on employment however can be further expanded by recognising that existing societal structures have embedded persistent inequalities and that, without corrective actions, structurally vulnerable groups are likely to be further disadvantaged through the transition to an environmentally sustainable economy, and ongoing processes of climate change and climate breakdown.

2.3. Understandings of Power

To effect a just transition, a good understanding of power relations is essential. Transforming norms and practices in order to enable liveable lives for all requires a comprehensive understanding of the various dimensions of power which can be summarised as the modalities, spaces, levels, and forms of power.

2.3.1. Modalities of Power

The first modality is ‘**power over**’. An analysis which is reliant on an understanding of power as ‘power over’ establishes a clear divide between ‘us’ the powerless, and ‘them’ the powerful. This results in a reductive understanding of power, domination, and subordination, in terms of a master/subject relationship. It implies that ‘we’, the powerless, are entirely innocent of harm and ‘they’, the powerful are guilty of harm (Allen, 2001). This ignores how we are all both powerful and powerless, dependent on intersecting power relations. Furthermore, it positions the ‘us’ as inherently powerless, leaving us with no substantive means of empowerment. Such an analysis orients ‘us’ to look to the State for protection from harm but State forms of retributive justice are reactive rather than preventive. Allen (2001) distinguishes between two types of ‘power over’, defined as constraining someone else’s options in a nontrivial way: 1) ‘Power over’ as domination wherein one constrains the options of others in a harmful way; 2) ‘Power over’ as a beneficent exercise, such as a parent’s power to

constrain a child's options for their own good. She further distinguishes between two forms of domination: 1) That which is exercised within the form of a master/subject relation, and 2) A far more fluid form of domination which is embedded in, and perpetuated through, cultural norms and social practices, institutions and structures.

'Power to' is related to our capacities to act. However, an overly simplistic invocation of the 'empowerment' of oppressed peoples to counter exercises of domination can efface the complexities of how power works. We may well be empowered to act in one way to counter our experiences of sexism, for example, whilst failing to recognise how those same actions uphold or reinforce other relations of domination and subordination, related to racism, for example. Allen (2001) cautions against conflating 'empowerment' with 'resistance' arguing that although empowerment is a precondition for resistance, it does not necessarily lead to the transformation of relations of domination. We may well be 'empowered' to act in ways that uphold or reinforce relations of domination.

A third modality of power is **'power-with'**, understood as the power we can enact in solidarity with one another in social movements. The feminist movement, the LGBTQI+ rights movement, labour and civil rights movements all draw on the power of collectives to progress their objectives (Allen, 2001; Gaventa, 2021).

A fourth modality of power is **'power-within'**. Building on the work of Paulo Freire (1997) this refers to the importance of 'conscientisation', understood as the development of a critical understanding of: 1) our own power-within, 2) the forces which shape dynamics of oppression and privilege (power-over) and 3) the possibility of actions (power-to and power-with) (Gaventa, 2021; Rowlands 1997; VeneKlasen and Miller, 2002).

More recently, and, with a view to better understanding transformative actions, Bradley (2020) has conceptualised the modality of **'power-for'**, as referring to:

the combined vision, values, and demands that orient our work. It inspires strategies and alternatives that hold the seeds of the world we want to create. 'Power-for' provides a logic to transformative power – motivating the sustained movement-building efforts that generate power to, with and within as building blocks for change (2020: 108 cited in Gaventa 2021: 5).

2.3.2. The Power Cube

A number of frameworks have been developed which link an analysis of the dimensions of power with organisational efforts to address unjust power relations (Gaventa 2006; Bradley 2020, Kashwan et al. 2019, Schutz 2019, Fung 2020). The 'power cube' developed by Gaventa (2006, 2021) outlines how the interrelationships of spaces, levels and forms of power can be analysed, and the possibilities of transformative action assessed.

Spaces for participation

Closed Spaces – Many decision-making spaces remain closed in the sense that officials (bureaucrats, elected representatives, experts) make decisions and provide services without the need for broader consultation. Civil society actors argue for greater public involvement, transparency or accountability in these closed spaces (Gaventa, 2006).

Invited Spaces – These are spaces into which ordinary people are invited to participate by government, non-governmental organisations, or supranational agencies (Cornwall, 2002). They may be institutionalised and regularised (e.g. seat at local government meetings) or once-off consultations (Gaventa, 2006).

Claimed/Organic/Third Spaces – These are spaces created by less powerful actors out of their common concerns. They may be the outcome of popular mobilisation and involve processes of debate, dialogue and contestation outside of institutionalised policy arenas (Gaventa, 2006; Cornwall, 2002).

These are very high-level categories. In reality, all spaces exist in dynamic relationship to one another; whoever creates the space is more likely to have power within it; social actors may have a lot of power in one space but relatively little power in others; and power gained in one space, through the development of critical consciousness and new skills for example, can be used to enter other spaces.

To challenge ‘closed’ spaces, civil society organisations may serve the role of advocates, arguing for greater transparency, more democratic structures, or greater forms of public accountability. As new ‘invited’ spaces emerge, civil society organisations may need other strategies of how to negotiate and collaborate ‘at the table’, which may require shifting from more confrontational advocacy methods. At the same time, research shows that ‘invited spaces’ must be held open by ongoing demands of social movements, and that more autonomous spaces of participation are important for new demands to develop and to grow (Gaventa, 2006: 27)

For transformative change to occur it must concurrently span closed, invited and created spaces each of which involves different, interrelated strategies, skills and resources.

Levels of Power

In a globalised world we cannot only focus on power at the local or national level. Gaventa (2006, 2021) outlines four levels of power:

- Global – formal and informal sites of decision-making beyond the nation state;
- National – governments, parliaments, political parties or other forms of authority linked to nation-states;
- Local – subnational governments, councils and associations at the local level;
- Household – the micro-level (2021:14)

Whereas Gaventa (2021: 14) notes that the household level ‘helps to shape’ what happens within the ‘public sphere’, a feminist lens would argue the inverse – that the public sphere is built upon and at the same time effaces relations of domination which are first structured at the household level (Zerilli, 2006).

Each of the levels – global, national, local and household – are interrelated with the challenge for action ‘not only how to build participatory action at differing levels, but how to promote the democratic and accountable vertical links across actors at each level’ (Gaventa, 2006: 28).

Forms of Power

Building on Lukes’ (1974) ‘three faces of power’ and the work of VeneKlasen and Miller (2002), Gaventa (2006: 19) presents three forms of power and discusses them in relation to civil society.

- **Visible power:** observable and recorded decision making. This level includes the visible aspects of political power – the rules, structures, authorities, institutions and procedures of decision making. **Civil society action** at this level seeks to change the ‘who, how and what’ of policymaking so that the policy process is more democratic and accountable, serving the needs and rights of people within planetary boundaries.
- **Hidden power:** agenda-setting. This is the ability to control who gets to the decision-making table and what gets on the agenda. Hidden power operates at multiple levels to exclude and devalue the concerns of less powerful groups. **Civil society action** at this level can work to build the power of social movements and new leadership, to influence

the shaping of the political agenda and increase visibility and legitimacy of civil society issues, demands and voices.

- **Invisible power:** shaping meaning itself. This relates to the embeddedness of harmful cultural norms which shape the boundaries of what is acceptable. As Gaventa notes 'processes of socialisation, culture and ideology perpetuate exclusion and inequality by defining what is normal, acceptable and safe. **Change strategies in this area target social and political culture** as well as individual consciousness to transform the way people perceive themselves and those around them, and how they envisage future possibilities and alternatives' (2006: 19).

Civil society actors often focus their efforts primarily at one form of power. One example is the use of public debate and evidence-based policy proposals to influence observable decision making processes (addressing visible power). Another example is the mobilisation of people and forms of collective action to get previously ignored concerns on the agenda and to get a seat at the decision-making table (addressing hidden power). A third example is awareness-raising campaigns which challenge harmful exclusionary norms and extend the boundaries of what is 'normal' (addressing invisible power). Strategies which link across each form of power – visible, hidden, invisible – are crucial to enable transformative change. Otherwise as Gaventa notes,

... a policy victory in the visible arena of power may be important, but may not be sustained, if those outside the arena are not aware that it has occurred and how it relates to their interests, or are not mobilised to make sure that other hidden forms of power do not preclude its implementation (2006: 30).

A policy victory for one group, which fails to take intersecting power dynamics into account, may also reinforce the domination and subordination of others (Crenshaw, 1991).

Summing up, an analysis of power and the development of effective strategies for action must take account of the interrelationships between various modalities of power (power-over, power-to, power-with, power-within and power-for); spaces of decision making (closed, invited, created); levels of power (global, national, local and household); and forms of power (visible, hidden, invisible). The same norms, practices, institutions and structures which constrain our possibilities to act are also the vehicles through which we must act. Our navigation of power relations, and our capacities to act, certainly provide for the possibility of resistance to relations of domination (and the transformation of norms and structures) but this is not necessarily the outcome of our actions.

2.3.3. An Intersectional Approach to Power Relations

Taking an intersectional approach to the analysis and transformation of power relations necessitates addressing 'invisible power' i.e. how meaning itself is shaped, making visible the embeddedness of harmful cultural norms, and transforming the structures and practices that derive from those harmful norms.

Within feminist theory it has long been recognised that the logic underlying binary thinking (masculine/feminine, universal/particular, marked/unmarked) and the resultant processes of identity categorisation are socially constructed, culturally and historically particular, and operations of power. However, many Global North feminists understood categories of difference other than gender (e.g. race, sexual orientation, dis/ability etc) as secondary to gender (Mohanty, 1991; 2003). This prioritisation of combatting sexism as it is experienced by middle-class white feminists, meant that many theorists and activists could not understand (or were actively indifferent to) how Black women, women within the LGBTQI+ community, or disabled women for example are gendered differently to able-bodied, straight, white women, and so how they experience sexism differently.

It is important to understand that to take an intersectional approach does not mean, for example, acknowledging that Black women suffer from both sexism (as white women do) and racism (as Black men do). There is some, but not much, commonality between white women's experiences of sexism and Black women's experiences of the *intersections* of sexism and racism. This is because white women and Black women are gendered differently through distinct norms and expectations. In the context of the US for example, historically white women were defined as weak, passive, irrational, in need of protection, and they were confined to the private domestic sphere. Historically Black women were subject to enormous levels of violence through enslavement and segregation. They were characterised as strong enough to do any labour; as perverted and/or sexually aggressive in order to justify widespread sexual assault by white men; and as fertile, to justify being a 'wet nurse' to white children and emotionally nurturing white slave owners (Collins 2000: 82 cited in Lugones 2007: 204).

Arguably the gendered and racialised norms which were embedded from the 17th century onward have not been significantly transformed. Straight, white, able-bodied men are still understood as the norm of 'the human'; boys and girls are still socialised in highly gendered ways; Black people of all genders are still stereotyped as animalistic and uncontrolled; and Black women's bodies continue to be hypersexualised (Lugones, 2007, 2010; Anyangwe, 2015). This is a function of histories, norms and structures. It is not 'the fault' of any individual, or group of people (Young, 2011). That said, in order to change course and enable livable lives for all, it is our collective responsibility to work to interrupt and transform norms and structures. If we are to transform norms that work to privilege some while subordinating others, we need to understand the origins of those norms, why they were (and are) deployed, and to what effects.

There are certainly commonalities across experiences of sexism, for example, but there are also important dimensions to experiences of oppression that are missed if we ignore how multiple dynamics of oppression intersect and work in concert. Indeed, we do active harm without such an analysis. As Crenshaw notes the 'failure of feminism to interrogate race means that the resistance strategies of feminism will often replicate and reinforce the subordination of people of color' (1991: 1252). Yet within some strands of feminism there remain significant levels of resistance to the development of a robust analysis of racism in particular (see Bilge 2013; 2020)

To take an intersectional approach is to understand that:

- Norms, structures and institutions create and reproduce socially stratified population groups i.e. our cultural, material and political realities assign us to groups which struggle for inclusion (e.g. experiences of domestic violence, workplace exclusion and the gender pay gap, the prohibition of same-sex marriage, inaccessible public transport, legacies of segregation and enslavement, discriminatory asylum processes) (Yuval-Davis, 1997; Collins, 2017)
- Public policy is already intersectional in the sense that it assumes and centres an able-bodied, full-time employed, heterosexual citizen with no significant caring responsibilities, as the primary unit of concern across all spheres of engagement. The intersection of whiteness, ability and good health, full-time labour force participation, citizenship, and no caring responsibilities, is valued.
- If we adopt an intersectional approach without a good understanding of both dynamics of oppression and dynamics of privilege, we risk replicating and reinforcing the subordination of others whose oppression we either cannot understand or to which we are indifferent (Collins, 2017; Crenshaw, 1991)

It is significant that intersectionality as a concept originated from Black women scholar-activists. By virtue of their social location, Black women have had to be attuned to multiple dynamics of oppression and privilege operative in any given space. Political practices such as 'flexible solidarity' (Collins, 2017) are an outcome of this attunement. Flexible solidarity is characterised by relationships of both compromise and contestation. It recognises the need for solidarity among groups. It also calibrates ideas and actions to keep in view the dynamics of oppression and privilege that are operative, and this work informs transformational strategies.

The concept of intersectionality was thus developed by Black women scholar-activists to address the forms that violence takes, within distinct yet intersecting systems of power. To take an intersectional approach is to understand and address the effects of violence enacted through, for example, racism, sexism, homophobia and class exploitation. Part of this approach is to understand how both dynamics of oppression and dynamics of privilege are operative and, in so doing, to work to not replicate and reinforce the oppression of others.

To take an intersectional approach is arguably to take an 'and/and/and' approach to categories such as race, gender and sexual orientation. Yes they are socially constructed (1); yes they are deployed in ways that privilege some and subordinate others (2); and yes we can work to transform them - those who are subordinated can subvert the naming process in empowering ways, and we can all resist the discriminatory narratives upon which processes of categorisation are based, alongside addressing the material inequalities that are the outcomes of those narratives, and of our histories (3) (Crenshaw, 1991).

3. Existing structural and spatial inequalities in Europe

It is crucial to acknowledge that the twin transition does not occur in a vacuum – quite the opposite, the shift towards an increasingly ‘green’ and ‘digital’ society takes place in societies that are already challenged by multiple intersecting inequalities. This needs to be accounted for. While this document does not aim at providing a comprehensive analysis of the state of inequality in Europe, there are certain patterns which require to be mapped prior to considering potential inequality impacts of twin transition on different groups. First, the most pronounced structural inequalities within the nation states should be noted. These produce systems of privilege and oppression/discrimination based on factors such as gender, socio-economic background (and, related to this, labour market position, wealth, income and educational status), family status, ethnicity and migration background, and age. These structures create an ‘unequal distribution of material resources, capabilities, and outcomes within or across societies and populations’, often measured through income or wealth, but also through other indicators such as health, education, decision making, power and various other metrics (Sweeney, 2018). Secondly, historical context is also of importance, namely an understanding of how previous crises and shifts created new inequalities or exacerbated existing ones, with some consideration also given to the way in which states responded to these systemic shocks with measures which either protected structurally vulnerable populations, or which resulted in the deepening of social divisions. Here, lessons from the two most recent crises, i.e., the financial crash of 2008 (and the subsequent recession), and the global COVID-19 pandemic, are interesting to be considered to inform our understanding of the dynamics between macro-level socio-economic shifts and structural inequalities.

Despite growing prosperity in Europe, structural inequalities continue to persist, both within and between the member states (McCaughey, 2024). Arguably, without acknowledging these, strategies developed within the twin transition will only allow to treat the symptoms and not the cause – particularly because the climate crisis itself is deeply rooted in structural inequalities between different groups in society and regions of the world (Arabadijeva, 2022).

This section will provide a brief overview of the different dimensions of existing structural inequalities in the European context, with the focus placed on inequality dimensions, and the impact of previous systemic shocks. While the analysis will present a ‘bird-eye’ view on these, examples from the FITTER six case studies will also be included to allow for a better understanding of the specific issues occurring on a national level. In relation to the dimensions of inequalities related in this section, two points need to be emphasised. First, the so-called ‘vulnerability groups’ are categorised under various definitions, also differentiating between national contexts. For the purposes of this analysis, we consider them as ‘structurally vulnerable’ to account for systemic issues rather than to shift the responsibility onto the individual. Second, we distinguish between different inequality grounds (such as gender or age) and inequality dimensions (such as education or labour market), yet it is crucial to also understand these are intersecting and interrelated on different levels, with multiple inequalities affecting individuals at the same time, and in different aspects of their lives.

3.1. Structural inequalities and vulnerabilities in the context of the twin transition

There is no agreed upon definition of vulnerability, nor an agreed upon criteria for identifying vulnerable groups, or a standard list of such groups in EU law or UN Human Rights Law. Typically, human rights bodies deal with vulnerable and disadvantaged communities on an ad hoc basis. (Chapman and Carbonetti 2011). EU equality law covers discrimination grounds based on sex/gender, race and ethnicity, age, sexual orientation, disability, religion or belief. In addition, individual states

adopt their own definitions of such categories, especially in the context of overlapping crises and the possible state response strategies. For example, the German government adapts their own classification of 'vulnerable groups' defined as 'individual population groups' including 'young and old people, women, men and non-binary people, LGBTQI+ people, people living in poverty, people with disabilities or other health impairments, refugees, people with a history of flight or migration or families separated as a result of displacement, single-parent households, people in other particularly vulnerable life situations' who are, however not treated as passive, but rather perceived as stakeholders necessary to include in disaster risk management (Federal Ministry of Interior and Community, 2022). Another example of more general approach to inequalities is one taking on a problem rather than a group perspective – for example, by understanding structural inequalities as influenced by housing crisis or healthcare crisis for Ireland (see Murphy, 2023a).

Income inequality, poverty, and social exclusion and how they affect different groups is also a cross-cutting theme when examining inequality and structural vulnerability. For example, Germany continues to have a significant gap between the top and the bottom income quintiles, with the top fifth usually 4.3 times higher than the bottom fifth. In addition, poverty (defined as income level lower than 50% of the median) stood at 16.1% in 2022. Among those most likely to fall into poverty were unemployed, mini-jobbers, East Germans, women, single parents, people with migrant background, single adults, and those with education at the equivalent of lower secondary. In Hungary, income inequalities increased over the past decade (with links to an introduction of single-rate income tax). This further intersects with other inequality grounds, such as gender, ethnic background or place of residence. For example, the gender income gap currently stands at 35%. Furthermore, parental background tends to influence children's school performance, while social class also influences overall social mobility. In Ireland, it has been estimated that around 13% of the population lives in poverty, and this intersects with other grounds such as age (poverty affecting children and older people).

This can be further linked to **socio-economic background**, otherwise referred to as 'social origin'. While not universally recognised by national equality legislation, socio-economic background has been argued to be an important driver of inequality and a source of discrimination linked, for example, to address, or accent – and also associated to education, income and occupational status. It has been argued that, while often intersecting with other inequalities, socio-economic background is a cause of social exclusion, preventing people from fully accessing employment, education or housing. When measured in economic terms, socio-economic background hinders social mobility, as research shows. For example, in the European OECD countries children from disadvantaged background earn on average 20% less than those with more favourable childhood, while socio-economic standing of parents also influences children's educational outcomes, leading to differences in job quality in later life (OECD, 2019)

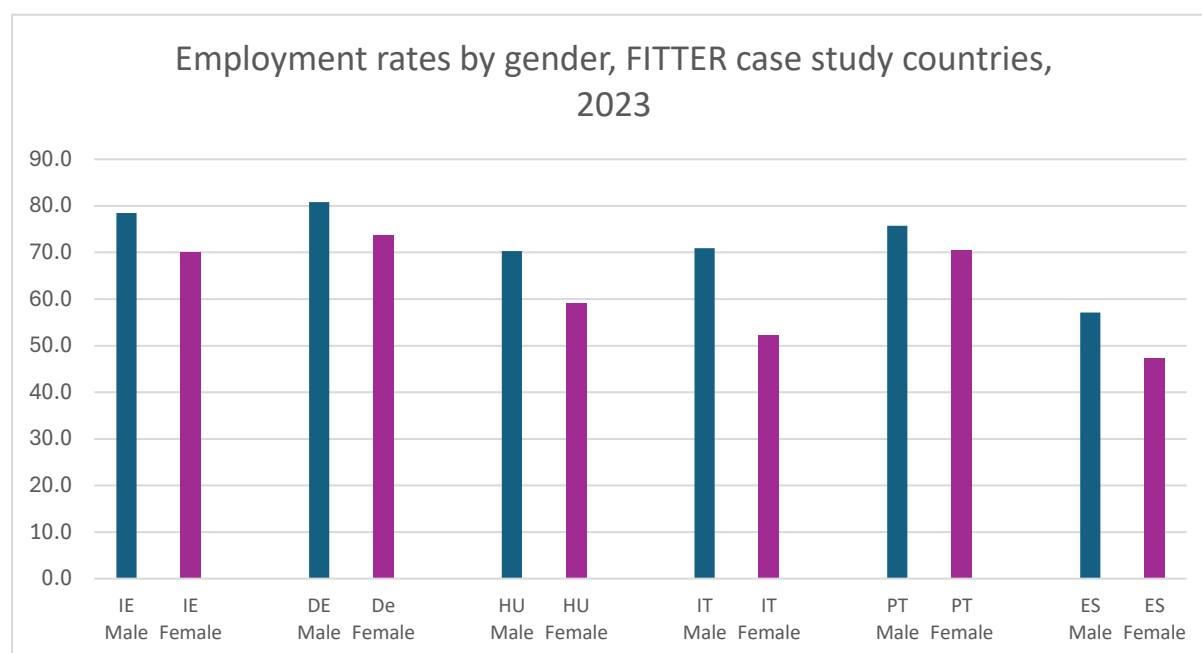
Socio-economic status, combined with labour market position, have been argued to be important factors in exacerbating inequalities during crisis situations. The financial crash of 2008 resulted in widening of income inequalities – most strikingly between working and upper middle class income groups (Moawad, 2023). The COVID-19 pandemic also affected those in low-paid positions as they were most likely to be located in sectors which were affected by the health-related closures. Thus, groups affected by job losses tended to be among the most vulnerable in terms of socio-economic status (Smit et al., 2023)

Indeed, in Germany, social inequalities emerging in the context of the 2008 financial crash had been paved long before the crisis (Kadritzke, 2009; Groh-Samberg, 2007) and the employment conditions of the working-class were affected in the most severe ways (Groh-Samberg, 2007). In Spain too, it was the lower income workers who were impacted at a greater extent, while in Portugal low-skilled

labourers were among those who experienced diminishing employment prospects during the global financial downturn. Likewise, in Ireland, a country where the construction sector was particularly impacted by the great recession, the unemployment rates soared among working-class occupations clustered around construction, which was the sector that suffered the most in the aftermath of the financial crisis (Bobek et al., 2018). Furthermore, those working in lower-paid jobs were most likely to suffer from employment loss in Ireland during COVID-19, with those working in sectors such as hospitality or retail being also impacted by the health-related lock downs. In Hungary, those of higher occupational status were in a more advantaged position during the pandemic, while those who were most likely to lose their jobs were among the most disadvantaged. In Portugal, groups with lower socioeconomic status experienced an increased risk of COVID-19 infection, combined with an increased risk of symptomatic and severe course of the disease. These inequalities were amplified by the differential impact of non-medical and non-pharmaceutical interventions, such as contact and mobility restrictions, differences in access to optimal care, and the unequal psychosocial, health, and socioeconomic consequences of the pandemic. Low-income workers were also among those who faced job losses and decreased working hours, while also encountering challenges related to accessing digital technologies.

Gender remains one of the most pronounced inequality dimensions in Europe, as traditional gender roles, in particular those related to caring duties, are still deeply rooted in social norms, thus influencing women's paid and unpaid work (Eurofound, 2022). Inequalities between men and women continue to be an issue – with substantial gender pay gaps, gaps in employment rates, disparities in political representation and under-representation in senior management as substantial points in the discussion on gender equality (Klasen and Minasyan, 2017). Existing gender inequalities strongly featured in our country-specific review analysis. For example, in Germany, young women, especially those with lower levels of education, are less likely to be in employment than young men. Women with tertiary education have the lowest income when comparing them to men with a similar level of education (OECD, 2021): they earn only 70 % of the income of men, although this gender gap is narrower regarding women with an upper secondary or post-secondary education (they earn 82% of the income of men). In relation to employment, available data from the 6 FITTER country reports clearly shows that female employment rates have generally risen on the national level, with approximately 10 percentage points over the past two decades. However, there are still significant differences between the countries, with Italy and Spain having female employment rates on a lower level. The following graph illustrates these differences for 2023:

Figure 3.1: Employment rates by gender in FITTER case study countries



Single parent households were also identified as being at risk of social exclusion. While the family situation is not only affecting women, single parent households in Europe are usually led by females – thus inequalities linked to family status are intersected with gender. This was highlighted as also often intersecting with monetary poverty in Germany. Single parents, as well as households with minors, were also identified as at risk of social exclusion in Spain.

Women were also unequally impacted by the two recent crises. The economic crash of 2008 impacted male employment more severely as the crisis was strongly linked with the downturn in construction sector, yet women were negatively affected by the subsequent cuts in public spending and investment in services (Karamessini and Rubery, 2014). The COVID-19 pandemic and the health-related lockdowns only exacerbated these inequalities. First, it amplified the existing gaps in care due to the closure of schools and childcare facilities (EIGE, 2022), and subsequently impacted women’s work. For example, international research demonstrated that women were more likely to either leave employment during the lockdowns, or at least reduce working hours (Hjalmsdottir and Bjarnadottir, 2020). Women were also overrepresented among key workers who experienced increased exposure to the virus (Blundell et al, 2020), and also overrepresented in those sectors most affected by closures (Cook and Grimshaw, 2021; Matthewman and Huppatz, 2020). Furthermore, the consequences of health measures and lockdowns also resulted in a reduced access to sexual and reproductive health services, increased domestic violence and overall disproportional effects on women livelihoods (Wenhan et al., 2020). The negative impacts of the pandemic on women were noted in the Irish country review, with particular reference to the unequal distribution of care (McKane, 2021). In terms of employment, the Italian, the Spanish and Hungarian case studies emphasised that the closures were pronounced in sectors related to tourism – such as hotels and restaurants – which are traditionally overrepresented by female workforce. In addition, in some countries such as Italy, the tourism sector has high prevalence of ‘undeclared’ labour, and this could have had negative impacts on accessing state supports during the closures.

Age is another important dimension of inequality – understood as inequalities linked with old age, but also considered as a factor across the life course, with socio-economic analysis also often turning

attention to issues such as child poverty, or the situation of young people. Examples include economic inequalities linked to old age, particularly in the light of a shift towards the privatisation of pensions (Ebbinghaus, 2017), and low pay work among older citizens. In this context, age also intersects with gender as women continue to have lower pensions than men, due to gender pay gap, labour market participation (often linked with gender division of care work) and the higher share of part-time work among women (Burkevica et al., 2015). On the other end of the spectrum, generational challenges affect young people and their labour market position when entering the workforce during downturns (Anders and Macmillan, 2020), or their housing tenure in the context of the widespread housing crises. (Eurofound, 2024).

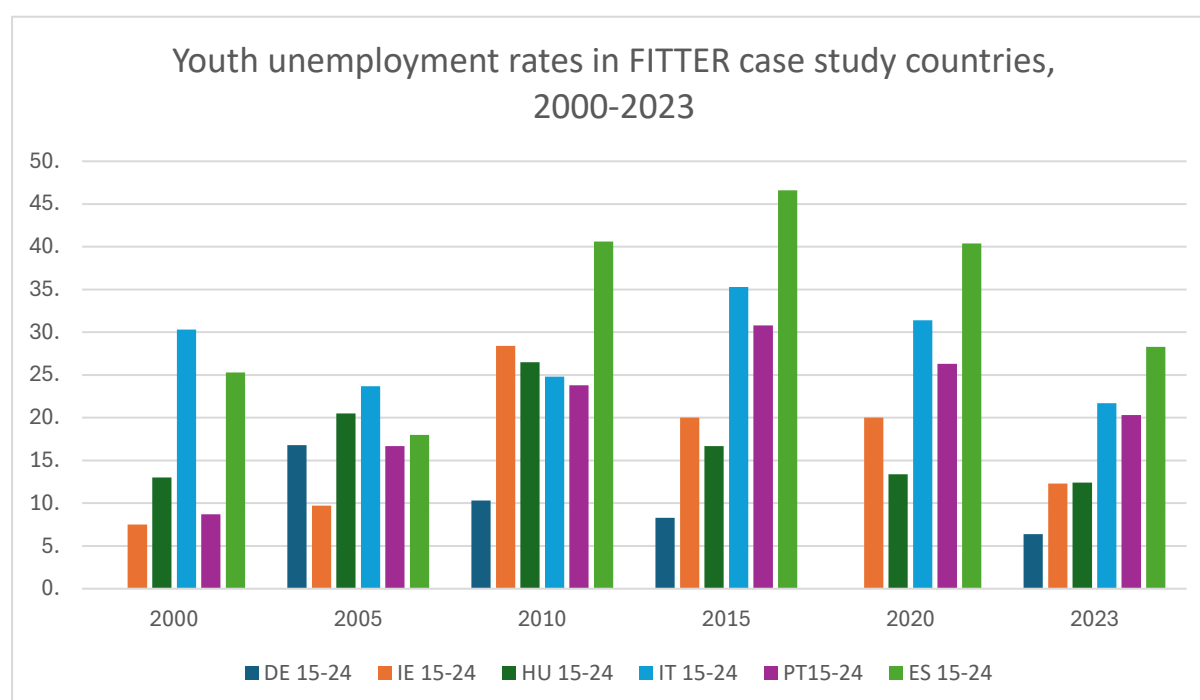
Our national reviews highlighted age as one of the grounds of existing inequalities. In Ireland, it was evident that age has an impact on labour market position. As demonstrated by available data, levels of employment among older age groups (55+) are relatively low, and particularly low (just below 50%) among those between 60-65, which suggests a reliance on state payments. There are also high rates of low pay workers in the older age groups (Collins and O'Dare, 2022). On the other hand, in Hungary, young people, particularly children, were highlighted as a group which requires further scrutiny, as the performance of Hungarian students lags behind the European average and the proportion of low-achieving students is higher.

The 2008 recession has not been usually discussed as having negative effects on the older population, with younger persons receiving more significant attention as a group impacted by this crisis. Labour market participation was of particular concern due to high youth unemployment levels, which, as studies shown, may have led to the so-called 'scarring effect' (Anders and Macmillan, 2020), namely the long-term impact of early issues related to finding suitable employment. For example, in Ireland, this crisis had a severe impact on young people as the total employment among under-25s' fell over by 50 percent between 2007 and 2011 (CSO, 2012) and emigration became a key strategy for many young people at that time (Moriarty et al., 2015). The Irish economic crisis also saw a rise in jobless households - many of them including children (NESC, 2013).

The COVID-19 pandemic proved to be especially difficult for older persons, who were subjected to mandatory health measures, thus suffering unprecedented levels of social isolation (Lazzari and Rabottini, 2022). In the context of the lockdowns, younger cohorts were also negatively affected, especially during the closure of schools, universities, and childcare facilities. Here, socio-economic background was a strong factor, as children from poorer backgrounds were more likely to experience difficulties in accessing the digital technologies necessary to support their learning and social connections (Ayllon et al., 2023). In Hungary, for example, students from lower social status had less opportunities of online learning, as they were lacking equipment and access to internet, and were often receiving lower levels of support within their families. Similar challenges experienced by children and young people were also noted in Ireland. The Irish review also highlighted that older people were amongst the most affected during the pandemic, due to 'cocooning' measures resulting in prolonged isolation periods, which impacted them in terms of access to health services. Reduced social contact to a minimum (or none) thus had a strong effect on both their physical and mental health.

In terms of employment of young persons specifically, data collected for the country reviews show a variation of youth unemployment among the 6 countries and across the time. The following graph shows the youth unemployment (15-24) in 6 time points between 2000 and 2023:

Figure 3.2: Youth unemployment in FITTER case study countries

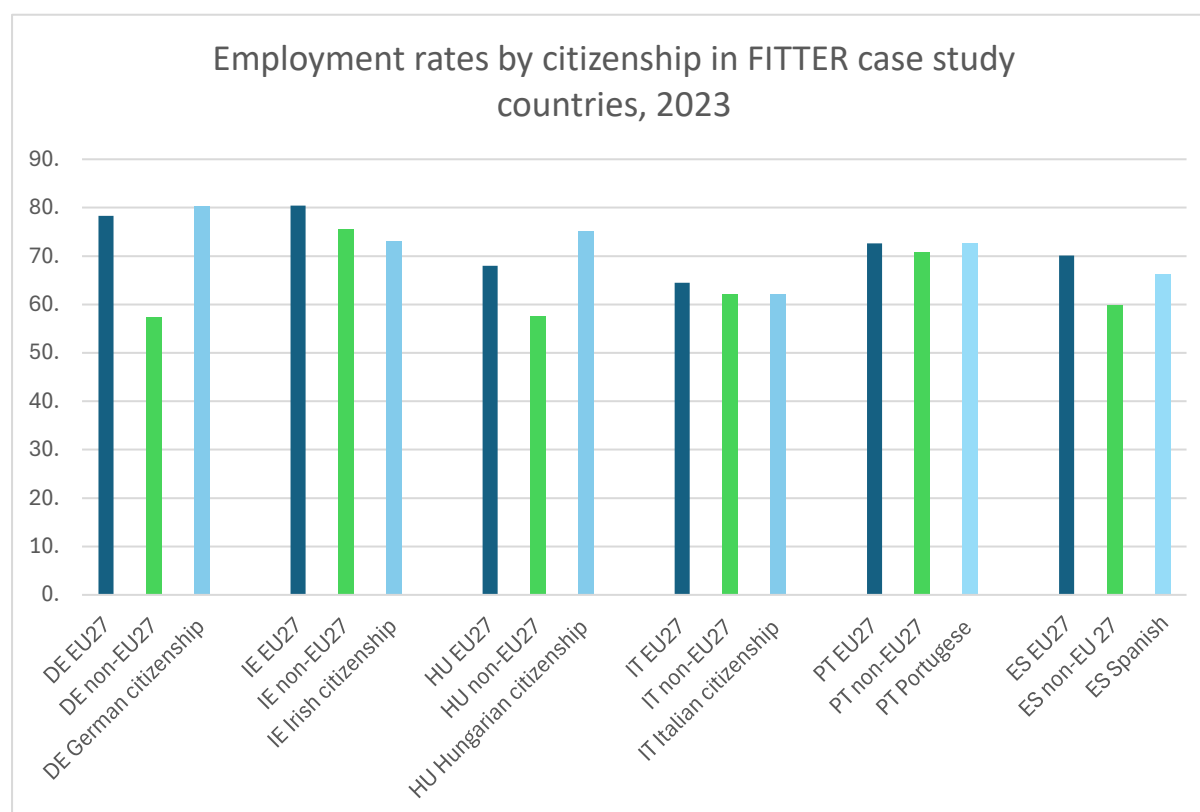


As can be clearly seen, in all the countries the 2008 crash had a negative impact on the unemployment levels of young people, as these increased sharply following the downturn. Interestingly, there is no evidence suggesting any strong impact of COVID-19 on youth unemployment rates. Nevertheless, economic shifts linked to the twin transition are likely to have a direct effect on these rates, so this needs to be further monitored, particularly in countries which recently had relatively high unemployment rates for those under 24 years of age: Italy, Spain and Portugal. Within this context, scrutiny should also be given to gender differences, with particular attention to such factors as levels of education, participation in STEM disciplines and the proportion of those Not in Employment, Education or Training (NEETs).

There is a strong correlation between **migration background** and inequalities. In Europe, migrants, when compared with indigenous population, are facing higher risk of social exclusion and poverty, especially if they originally came from a non-EU country (Lelkes and Zolyomi, 2011). Migrants often are placed in low pay sectors of the host country labour markets (Castles, 2015), excluded from political participation and segregated by the mode of entry which determines their civil and social rights (Kofman and Sales, 1998). Furthermore, ethnic background, which is not synonymous with migration background, also shapes inequalities, as health outcomes, education and employment vary significantly between ethnic minority groups (Modood et al., 1997). Migration status also intersects with gender, as female migrants faced multiple vulnerabilities, particularly in the context of migration and care work and domestic work (Fleury, 2016).

In relation to employment, data collected as part of the FITTER country-specific reports shows that citizenship status may be correlated with employment rates, however, the relationship is not as straightforward as in cases of other inequality grounds. The following graph illustrates the relationship between citizenship status and employment rates in the six countries:

Figure 3.3: Employment rates by citizenship in FITTER case study countries



As this graph demonstrates, in most cases, employment rates tended to be the lowest among those with a citizenship from outside the EU – with the exception of Ireland and Italy. However, rates of employment for those citizens from any other EU 27 Member States were often higher than those having a citizenship from the reporting country, particularly in Ireland and Spain. Given this, more scrutiny is required to explore the potential vulnerabilities and privileges of diverse migrants in employment, also suggesting that an intersectional approach is required to allow for other factors to be examined.

In relation to **migration status**, this was identified as a background factor influencing education of young children in Germany. e.g. by facing racism or by facing language barriers. However, it seems that migration status does not have a huge influence when people with a migration background are already employed (OECD, 2021). In 2019, full-time employed adults not born in Germany, with an educational level below upper secondary level, earned 95% of the income of those born in Germany with the same educational level (ibid.). The figure for adults with an upper secondary or post-secondary non-tertiary qualification was 93%, and 89 % for adults with tertiary education (ibid.). Employment rates are also significantly lower for non-EU migrants, compared with those for the native population and to those coming from other EU countries. In Ireland, migrants tend to be concentrated in lower-paid sectors, such as hospitality or retail, which affects income inequalities. Studies show that the global financial crises had a negative impact on migrants as the unemployment levels of persons born outside of their country tended to increase twice as fast as those of native-born workers in the OECD countries (Castles, 2015). They were often then trapped in ‘networks of instability’, while facing barriers with accessing welfare state support which is determined by the legal status (Avato et al., 2010). The COVID-19 crisis also disproportionately affected migrants and ethnic minority groups. Ethnic minorities were often more exposed to the virus itself, and they were also over-represented among key workers thus making them more exposed to health risks (Blundell et al., 2020). On the

other hand, as migrants often have less stable jobs, they were more vulnerable to the economic shocks related to the pandemic (OECD, 2022a).

For example, in Portugal migrants were among those who were more vulnerable to job losses, decreased working hours and economic uncertainty. Barriers in accessing health services were also noted in the Spanish review, as language barriers hindered accessibility of telemedicine. In Ireland, migration status was of importance during lockdowns as those living in Direct Provision Centres accommodating asylum seekers were experiencing inadequate living conditions due to the lack of physical space.

While these factors were among the most noted inequality grounds in the 6 FITTER country reviews, it should also be emphasised that **disability, sexual orientation and gender identity** are other factors that would require further scrutiny in the future analysis of the effects of transition on existing inequalities to be undertaken within the project. **Location** is also a factor to be taken into account. As will be discussed in the next section, spatial variations are important for the analysis at the European level, particularly in relation to variations between countries and between regions within nation states. In addition, **rural-urban** divisions related to households should also be explored as a source of structural inequalities. For example, in Ireland rural communities have been disproportionately affected by the two previous crises (Faulkner et al., 2019). In general, the rural-urban divide poses questions for equality issues in relation to the twin transition, particularly in terms of accessing digital infrastructure or green transport infrastructure, and can be further related to transport poverty, discussed in section 4.

4. Key academic debate on twin transition: shifts, impacts and inequalities

In this section, we explore inequalities linked to different aspects of the twin transition by providing an overview of the existing literature. However, the literature in relation to some of these inequalities is too fragmented and thus a more complex analysis will be needed to develop a conceptualisation of a 'just' twin transition.

4.1. Green, digital and twin transition: main trends and relationship with social and structural inequalities in academic debate at the European level

Within the EU, the discourse of sustainability and sustainable development is framed around the European Green Deal (EGD) and focuses on the green economic growth, low-carbon infrastructures and technological solutions (Hereu-Morales et al. 2023; Eckert and Kolevska 2021; Bauhardt 2022; Mastini et al. 2021; Dunlap and Laratte 2022). It has been argued that while the EGD is deemed a transformative masterplan in terms of public planning for sustainability, and is clearly oriented towards achieving the climate objectives of the UN's Paris Agreement and towards embedding the sustainability vision of the 2030 Agenda, it may not be fit to fulfil its purpose as the overall focus on economic growth does not acknowledge sustainable development goals not directly connected with the economy (Hereu-Morales et al. 2023: 11). This further links with the academic counter-debate identifying sustainability as connected to degrowth, with principles including minimising the human impact on the environment, reducing the economic activities and putting the emphasis on social justice and equality (Eckert and Kovalevska 2021: 19). This parallel debate, however, also encounters difficulties as the corporate and industrial interests tend to block relevant legislative shifts, thus altering European socio-economic values, with the importance of 'understanding the social impact of progress and development, and of the present relationship between humans and nature.' (Eckert and Kovalevska 2021: 19). Moreover, the academic and activist debate concerning degrowth is manifold, although it can be considered an umbrella concept rather than a political strategy, as it highlights the variety of social struggles connected to a green transition but has not yet found its way into policy impact (Mastini et al. 2021: 3).

One major concern in the debate bringing together EGD and de-growth is renewable energy infrastructures. Some voices consider the presence of so-called 'renewable energy' infrastructures in landscapes and ecosystems as colonising (Dunlap and Laratte 2022), so the question is: 'environmental justice for whom?' This question surrounds distributional justice of benefits and costs of the green transition and relates to supply chain violence and justice, mostly affecting countries in the Global South due to the material extraction required for the green transition at large and especially the clean energy transition and electric mobility (Dunlap and Laratte 2022; Mastini et al. 2021). What would such a Green New Deal without growth look like is exemplified by work by Mastini et al. (2021:7) which highlights that, in the light of reducing material extraction and energy use, GDP growth will automatically decline, and this can be prepared by reducing working hours, distributing wealth through a green tax reform and environmental and wealth caps.

Overall, climate action and its societal implications are often discussed through a labour market lens, as well as wider impacts on local and national economic systems. For example, one of the issues featuring in the literature is the fossil fuel dependency of some of the SMEs, particularly in relation to identifying core actors participating in the change. In this case, and in relation to communities, it has been emphasised that environmental damage cannot be left unaddressed by economic actors nor to be transferred to the public pursue (Healy and Barry, 2018; Just Transition Alliance, 2018). In addition, the debate on the green transition and communities also refers to so-called 'carbon lock-in' (Unruh,

2000) and its different dimensions: infrastructure and technology, behavior and institutions (Goldrick-Kelly, 2021).

In relation to labour markets specifically, the transition towards sustainability requires changes in the production and consumption of goods, with inevitable consequences for jobs and skills demand (Rollnik-Sadowska, 2023). It has been noted that the green transition can boost employment and support 'job-rich recovery'. However, these will require investments, particularly in terms of skills, as well as the right mix of incentives (Revoltella, 2020). Moreover, if not managed well, the transition to a low-carbon economy may result in job losses and cause hardship, particularly to the communities that currently rely on carbon-intensive activities (Atteridge and Strambo, 2020), with the Polish coal basin in the region of Upper Silesia often provided as an example of such communities (Jonek-Kowalska, 2015).

Studies anticipate a limited impact of the green transition on employment rates overall (Malerba and Wiebe, 2021; Vandeplas et al., 2022), with shifts 'likely to occur between sectors, firms, occupations and regions' (Vandeplas et al., 2022: 5). As a result, the positive and negative repercussions and implications of the green transition on the population may be unevenly (and inequitably) distributed. In addition, the working conditions of new jobs are not guaranteed to be adequate. New jobs may be created in sectors with high in-work poverty rates and low social protection, so a proactive policy will be needed to ensure fair wages and adequate working conditions (Malerba and Wiebe, 2021). Studies have focused on the characteristics of 'green jobs' and 'green skills', anticipating a shift in demand that will affect vulnerability to job loss (Urban et al., 2023; Vandeplas et al., 2022; Vona, 2021). Nonetheless, there is a lack of an agreed, unified definition of 'green jobs' which results in a difficulty in separating 'green' jobs from other jobs (Eurofound 2023b). The ILO-UNEP defines 'green jobs' as positions in agriculture, manufacturing, construction, installation, and maintenance, as well as scientific and technical, administrative, and service-related activities, that contribute substantially to preserving or restoring environmental quality, including 'jobs that help to protect and restore ecosystems and biodiversity; reduce energy, materials, and water consumption through high-efficiency and avoidance strategies; de-carbonise the economy; and minimize or altogether avoid generation of all forms of waste and pollution.' (ILO-UNEP, 2008, 35-36). This definition identifies 'green jobs' are those usually located in sectors dominated by men, while jobs conducted mainly by women (such as care jobs), are often 'invisibilised' in the context of the green transition (Culot and Wiese, 2022). This issue has been discussed as critical in relation to the importance of care work and its gendered nature, with calls to include a variety of caring roles as 'green' jobs (e.g. Klein, 2022, Culot and Wiese, 2022).

In relation to skills, the EU ESCO system identifies certain skills as 'green' in order to support the green transition in the context of EU labour market transformation (Koundouri et al., 2023: 20). In terms of the skills level, the forecast is set to anticipate growth in the medium-skilled jobs (technicians and specialised level, according to the ISCO classification), yet to date the job creation has occurred on the biggest scale among the high-skilled professionals (Eurofound, 2023a).

In relation to energy interventions, these must go hand-in-hand with tackling existing power asymmetries, to ensure that inequalities stemming from structural, embedded dynamics are not replicated in new energy regimes (Johnson et al., 2020). Going beyond approaches that assess the vulnerability of specific sectors or occupations, some studies underline that the adverse impact of the green transition is likely to be concentrated in specific countries, regions and communities, specifically those that have historically relied on high-carbon industries (Gambhir et al., 2018; McDowall et al., 2023). This has motivated research on spatial justice in the EU context, seeking to prevent the perpetuation of socio-economic disparities among European cities and regions (Walsh et al. 2021).

Furthermore, while the focus of the debate on the green transition is often on the financial consequences of the shifts, other consequences should also be accounted for, such as social isolation and the deterioration of mental and physical health and life expectancy (Della Bosca and Gillespie, 2018; Ostry, 2009, Pini et al., 2010). It is also crucial to understand that while job losses in many cases affect men, studies also show that a significant burden is transferred to women (Haney and Shkaratan, 2003; Olivier, 2012). It should also be noted that the focus on employment in the context of the green transition does not give visibility to some vulnerable groups which will be affected (e.g. disabled or elderly) and who are often missing from the green economy initiatives (Foundation ONCE and ILO GBDN, 2023).

The digital transition, and its impacts on society has also been gaining increased attention among scholars. In general, there is a consensus that the transformation towards digital was accelerated, and, to an extent, enabled by the COVID-19. During the crisis, European governments had to 'overcome a cognitive bias' in relation to accessing technology, thus providing a shift towards universal access to digital technologies rather than perceiving them as a 'luxury' only afforded by developed economies (Grillo et al., 2023:7). This also led to a shift in the general understanding of digitalisation, with an emphasis on the need for adapting to the 'new normal' in the post-pandemic world (Revoltella, 2020). Among other issues, it is the importance of the interaction between digital technology, employment and education. In this context, the technological revolution is regarded as radically redefining how work is conceptualised and performed, how value is created and how it is distributed within economies (Brynjolfsson and McAfee, 2011). Digitalisation and digital transformation affect both business models and job content, thus influencing employment futures, mainly through the expansion of automation, artificial economy, and the platform economy (Eichorst and Scalise, 2023). One repercussion relates to inequalities in financial compensation as technological change is argued to be a driver of increased income inequality over recent decades, due to a skill-bias in wage distribution and widening market income inequality (European Commission, 2019). While it has been noted that digital and ICT skills can improve labour market opportunities (Pichler and Steher, 2021), specific segments of the population are likely to remain at a disadvantage in acquiring these skills, which may have negative consequences for the employability of vulnerable populations (Lyons et al., 2019).

Considering this impact, current inequalities in relation to basic digital skills between regions and groups have been noted (Eurofound, 2023a), particularly in relation to gender and age inequalities, with women generally having lower ratio of digital skills than men, and older citizens possibly facing additional barriers in relation to digitalization (Eurofound, 2023a; Pirhonen et al. 2020). The variation in digital skills is thus emphasised. For example, studies have shown that, on average, 54% of adults in the EU have the basic digital skills, well below the target of 80% set by the European Pillar of Social Rights Action Plan (Eurofound, 2023a) Finally, a different perspective on societal impact can be adopted when looking at the digitalisation and automation shifts with potential effects on job losses (Revoltella, 2020), for example, in the retail sector or retail banking, both containing traditionally feminised low to middle skill jobs.

The concept of the digital divide is often used in this context, defined as a gap between the privileged and advantaged by the new technologies and those who lack access, skills, resources or abilities to seize its full potential (Calzada et al., 2021; Caragliu and Del Bo, 2023; Kitchin et al., 2019; Rogers, 2001; Van Deursen and Van Dijk, 2019). The conceptual evolution of the digital divide can be illustrated with three levels of digital inequalities: from the initial dichotomy between the 'have' and 'have-nots' access to Internet, to inequalities in digital skills and digital capital, and to the social and cultural benefits of using digital technologies (Ragnedda, 2017). Older people, people with disabilities, low-income people, women, ethnic minorities and indigenous communities are some of the vulnerable

groups that are most likely to be left behind by the advance of the digital technologies and the new digital economy (Nijman and Wei, 2020; Tsatsou, 2022).

Ageism and ableism are reproduced in most digital experiences and become important barriers for older people and people with disabilities (Kolotouchkina et al., 2022; Kolotouchkina et al., 2023; Nahafi et al., 2022). In terms of age, the literature utilises the concept of the 'grey digital divide' (Mubarak and Suomi, 2022) to describe difficulties with digital systems experienced by senior citizens. Research has shown that the likelihood of using the Internet decreases by 8% each year after 65 (Friemel, 2016). Technophobia, the perception of uselessness of digital technologies, the lack of digital literacy, low self-esteem and self-confidence in the learning process, as well as limited hearing and sight and other age-related conditions are some of the important barriers of digital inclusion for this age group (McDonough, 2015; Rosales and Fernández-Ardèvol, 2020). However, adapting an intersectional perspective is crucial here as certain groups of older adults as, for example, financially secure retirees who used computers in their working lives, can be more privileged in terms of accessing and using digital tools (Fang et al, 2018). For people with disabilities, who account for 15% of the global population (World Health Organisation, 2011), the most critical digital barriers are related to the low effectiveness of accessibility tools and assistive devices, personal safety and security online, as well as the cost of adaptive technologies (Lussier-Desrochers et al., 2017; Tsatsou, 2021). The digital divide is also negatively impacting education of students with disabilities, due to barriers to the provision of services to address special education needs (Golden et al., 2023).

The digital divide can also act as an obstacle in accessing health services, with people experiencing homelessness, particularly Roma and Traveler communities, and older people identified as at risk of exclusion from digital health infrastructure (Heaslip and Holley, 2023) due to a gap in participatory design practices for technology services aimed at senior citizens (Riessenberger and Fischer, 2023). Digitalised health services can also have a negative impact on women, especially those with ethnic minority background or lower socio-economic groups, as these are often underrepresented in digital health research (Figuerola et al., 2021). Members of low-income households, older persons and those with ethnic minority background are also associated with lower uptake of video consultations and access to digital health services, as access is hampered by the insufficient level of digital skills or local language skills (Husain et al., 2022).

Moreover, in the context of persisting economic and political austerity, war conflicts and climate change, displaced migrants and refugees face significant digital barriers. While digital technology has become a key enabler of their social inclusion and community cohesion, the digital divide of this vulnerable group is further compounded by low access affordability, low digital literacy, and language barriers (Alam and Imran, 2015). The lack of digital representation of vulnerable groups, their inability to contribute to digital content creation and therefore to leave their digital footprints, lead to their online invisibility and reinforce existing offline social inequalities (Isin and Ruppert, 2015). Intersectional challenges posed by the digital divide are also part of an 'inequality loop' where those already marginalised are limited in accessing digital technology as tools for social inclusion, thus being further marginalised (Ragendda et al., 2022).

Within the emerging debate on the twin transition, one of the central questions is whether the green and digital transitions will go hand in hand and create synergies, or whether the twin transition will be unbalanced due to tensions between policies aimed at achieving either digital or green transition goals (Fouquet and Hippe, 2022). For example, there is a possible complementarity between digitalisation and energy efficiency, with the former positively impacting the latter, or even accelerating, if specific policy frameworks are applied consistently (Hedberg and Sipka, 2020). However, some studies have shown that introducing advanced digital technologies does not always lead to more sustainability

(Best et al, 2020). For example, digitalisation may facilitate remote work, hence reducing carbon emissions related to transport, but at the same time increase emissions due to increased residential energy-use or an increased production, use and disposal of ICT (Lange et al 2020, Andrae and Edler, 2015; Belkhir and Elmeligi, 2018; Malmmodin and Lundén, 2018). As such, the interconnectedness of technological advancements and environmental sustainability requires further scrutiny. In terms of impact on jobs, there will be requirement to adapt skills, since jobs created through the twin transition will be different to the ones lost, with a strong shift towards higher skills (EIB, 2018).

Box 4.1: Key points from academic debate in FITTER case study countries

Green, digital and twin transition debate in the six FITTER case study country studies

Some aspects of the twin transition and societal impact were covered in the academic debate in the 6 country studies on a national level, although significant gaps were also identified. For example, a German meta-study demonstrated that a priority is given to the environmental impact of the transition, while the social dimension is very limited (Hofmann et al, 2023). While skills required for the twin transition have been explored (e.g. Bertelsman Stiftung, 2024), the negative effects on specific groups would require more scrutiny. Similarly, in Hungary, the academic literature on the twin transition was found lacking a specific national focus. In Spain, the labour market impacts of the twin transition feature prominently in the literature, highlighting the importance of training and education. However, gender aspects are also covered, particularly in terms of the inclusion of women in the activities related to the green and digital shifts. In Ireland, the main emphasis has been put on the regional inequalities (e.g. Mercier, 2020) or on specific sectors, such as agriculture (Mercier et al., 2020), although there is also a criticism towards the current approach to the twin transition for lacking a proper recognition of the role of care in the transition process (Carolan, 2024; Dukelow et al., 2024). The Irish government approach has also been criticised for its likeliness to exacerbate inequalities related to groups absent from the decision-making process, such as women, Travellers, disabled, older people and other possible structurally marginalised groups (Dukelow et al., 2024).

In addition to the common points mentioned above, the literature review at the national level also reveals a shared concern for the need for comprehensive policies that address social, economic and environmental challenges holistically. In the Spanish literature, Miñarro (2022) shows how the fast fashion business model negatively impacts the environment and the working conditions of textile workers, highlighting the connection between environmental degradation and social inequality. Other studies from this country advocate strategies to promote training and education in emerging sectors such as green and digital jobs, recognising the importance of an integrated approach that considers both environmental protection and job creation and gender equity (Alvarez Cuesta, 2023; Barcena et al., 2022). Alvarez (2023) highlights the importance of education and training programs that prepare employees for future jobs in areas such as green technology and the circular economy. This includes initiatives to promote women's participation in STEM fields (science, technology, engineering, and mathematics) and leadership roles in emerging sectors. Finally, the literature in Ireland identifies barriers to the implementation of a Just Transition. For example, the hegemony of discourses of eco-modernism and green growth (McIlroy et al, 2023), with the assumption of growth as an inherent good conflicts with an understanding of planetary boundaries, have been discussed. Furthermore, it has been noted that politics and policy put emphasis on feasibility, combined with centralised decision-making structures, which lead to top-down planning and operationalisation of 'business-as-usual' models of adaptation to climate breakdown (Revez et al, 2022; McIlroy et al, 2023).

4.2. Twin transition and spatial inequalities in the academic debate: key lessons from regional variations

It has been noted that adjustments associated with twin transitions will vary ‘greatly’ between regions and local assets, such as business clusters, skills available, and the social fabric, will play a crucial role (Maucorps et al., 2023). Regions with higher exposure to the risk associated with the twin transition tend to be poorer, less densely populated and characterised by labour markets that include some structural difficulties (Revoltella, 2020). Furthermore, carbon-intensive regions which are to be transformed are already challenged by impacts of globalisation and automation (Atteridge and Strambo, 2020; Snyder, 2018). This is, for example, the case of regions located at the external EU borders in countries like Bulgaria, Poland and Romania, which are mostly agricultural and display ‘low levels of readiness’ for the twin transition (Maucorps et al, 2023). The final report of the EU project ENTRANCES identifies significant socio-economic challenges across several European regions designated as Coal and Carbon Intensive Regions (CCIRs). First, the well-being and quality of life for the aging population is pointed out as a priority, which includes access to essential services such as healthcare, affordable housing, and transportation. Second, many CCIRs struggle to provide attractive opportunities for young people, leading to difficulties in retaining them. Limited job prospects, lower quality of life, and cultural offerings compared to larger urban areas contribute to this challenge. Third, high levels of unemployment and underemployment are significant socio-economic issues within CCIRs. These challenges are often intertwined with broader economic and environmental trends such as decarbonisation. Fourth, in many CCIRs there exists an overreliance on a single industry, such as coal mining or oil production. This dependence poses economic and social challenges, hindering the transition to a low-carbon economy. Encouraging re-industrialization and diversification of local economies is essential (ENTRANCES 2023).

In relation to the **green transition**, investments in renewable energy and energy efficiency tend to be concentrated in wealthier regions with stronger financial resources and more developed regulatory frameworks. This creates a ‘green divide’ where poorer regions struggle to adopt sustainable practices and transition away from traditional industries. According to OECD (2023), green skills shortage across the OECD is holding back growth in sustainable development jobs and could risk reaching net zero by 2050. Significant differences within countries are highlighted in this context and, as a general pattern, the share of workers in green-task jobs – defined as jobs where at least 10% of tasks directly supports sustainable development – grew just 2 percentage points across 30 OECD countries over the last decade, from 16% in 2011 to 18% in 2021. If urgent action is not taken to boost skills, the green transition could deepen inequalities and hinder progress towards 2050 net-zero goals. Typically, capitals such as Paris, Stockholm or Vilnius have a larger concentration of highly skilled workers and a larger share of green jobs (as high as 30 %). In remote areas, this figure can sink to 5%, exacerbating a social divide (OECD, 2023). The ‘green divide’ on the regional level also brings the issue of the need to upskill those living in regions which are suffering due to the loss of carbon-intense jobs, particularly in some of the regions of Chechia, Germany, Hungary and Poland (European Committee of the Regions, 2023).

The **digital divide** is another dimension of spatial inequality. First, there are important differences across the European states, with Northern and Northern-Western part of Europe generally found as having higher proportion of individuals possessing basic digital skills, and with Southern-Eastern Europe located at the lower end of the digital skills coverage (van Kessel et al., 2022). In this regard, Netherlands and Finland have been identified as leaders while Bulgaria and Romania considered to be lagging behind (European Commission, 2021). This reflects the overall historical patterns of development in Europe and can be also correlated with the household’s access to broadband

(Lucendo-Monedero et al., 2019). Urban-rural divisions are also important to consider when inequalities in access to digital technologies are examined. These also overlap with differences within the EU as countries in the North-West tend to perform better in terms of bridging the urban-rural gap (Feurich et al., 2024).

5. Twin transition in four FITTER sectors

5.1. Agriculture

Agriculture is a strategically important sector of the economy and holds a substantial social and cultural significance. Its activities also have significant implications for nature, air, water and climate. Within the EU, the sector is undergoing restructuring away from a productivity focus on intensification towards more environmentally friendly farm practices. It has been identified as a sector particularly vulnerable to the impacts of the green transition, due to its carbon emission and high path-dependency which can be linked with being tied to markets and contracts, capital investment or skills availability (Knickel et al., 2017; Sutherland et al., 2012). Furthermore, it is crucial to note that the agriculture sector is particularly dependent on nature and environment, and thus itself depends on climate sustainability (McCabe, 2019). Across Europe, farmers experience growing competitive pressures and face challenges related to labour shortages and increasingly rely on migrant workforce, in some cases of informal nature (Cohen et al., 2022). However, it is important to recognise that transition in this sector will not only affect farmers but will also require changes in consumption (Hebinck et al., 2021) and will thus challenge cultural models of eating well (Kaljonen et al., 2020).

Despite the recent emphasise on the role of women in agriculture, the gender imbalances in this sector continue – for example, with regards to farm management, as women in the EU on average represent 30% of farm managers (Fanelli, 2022). Furthermore, women are underrepresented in the EU agriculture policymaking (Bas-Defossez, 2021). In the European context, women in rural areas are also more likely to be unemployed than men, as well as over-represented in informal and vulnerable employment, although this is not specific to agricultural sector (Casares Guillen, 2021). Farms are also more likely to be led by men, with most of the regular agricultural workers in the EU involved in family farms, and the farm managers also located in the older age brackets, with the majority over 55 years old (Eurostat, 2024). Concerns are also raised regarding the income inequalities in the sector, as the poorer households are more likely to foot the transition bill, with small, family-owned farms facing new set of environmental restrictions (Leitheiser et al., 2022; Van der Ploeg, 2020).

However, it should be noted that recent studies show that women and younger farmers are more involved in sustainable agriculture, with younger female farmers more oriented towards environmentally sustainable practices compared to younger male farmers (Unay-Gailhard and Bojnec, 2021: 72). Nevertheless, more caution is needed with regards to conclusions drawn from these findings. International evidence suggests that gender continues to be an issue even in sustainable farming. For example, a Canadian study showed that the majority of organic farms are conventional in their gender relations with women continuing to take on primary responsibility for care work and undertaking farms jobs as required to supplement farm income (Hall and Moggyroody, 2007). This highlights the importance of scrutiny in examining the actual farm family orientations and practices to transform gender relations.

BOX 5.1: Six country reviews and the transition in Agriculture

Germany is one of the largest agricultural producers in the European Union, and the food and agriculture sector plays a crucial role in food security, rural development, and environmental sustainability. There is a strong focus on reducing energy consumption and GHG emissions in agriculture, and the use of innovative, energy-efficient agricultural technology is becoming increasingly important. Digital technologies such as precision farming enable optimised water management and more efficient cultivation. There are also regional differences - for example, northern regions are known for cereal cultivation and dairy farming, while southern regions specialise in wine and fruit production. Seasonal migrants (potential structurally vulnerable group) also play an

important role in German agriculture, with workers coming predominantly from Poland and Romania. In **Hungary**, the share of agricultural production in the economy is relatively high, due to beneficial lands and historical path dependencies. Specifically in relation to climate change, farmers are increasingly affected by weather anomalies associated with climate change as crops often suffer from late frosting, hailstorm, or extreme drought. Several social groups have been identified as vulnerable to the transition, and these included farmers threatened by climate events and physical workers conducting manual tasks who are threatened by automatization.

In **Ireland**, the agri-food industry remains a key component to the country economy and the agriculture is one of the oldest and largest indigenous exporting sector (Department of Agriculture, Food and Marine, 2023). In relation to farming specifically, most farming activities in Ireland take place on family-owned farms (up to 99.7%) with few employees (O'Connor et al., 2022). Changes to the sector are likely to have a strong impact on farmers and on rural communities, with its nature being threefold: environmental, economic and social; however, it has been emphasised that this is not a transition out of agriculture but rather a shift in land use and agricultural resources use (NESC, 2023).

In **Italy** the agri-food sector plays a fundamental role in the twin transition, particularly in terms of the environment, as this is among the sectors with the highest GHG emissions, but also one of the economic sectors most affected by the consequences of climate change. Italy's agri-food sector performs well in terms of biodiversity protection and environmental attention. The strong roots of the Italian agri-food offer in the territories, specialization in high value-added and low-impact products, such as wine production, lead to the achievement excellent results in terms of GHG emissions. The agri-food sector is also well positioned from the digital transition perspective. In relation to gender, farmers operating in rural areas, small organic farms, appear as owned by women however this is a 'fictitious' ownership due to taxation advantages. Gender inequalities are also pronounced for women with children due to scarce presence of childcare facilities in rural areas.

In **Portugal**, agriculture is in the process of shifting towards more sustainable and resilient methods, propelled by digitalization and agroecological techniques. This transformation offers employment opportunities in precision agriculture, organic farming, and sustainable land management practices. Nevertheless, there exists a deficit in skills related to digital agriculture technologies and sustainable farming practices, which necessitates addressing through training and educational initiatives. Vulnerable demographics, like small-scale farmers and rural communities, may encounter difficulties in accessing resources and technology, emphasizing the significance of inclusive agricultural policies and supportive mechanisms.

Spain is also moving towards a more sustainable production model. Innovation and the adoption of new technologies will be encouraged to improve resource efficiency and reduce emissions and waste. In the fisheries sector, efforts are being made to manage the exploitation of resources, promote environmentally friendly technologies and step up the fight against illegal fishing.

5.2. Construction, building and housing sector

The construction sector comprises an important part of the EU economy. It accounts for around 6% of the EU Gross Domestic Product (GDP) and employs around 13.5million people – 6.6% of total employment in the EU27 in 2021. However, the construction sector has been heavily affected by the financial crisis (2007-2008) and besides in Hungary and Romania, the share of employment has declined in all EU Member States. Additionally, the sector experiences widespread labour shortages (Fondazione Giacomo Brodolini 2023). Studies demonstrated that there are significant challenges in

retraining and reskilling the existing workforce as well as making the working conditions more attractive for new workers (Eurofound 2023c).

As the EU has concretised its aims for a green transition within the Fit for 55 Package, it is estimated that the policies included in the framework will have a modest positive impact on employment growth. Construction is set to drive most of the net employment growth, with a concentration in low-mid and mid-income jobs (Eurofound, 2023c). The EU strategy 'A Renovation Wave for Europe: Greening our Buildings, Creating Jobs, Improving Lives' includes a plan to address energy poverty, improve the worst performing buildings, renovate public buildings and decarbonise the heating and cooling systems. The required skills from professionals will therefore concentrate on building renovations and heating expertise, with labour needed to, for example, improve insulation of the existing housing stock, or to convert heating and cooling systems to more efficient ones (Eurofound, 2023c). In terms of skill shortages, and closing the skill gap in the sector, one of the solutions proposed is to increase the involvement of the industry 4.0 technologies (e.g. data analytics, AI, robotics and automation), which, one hand, should increase the health and safety of workers, but, on the other hand, may pose risks in terms of data privacy and cyber security (Turner et al., 2021). In terms of inequalities in employment in this sector, the main concerns are related to the gender profile, as women were traditionally underrepresented in construction, and age group, as the sector has not been attracting younger workforce (Mella and Werna, 2023).

While the restructuring of the sector should bring benefits to the wider society, there are some risks related to vulnerable groups. For example, it has been suggested that there should be more focus on the threat of labour exploitation in the construction sector with a 'routine' breach of employment legislation, often affecting migrant workers, often intra-EU and posted workers (Davies, 2021). At the receiving end of the shifts within the sector, namely the occupiers of the homes, several inequalities have been also highlighted. While the One-Stop-Shops constitute an important solution towards closing the information gaps for homeowners, the assumption that a homeowner is motivated to renovate and to save money has been identified as too simplistic (Bertoldi et al., 2021). Furthermore, the One-Stop-Shops are focused on homeowners and therefore investments are less likely to be done in ways which positively affect tenants in private rented accommodation or social housing, among others (Economidou and Bertoldi, 2015). Lower-income households are less likely to be homeowners, and also have limited access to credit, which is often needed to invest in greening and retrofitting (ILO, 2022). Some authors also refer to the processes linked to the large-scale renovations targets as to 'low-carbon gentrification' (Bouzarovski et al., 2018). When understood as a form of the green gentrification, leading to involuntary displacement of historically marginalised groups, such as working class, low-income, racialised minorities, youth and older residents, it leads to the loss of social networks, social capital and community cohesion, eroded sense of place and identity, socio-cultural exclusion, increased criminal activities, and barriers in accessing healthcare services (Anguelovski et al, 2021). This can be further related to a so-called 'renoviction' (combing renovation and eviction together), which is usually discussed in the context of profit-oriented renovation practices (Polanska and Richard, 2021) but should also be included in the debate on the current EU renovation plans. Moreover, within this context, it is crucial to acknowledge that housing should be understood through the human rights perspective, especially in the light of the ongoing housing crises. While the EU has no direct competency in housing policy, its norms and policies can have an important role in shaping housing conditions, including affordability. In particular, the Recovery and Resilience Facility, and its focus on renovations, has been critically analysed as challenging in terms of housing inclusion, while more attention should be given to the enabling the expansion of affordable not-for-profit housing and to empowering public authorities to regulate the private housing sector (Delclos and Vidal, 2021). Finally, changes in the build environment in the context of twin transition can be also viewed from the

point of view of the so-called smart cities infrastructures. As noted by Criado Perez (2020), despite data availability, gender continues to be not a priority in the city design, which are argued to be designed by men and contain barriers for women and gender queer people (Kern, 2021). There are also insights showing that digital divide in smart city infrastructures can lead to a potential marginalization of disabled people, due to the issues with accessibility and digital literacy (Kolotouchkina et al., 2022).

BOX 5.2: Six country reviews and the transition in Construction

In Germany, the building and housing sector contributes significantly to Germany's economy and faces challenges related to urbanization, affordable housing, and sustainability. Despite the shifts in the building technologies, the German construction sector remains rather sluggish due its low annual rates of change. In general, owners represent most of the EU population in terms of tenure status. In terms of regional differences, urban centers experience high demand for housing, leading to rising rents and housing shortages. Rural areas may have lower housing demand but face issues like aging infrastructure and population decline.

The **Hungarian** construction sector has been fueling economic growth over the past decade. Nevertheless, housing remains a problem for a significant portion of the population with prices rising rapidly in comparison to neighboring countries. The state does not participate in the market with low-priced rental apartments, but there are various financial support mechanisms for citizens, although these are tied to having children. There is a high need for the modernisation of energy systems of buildings in Hungary as their general effectiveness is low. Yet, the government is reluctant to offer such programmes with the exception of PV installation. The sector lacks renewable energy integration and smart homes related skills.

In Ireland, the construction sector has been of particular importance to the national economy and the labour market. An important feature of the Celtic Tiger, the construction and property related credit bubble impacted the Irish situation during the Global Financial Crisis. Current employment in the sector can be characterised by a high proportion of male workers and relatively high number of self-employed workers, often precarious. Housing remains a highly contentious issue in Ireland in the light of the lack of availability, affordability and the rise in homelessness thus changes towards green transition in housing may negatively impact vulnerable groups due to the lack of sufficient funding. Despite the grants available, significant investments are still required from individual citizens (homeowners) while the prices of new **houses** have been steadily rising thus excluding a large proportion of society from homeownership.

In the last years, **Italy** has reached the most significant growth in the construction sector compared to other major European countries. The sector has been affected by significant national interventions carried out within the broader framework of policies and interventions implemented at the European level. In relation to twin transitions, green skills are already considered strategic mainly for profiles related to construction and housing redevelopment. Concerning potential inequalities, migrants and illegal workers could be included in vulnerable groups as a result of the expansion of the construction market, as the need for recruitment has been, in some cases, associated with unfair working conditions. While the labour market in construction is mostly male-oriented, and thus characterised by gender as an inequality, digital and green development will create jobs which may be attractive for women.

The construction sector in **Portugal** is placing growing emphasis on sustainability, with a rising request for green building techniques and energy-saving technologies. To encourage sustainable construction practices and materials, including recycled materials and passive design principles, building codes and regulations are being revised. In relation to the twin transition within the construction sector, an

increasing emphasis is placed on sustainable building techniques and environmentally friendly construction materials. This shift presents job prospects in green construction and energy-efficient retrofitting initiatives. Nonetheless, a gap exists in expertise regarding green building technologies and sustainable construction methods, necessitating training programs and vocational education. Vulnerable demographics, such as low-skilled laborers in construction, may encounter displacement resulting from automation and technological progress, underscoring the significance of retraining and supporting initiatives.

In **Spain**, the annual rate of change of the housing price index stands at 4.2%. With regards to the twin transition, in the household and construction sector, the promotion of a circular economy is a key strategy to promote responsible production and consumption, in line with Sustainable Development Goal 12. Additional measures will be taken to facilitate urban regeneration, with investments in the rehabilitation and renewal of buildings and housing to achieve climate neutrality.

5.3. Energy

Energy can be described as systems of power, as they determine who enjoys the benefits of energy enabled technologies, and who bears the resulting costs and burdens. When considering energy justice in the framework of just transition, the aim is thus to ‘ensure that every individual, regardless of their location, has access to secure, affordable, and sustainable energy resources’ (McCauley et al, 2013, cited in Kashour, 2023). A just energy transition is thus one that acknowledges and mitigates the disruptions in people’s lives that it is likely to cause (Biswas et al., 2022), countering inequalities alongside abolishing fossil-dependent energy systems (Larsson and Abrahamsson, 2021). Furthermore, besides this consumer-directed understanding of energy justice, the concept addresses the representation and recognition of diverse social groups as active energy producers, prosumers, and decision-makers, hence, to guarantee a fair treatment of people and communities in decision-making processes related to energy (Kashour, 2023).

Nevertheless, several social groups are not represented in energy policy and decision making, neither in the private sector nor in the creation of energy technologies. For example, women’s decision-making power in this sector, on a global scale, is limited (Clancy and Roehr, 2003; Feenstra and Clancy, 2020; Petrova and Simcock, 2021) due to structural inequalities, gender stereotypes, unequal financial resources and gender pay gap, or the unequal distribution of unpaid care work (DESTATIS, 2023a; DESTATIS, 2023b). It has been estimated that, despite constituting 39% of the global workforce, women only account for 16% of those employed in traditional energy sector with the gender pay gap at the level of 20% (IEA, 2019); however, the female share of employment tends to be higher in the renewables industry (Czako, 2020). In terms of employment, it has been argued that women still face barriers in entering both, conventional and renewable energy sectors, with perceptions of gender roles as well as the workplace culture, among the most important (Czako, 2020). Women in this sector also face barriers in promotion, advancement and leadership (BCG and WoNY, 2018). There are also leadership inequalities as children with migration background, as well as those coming from households with lower-level of academic education, are less likely to be represented in the leadership positions in renewable energy companies (e.g. Genkova et al., 2022). Finally, there are concerns related to the age profile as the workforce in this sector tends to be aging, especially in the conventional energy sector (Czako, 2020).

One of the key concepts related to energy and inequalities is **energy poverty** defined as ‘a situation in which households, or individuals, cannot attain and/or use energy services required for good health, wellbeing, and the ability to fully participate in society’ (Simcock et al, 2021:2), for example, in terms

of inability to sufficiently heat a home or to afford the cost of other necessary household energy services (Redemaekers et al., 2016). It can be shaped by existing inequalities: for instance, lone parents are more likely to anticipate difficulties in paying bills compared to the general population (EIGE, 2023). While green energy can be essential in addressing the immediate and long-term challenges of energy poverty (Carfora and Scandurra, 2024), several groups have been identified as vulnerable to energy poverty in the context of green transition the Global North. These include: low-income households, people from ethnic minorities and indigenous people, immigrants and those with low proficiency in the local language, older people, disabled persons, women, young people and full-time students, single-parent families, socially isolated people and those living alone, as well as large households and multi occupancy households (Jessel et al., 2019; Middlemiss, 2022). These factors interact and exacerbate patterns of spatial inequality, specifically in terms of socioeconomic status, race, and neighbourhood factors (Jessel et al, 2019). Intersectionality also needs to be taken into account as, for example, single-parent households are also likely to be led by women, often from ethnic minority background, and of low income (Middlemiss, 2022). In addition, research has shown that poorer households tend to spend a higher share of income on energy, compared to richer households, especially in the context of rising energy costs, which can amplify energy poverty (ILO-OECD, 2022) as is usually the case for all utilities (Fankhauser and Tepic, 2007). In order to assess primary energy poverty, a set of indicators was developed at a European level, according to which a household is facing energy poverty if:

1. Its share of income spent on energy services become larger than twice the national median
2. Its income after energy costs drops below poverty line and the share of its income spent with energy is above the national median
3. Its energy spending is lower than the half the national median energy expenditure
4. In cannot warm the house during cold season

(Source: Redemaekers et al., 2016).

Finally, the recent recast of the Energy Efficiency Directive (EED) (European Commission, 2023) includes updated and more comprehensive interpretations of energy poverty with a cross-cutting approach that integrates among others the impacts of economic and health crises into the understanding of energy poverty. It progresses regarding the uptake of intersectionality by recognizing the diverging impacts of it on different societal groups, such as women, people with disabilities, older persons, children and racialised people (Clancy, et al., 2022).

BOX 5.3: Six country reviews and the transition in Energy sector

In **Germany**, the energy sector is historically influenced by fossil fuels, especially by coal. Together with the USA and Great Britain, Germany accounted for 80 per cent of global coal production around 1900 (Quaschnig, 2021). For decades, hard coal was one of the most important and most subsidised energy sources in Germany, with its own hard coal deposits, particularly in the Ruhr region and Saarland, which secured the energy supply and thus led to independence from energy imports (Burck et al., 2016). However, the German energy system has undergone substantial and structural changes, particularly with regard to the electricity supply - since the introduction of the Renewable Energy Sources Act, the share of renewable energies in electricity generation has been steadily increasing (Federal Government, 2023). The share of women in the energy sector was at 22% for oil and gas, 21% for wind energy, 40% for Photovoltaics (PV) and 32% for renewable energy in total in 2021 (BMWK, 2023). The proportion of women in management positions in the energy sector is still growing very slowly, but steadily reaching 15.5% in 2021t

Hungary has been largely reliant on Russian imports of gas and oil (pipe system comes through

Ukraine) and its only nuclear plant is also built on Russian technology thus further adding to the lists of vulnerabilities. These risks manifested in 2022 with the Russian war on Ukraine and following EU policies. PV remain the focus of the government's efforts, various monetary incentive programs have been launched from 2021 onwards to motivate citizens to establish PV generation at residential buildings. However, experts warn that grid congestion remains a constraint ahead of PV expansion. In terms of employment, specific vulnerable groups include people employed at coal and gas plants, as well as some categories of technicians and engineers and petrol station operators. Skill gaps concern energy storage, waste reuse and circularity.

Ireland is heavily reliant on energy exports and the fossil fuel dependency is also high (SEAI, 2023). It has been highlighted that the policy discussion on energy has been largely focused on electricity, with a neglect on the decarbonisation of transport and heat (Yue et al., 2020). In relation to the twin transition, renewable energy has been on the increase and wind has been one of the most successful usages of renewals in the production of electricity (CSO, 2019).

In **Italy**, the energy crisis due to the conflict between Russia and Ukraine has further accelerated the need to change supply sources and to advance along the long path of decarbonization necessary to contain CO2 emissions. With regards to twin transition, the energy sector is affected by various factors of change in terms of demand for skills. In relation to inequalities, households in energy poverty (such as people living in islands, small towns, suburban areas, southern regions, families with an unemployed parent/s, households using gas as a fuel, people in economic poverty or economic vulnerability, single women, single-parent families, older citizens) have less opportunity to access alternative sources and face also the prices increasing, due in part to the European Emissions Trading System (ETS) which increases the costs of CO2 emissions. In addition, also SMEs operating in sectors that are energy intensive entail difficulties in this sense.

Portugal is swiftly shifting towards renewable energy sources, especially wind and solar power. Substantial investments in renewable energy infrastructure have resulted in enhanced energy self-sufficiency and decreased carbon emissions. Additionally, the movement towards decentralization and the adoption of smart grid technologies are gaining momentum, facilitating more effective energy distribution and usage. In relation to twin transition, Portugal is expediting its transition to renewable energy sources, generating employment prospects within the renewable energy sector. Nevertheless, addressing skills gaps is imperative to fulfil the demand for proficient workers in renewable energy technologies. Vulnerable demographics, like low-income households, might encounter obstacles in accessing renewable energy solutions due to financial constraints, underscoring the necessity for inclusive policies and support systems.

In case of **Spain**, the economic and energy dynamics in 2022 were affected by the Russia-Ukraine conflict, which influenced the European and Spanish context. In terms of the twin transition, actions in the energy sector are highlighted, including the promotion of renewable energy in production and consumption. The focus is on reforming the electricity market and promoting energy savings and efficiency. It will also promote innovation and interconnections to strengthen strategic autonomy and security of supply.

5.4. Transport and mobility

The transportation sector is critical to European business and supply chain, accounting for around 5% of the EU GDP and employing more than 10 million people. Transport also accounts for almost a quarter of Europe's GHG emissions, so low-emission mobility is essential to achieve a broader goal of shift towards low-carbon economy (European Commission, 2016). It has also been pointed out by the International Transport Forum and FIA Foundation (2016), that there is a need to strengthen the

awareness of the gender, transport, and climate policy nexus. Transitions in transport and mobility affects people in multiple ways, as shifts are having an impact on those directly employed in transportation (e.g. road transport, rail, air and maritime transport), those who are involved in manufacturing of vehicles (e.g. cars) and those who are end users of the different modes of transport.

In terms of employment, challenges include a move towards sustainable fuels, with a goal to reduce GHG emissions. Employment in the sector presents high inequalities in terms of gender and age, as it tends to employ older male workers; hence attracting younger and female workers is one of the major challenges (Broughton et al., 2024). Gender balance is argued to be difficult to improve with women significantly underrepresented in the sector, which is also traditionally characterised by workplace violence (ILO, 2013). As shown by other studies, women constitute around 22% of the total transport workforce, and suffer barriers such as misogynistic attitudes, male dominated hierarchies and gender pay gap, among others (Woodcock and Rosenau, 2023). Male dominance also affects the decision-making process in the design of transport and of vehicles, therefore resulting in gender inequalities at the user level (Woodcock, 2023).

While access to transport can be an important tool to overcome deprivation and reduce inequalities, transport poverty needs to be addressed. Transport poverty can be defined as ‘the inability to attain a socially- and materially-necessitated level of transport services’ due to affordability, mobility or access (Simcock et al., 2021). The cost of transportation tickets, for example, can strongly influence individual choices. When they are high compared to available income (all other characteristics such as accessibility, adequacy and availability being constant), individuals’ mobility and access to services (e.g. healthcare, education) and opportunities (e.g. employment, cultural events) might be limited (Kuttler and Moraglio, 2021a). Lack of accessibility to transport is therefore a factor that intervenes in reinforcing poverty and social exclusion. In terms of structural vulnerabilities and its links with transport poverty, studies found that these often overlap with groups identified as at risk of energy poverty, including those with low income, people with pre-existing health conditions and disabilities, lone parent families, older citizens, women, and those with ethnic minority background, with those in urban settings more likely to experience energy poverty, while suburban households more exposed to transport poverty (Koukoufikis and Uihlein, 2022).

However, affordability, mobility or access are not the only factors behind the ways in which people access transport services. For example, research on active transport shows that the benefits of walking and cycling are not distributed evenly across the populations, as socio-economic and cultural factors play an important role in this context (e.g., underrepresentation of disabled people in active transport journeys, men cycling more than women) and that intersectional lenses need to be adopted to analyse these interconnections (Yuan et al., 2023).

Finally, the push towards electric vehicles for private usage also requires more scrutiny in relation to potential benefits and costs in relation to inequalities. The shift towards electric vehicles will result in substantial job losses and the need for re-skilling and upskilling (Broughton et al., 2024). The price of electric cars also remains high and may be unaffordable for certain segments of society. This is despite subsidies available, as these are often conditional upon the upfront payment made by the household, thus requiring significant financial capital as an investment, before the subsidy is applied – similarly to the retrofitting upgrades in the built environment (Dwarkasing, 2023). Indeed, international research shows that inequalities in availing of electric cars are prevailing. For example, in the US, most of the electric vehicles’ buyers are male, high income, high educated, homeowners, with access to charging points at home (Hardman et al., 2021). This issue would require further study, particularly in the context of future penalization of fossil fuel ran vehicles, as this can potentially affect groups already at risk of transport poverty and result in their further marginalisation.

BOX 5.4: Six country reviews and the transition in Transport and mobility

Germany has a well-developed transport infrastructure, including an extensive network of highways, railways, and waterways. The transport sector is vital for facilitating trade, connecting regions, and supporting economic activities. The transport system is dominated by the car industry. The German population's choice of means of transport shows a strong preference for motorised private transport (MIT), primarily the car (infas, 2023). Due to the focus on the car industry and a strong car lobby, transport policies in Germany were not centered around a public railway system. Urban areas like Berlin, Hamburg, and Munich have comprehensive public transportation systems, while rural regions rely more on private vehicles and face connectivity challenges. In relation to employment, the Institute for Employment Research (IAB) provides insights into employment trends in the transport sector, including shifts towards automation, electric mobility, and logistics services, see for example Mönnig (2018).

In Hungary, MÁV, the public train company is responsible for service provision, maintenance of trains and the track system. However, the service is currently considered to be unreliable with worsening quality of tracks and general shortage of funding. There is a system of public buses operational between municipalities across the country and these are more reliable and modern than trains, but are less sustainable. As for transportation in cities, the public transport in Budapest offers good value for money, buses are reliable, and there are also various tram lines and metro lines. However, the major mode of transportation in Hungary is car travel. The highway network has been spreading through the country in the last decade, but blind spots remain, and the system is very much Budapest-centered; the same also applies to the train network. These are partly due to drastic change of borders in the 20th century. . The groups identified as vulnerable to twin transition include care mechanics, combustion engine car owners, logistics professionals, and airport ground handling personnel, as well as rural population in general.

In Ireland, the reliance on private cars is similar to the rest of the EU and the usage of public transport and active modes has increased over the recent years (Department of Transport, 2022). Nevertheless, private car energy use dominates and accounts for nearly 40% of transport energy use in 2022 (SEAI, 2023). Regarding the green transition context, transport is the second-largest source of GHG emissions in Ireland, after agriculture (CIE, 2022). While the government emphasises the decrease in CO₂ emissions in the sector, it has been noted by the OECD (2022b) that Ireland's 'electrification and fuel efficiency improvements in vehicles are insufficient to meet Ireland's ambitious target' set to half the emissions in transport by 2030 and a behavioural change towards more sustainable models, as well as the travel reductions, will be needed. Public transport remains an issue highly contested in the public debate, in particular in relation to the strong emphasis on roads rather than rail transport, high dependence on aviation and the urban transport developments regarding affordability and environmental impact.

In Italy, in all forms of transport, the demand levels have almost recovered or exceeded the values of 2019. As for the rail system, the demand levels have reached and sometimes exceeded the values of 2019. IC/ICN services in the fourth quarter of 2023 are at the same levels as those of 2019. For local transport, with reference to Trenitalia's regional rail transport, demand levels are still significantly lower than the values of 2019; with passenger traffic 15% lower than that of 2019 (compared to a -11% recorded in the third quarter of 2023). With reference to road-based public transport, passenger traffic is still 24% lower than the 2019 values (ten percentage points less than recorded in the third quarter of 2023). Finally, for air transport the demand levels have recovered and in part exceeded the values of 2019; specifically, with national and international flight offerings in line with 2019. Passenger traffic is 4% higher (in line with the third quarter of 2023), while freight traffic is down by 1% compared to 2019 (one percentage point lost compared to the third quarter of 2023).

Portugal's transportation sector is evolving towards electrification and sustainable mobility alternatives. The focus is increasingly on electric vehicles (EVs), bolstered by governmental incentives and the expansion of EV charging networks. Moreover, there's a movement towards integrated and multimodal transportation systems, incorporating public transit, cycling infrastructure, and shared mobility services, to alleviate congestion and emissions in urban centers. In relation to twin transition, the Portuguese transportation sector is in the midst of a shift towards electrification and sustainable mobility solutions. This evolution presents employment prospects in electric vehicle manufacturing, infrastructure expansion, and maintenance. Nonetheless, there's a requirement for upskilling and reskilling initiatives to provide workers with the essential competencies for positions in electric vehicle technology and intelligent transportation systems. Vulnerable communities, like rural areas lacking adequate public transportation access, may encounter mobility hurdles necessitating inclusive transport policies.

Transport in Spain, which accounted for 29.6% of national emissions in 2019, must be decarbonised to achieve climate neutrality by 2050, in line with European and national policies. Despite the increase in emissions, measures are also being taken to tackle local pollutants to improve air quality. The Climate Change Report underlines the importance of transport in GHG emissions and the need for action to reduce them. It highlights the efficiency of multimodal and intermodal transport to improve competitiveness and progress towards decarbonization. The transition to more sustainable mobility with measures to renew the vehicle fleet with less polluting options. Emphasis will be placed on the electrification of transport, in particular the development of electric vehicles and improvements to rail transport. The promotion of the use of public transport, intermodality and the creation of low emission zones in urban areas are mentioned.

6. Twin transition in public debate: analysis from the six FITTER case studies

This section considers public debate related to twin transition, social justice and inequalities in the six FITTER case study countries. These were compiled by the national case study researchers following standardised guidelines, with several points explored. Specifically, the section explores the issue of the transition as discussed in the main political debate, as well as the industry, trade unions and the NGOs. More specifically, four FITTER sectors are part of the analysis, as well as the inequality grounds as relevant inequalities. It needs to be emphasised that the analysis does not aim at providing a complete overview of the debate in each country, but rather focuses on mapping of such discourses, with most recent media outlets, as well as the organisational and official websites explored for this purpose. The section starts with a table which provides a summary of main points identified through the analysis. The following sub-sections will involve a more detailed thematic description of the specific points selected during the data collection process.

TABLE 6.1: Main points in the public debate by actors and country

	Mainstream political debate	Industry	Trade unions	NGOs
Germany	Climate protection that creates work, digitalisation that support sustainability	Green and digital transition discussed strictly in the business contexts	working conditions, safety and updating infrastructure	Some voices calling for global justice and intersectional perspective; issues around digitalisation
Hungary	Very limited political debate on twin and just transition	Skill shortages	working conditions and skill shortages	very limited discussion on just transition, focus on other societal issues
Ireland	Transition as an opportunity, just transition as a strong component of official strategies	Very limited engagement with social justice concerns, business as focus of the discussion	working conditions as part but focus beyond labour market	Climate justice, inclusive social dialogue on climate change, some recommendations for just transition, care economy
Italy	Skepticism and concerns around employment and economy but little discussion on justice and equality	Some commitments to social justice	working conditions but also fair transition beyond labour market	Digitalisation highlighted as an important issue; transition to new development models
Portugal	Focus on labour market and social protection	Focus on employment with some equality issues (gender) addressed	Skills and workforce and skill gaps	General protection of structurally vulnerable groups

Spain	Emphasis on social justice, equality and inclusion	Workforce shortages	transition in conjunction with social protection, industrial supports and regional development	Some issues related to structurally vulnerable groups, including accessible energy
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1.1. Government

1.1.1. Twin transition: main government strategies

Mainstream political discourse on twin transition in the 6 case study countries is dominated by a number of themes which can be related to equality issues. First, there is the need to protect employment and to ensure that skills shortages are addressed. Second, synergies are envisaged between digital and green with the former acting in support of the latter. For example, in **Germany**, the motto of one of the coalition parties – the SPD – is ‘Climate protection that creates work’, while the Green party, also in the government, proclaims that digitalisation must be actively shaped in terms of sustainability, and actively promotes the potential of digitalisation for climate protection and for the reduction of consumption and the CO₂ emissions. In **Ireland**, while the government acknowledges challenges related to the twin transition, it perceives it as an opportunity, with ‘just transition’ at the core. It outlines goals for ‘Better quality of life for all’, in the context of post-COVID-19 recovery. The focus is to respond to public demand for change in housing, health, climate action and quality of life.

In relation to skills, in **Portugal**, the “National Priorities 2024” document sent to the European Commission indicates the need for reskilling and upskilling the workers in the context of the twin transition. The insufficiencies indicated in the document are the lack of active employment policies to comply with the recent labour market demands, lack of adequate social protection and compensations for workers and companies requiring costs for the transition.

Digitalisation was part of the mainstream political discourse in **Ireland**. In relation to the digital divide, the government acknowledged that the pandemic had an acute impact on those already experiencing disadvantages, particularly in terms of education. Nevertheless, the government has a positive view on the digitalisation, arguing that Ireland will ‘harness the power of digital to tackle the big challenges we face in the coming years – achieving our climate goals, driving balance and inclusivity, and enhancing productivity’, although some groups (e.g., older cohorts and disadvantaged socio-economic groups) may be left behind and thus in need of support (Government of Ireland, 2021).

Finally in **Italy**, the government official position on building efficiency was expressed at the Ecofin council on the final approval of Energy performance on buildings directives, where Italy, along with Hungary, voted against it. Recently, the national Chief of government, Giorgia Meloni, in her closing speech in the Fratelli d’Italia party *kermesse* addressed two mentioned key issues for the twin transition: building energy efficiency and stopping diesel and fuel vehicles productions. As for green buildings, the head of government pointed out that the ‘green buildings’ directive does not consider the cost to citizens for building efficiency and that the approved rules consider every case in Europe in a single way (e.g. A wooden house in the Finnish tundra and a house in a village in Sicily), without considering local specific needs.

As for the transition to electric in automotive sector, according to the policy expressed in the permanent representatives committee (Comité des Représentants Permanents), Giorgia Meloni affirmed that the transition to electric cannot be the only way to decarbonise transportation. The

premier also considered that the main major electric foreign manufacturers do not respect the environmental constraints in force for Italian manufacturers. No further arguments were found on this last point regarding social justice and equality, including with reference to citizens and workers. It can be referred, however, that the government represents concern about the effects of the measures on the national economy and, presumably, on workers in the sector.

6.1.2. Just transition

In relation to the just transition specifically, the official New Green Deal published by the **Irish** government contains 'just transition' as a separate item, pointing to both challenges and opportunities related to anticipated social change. Vulnerable groups are noted in this context with a focus on employment. In addition, the just transition principle includes establishing a Just Transition Commissioner and ensuring that financing is available to grow the size of the Just Transition Fund. A New Engagement Model with citizens, sectors and regions is also outlined, with an assurance for 'protection for vulnerable families and communities least equipped to make the transformation'. Finally, a New Social Contract is outlined, with supports for vulnerable groups based on the 9 equality grounds, although these are not discussed in relation to the twin transition. In **Spain**, the official government line is that the twin transition represents an opportunity for the Spanish economy, but also a challenge in terms of social justice. To ensure an inclusive recovery, a plan is being implemented to invest 100 million euros in environmental, digital and social services in the municipalities and territories in transition. This plan aims to create new service models that would contribute to a more equitable future strategy, and to preserve, recover and attract population to disadvantaged areas. It also focuses on environmental management and the circular economy, developing digitalisation tools to ensure a secure and accessible supply chain of materials, which could particularly benefit marginalised communities.

Spain's Recovery, Transformation and Resilience Plan emphasises social justice, equality and inclusion in the area of the twin transition. The plan is divided into four main pillars, one of which is dedicated to social and territorial cohesion and gender equality. Steps are being taken to reduce disparities in technology and socio-economic status, to promote gender equality in the digital realm, and to increase digital skills in society, with a specific focus on the education and empowerment of women and ICT professionals. It also seeks to ensure that digital public services are accessible to all, promoting inclusion and equal participation in the digital society. Finally, the Spanish Housing Rehabilitation and Urban Regeneration Plan, which is part of Spain's Recovery, Transformation and Resilience Plan, has been identified as essential for promoting social inclusion and improving the quality of life of the population. The strategy is positioned as an attempt to reduce economic inequality, in addition to its direct impact on accessibility and gender equality. The creation of employment opportunities in the construction sector, driven by rehabilitation and urban development activities, is expected to contribute to this goal by providing employment opportunities to currently marginalised segments of the population. The plan aims not only to improve housing conditions, but also to revitalise disadvantaged urban areas.

6.1.3. Opposition parties, counterarguments and other voices from the political actors

The narrative in **Germany** is frequently fed by fears from the conservative wing (FDP) as well as the right-wing parties that the green transition will harm the middle income of Germany and its SMEs. This has tangible effects with, for example, the influence of the media debate on 'harming the middle-income class on the amendment the Building Energy Act' (Jost et al., 2024)

Interestingly in **Hungary**, there is a lack of mainstream political debate on the twin transition with the exception of digitalisation and the skills related to moving education online. There are also political pressures to actively attract batteries production to the country to keep the car industry afloat. These investments are granted a 'special status', assigned to investments deemed to be of special national interest, which may result in neglecting protectionary measures towards the protection of the environment.

In **Ireland**, most of the opposition parties criticised government for their programmes on the twin transition and offered their own strategies, with some commitments to just transition and social justice. However, while climate change remains an important topic on the political agenda in Ireland, recent voices have been warning that, in the light of changing political climate on the ground, politicians are 'running scared from the vocal minority', with a danger of downplaying climate change in the future government - despite no signs of an upsurge in anti-climate action sentiment in Ireland (White, 2024). Furthermore, relatively less attention is paid to the digital transformation, which is considered a fact rather than a goal, and a positive view is adopted in this context. The Just Transition is a strong feature in most of the documents, yet the discourse revolves around general statements, while more concrete actions proposed would require more scrutiny.

In **Italy**, opposition parties are mostly in support of the green transition. In contrast to the opinions held by the government, for the main opposition party, Partito democratico, the green deal is 'the only choice to boost the economy, to reduce costs of energy for businesses and citizens, fighting speculation. According to declarations of the Partito democratico, new mobility and local public transport should be introduced, accompanying those who struggle the most, without lowering the ambition. Among the main opposition parties also Movimento Cinque Stelle, Europa Verde-Verdi e Sinistra Italiana, standing with the vision expressed also in past years, support the green transition and welcome European policies in this direction.

6.2. Industry: four FITTER sectors

In relation to twin transition and inequalities, a relatively low level of engagement was recorded in the six FITTER cases studies from the industry stakeholders. For example, in **Germany**, there are separate discussions related to green and digital transformation, with a strict binding to the business contexts. Furthermore, multiple companies engage with what can be characterised as 'green washing' through additional donations via their foundations, e.g. by sponsoring education programmes, or compensating for their emissions. In **Hungary**, the main concerns raised by the industry were mainly linked to the skill shortages and the need for more qualified workforce. This issue has also been raised within the energy sector in **Spain**, where there financial viability and availability of skilled labour were identified as a concern.

In **Ireland**, the increases in the energy costs for households have also been raised, for example, in relation to mortgaged households. However, the emphasis coming from the financial sector (Central Bank, for example) is placed on 'behavioural changes' and implementing changes among these households (Adhikari et al., 2023), thus placing the responsibility on the individual rather than the State. Regarding the just transition and vulnerable groups specifically, the Irish analysis shows limited engagement of industry groups with social justice concerns in relation to economic transitions. For example, the main message from Enterprise Ireland focuses on making businesses more sustainable, with business, customers and the planet listed as priorities¹. The Energia Group, one of the main suppliers of electricity and gas to Irish households, commits to the just transition in their annual report for 2023 by 'providing a secure, affordable and clean energy supply to homes and businesses as well as offering accessible energy services and products to ensure no one is left behind'.

In **Italy**, the CGIL FILLEA manifesto include points related to equality, including calls for urban regeneration, addressing social regeneration and participatory approaches in transformation of neighbourhoods and cities. They also call for a revitalised workforce, beyond creating new jobs related to people's care and the environment, involving retraining, and educating in new construction techniques and materials. In the energy sector, the ENI group published a focus report on just transition initiatives. The document mentions the wish to put the focus on the person, to implement a fair and social transition. ENI works so that the decarbonisation process offers opportunities to convert existing activities and develop new production fliers with relevant opportunities for workers, economies, and communities in the countries in which the company operates. Further to that, another important actor, SNAM is committed to ensure a just transition by respecting inclusion and equality principles, with a view to developing future job prospects, while ensuring new opportunities within the company and protecting people's safety, training, development, and well-being.

Finally, in **Spain**, construction sector companies are grappling with integrating inclusivity into their hiring policies and ensuring the affordability of energy-efficient housing for marginalised communities. Initiatives such as inclusive hiring practices and promoting gender diversity aim to address long-standing inequalities within the sector. In addition, companies are exploring affordable housing initiatives that benefit disadvantaged communities and contribute to greater equity in access to decent housing. The transportation sector in Spain faces concerns about job displacement as it transitions to cleaner technologies. While retraining programs exist, ensuring equitable access to these opportunities remains a challenge. In addition, implementing inclusive urban mobility policies to address socioeconomic disparities is critical. Initiatives such as affordable public transportation and improved accessibility for underserved communities are essential to promoting social inclusion and equity.

6.3. Trade Unions

It comes as no surprise that trade unions in the FITTER case study countries are mainly concerned with issues related to employment and working conditions in the context of twin transition. In **Germany**, most organisations reviewed are concerned about training workforces and supplying trained workforces in response to the transition. Depending on sectors, concerns vary. The energy sector is concerned about the electricity price not to lose its competitiveness in the world market. The construction sector is focusing especially on the safety of employees in the use and interaction with machines. The transport sector (German trains and railways) is strongly demanding more infrastructure funding for the long-term use of trains due to their deteriorated and aged facilities and a huge amount of finance is required for repairment and renovation. Furthermore, one of the German unions representing service workers across all sectors, the ver.di, considers digitalisation as an important driver changing working conditions, and notes that digitalisation should be used to improve working conditions in the care sector (e.g., social work) and to promote democracy. In the energy sector, unions demand to stabilise the supply of skilled labours and expand the range of workforce with an investment. They also highlight that the industry should become more 'feminine' and that it should empower the female workforce to be more active in the sector. In **Hungary**, the debate within unions mainly revolved around wages, with shortage of labour in various sectors (including construction and transportation) as a baseline. More specifically to the twin transition, and in the context of the Hungarian government pressures on production of car batteries, significant health and safety risks faced by workers in battery production have been highlighted, with exposure to hazardous chemicals, high-speed machinery and extreme working conditions among the issues raised. In **Portugal**, unions stress the need for a skilled workforce which is crucial for the development and maintenance of energy infrastructure, as well as for innovation in renewable energy technologies.

In **Italy**, the General Confederation of Italian Workers (CGIL) highlights the need for more protection for high-quality agricultural production and field workers, who are often exploited. According to the union, the current model of agricultural production is unsustainable from an ethical and environmental perspective, as well as from a purely economic perspective. The National Observatory of the CGIL reports and analyses working conditions in the construction sector, also exploring issues concerning the transitions and related policies. Despite new measures introduced by the National Recovery plan and first results achieved after the entering into force of the new public procurement code (1 July 2023), criticalities remain: undegrading work, migrant working conditions, which are defined ‘the main victim of old and new forms of exploitation, *caporalato*’ and, even when regularised, they are subject to additional forms of ‘discrimination’. In the energy sector, the Italian trade unions go step beyond the focus on the labour market and call for the following issues to be addressed: a fair and equitable transition by ensuring tools and resources to offset the costs of transition for consumers and businesses and to fight energy poverty by ensuring a sustainable cost of energy; affordability and social accessibility to climate-compatible solutions through specific campaigns to guide citizens toward responsible consumption; a ‘labour-oriented’ transition to maintain and develop employment levels in quantitative and professional terms with resources and investments tied to job creation and re-employment opportunities. There are also concerns related to age discrimination, for example in the transport sector, with older workers made redundant due to their inadequate skills set.

A wider understanding of the transition was demonstrated in case of the **Irish** and the **Spanish** trade unions. In **Ireland**, as part of the ‘Just Transition Alliance’ major trade unions (ICTU, SIPTU, Fórsa, Unite the Union) have focused their campaigns on the need for substantive government supports, including the establishment of a ‘Just Transition Commission’.

In support of these calls, the alliance draws on various commitments established by the UN and ILO Just Transition Guidelines. However, they do not reference ‘green growth’ - instead focusing on the need for the development of ‘a more sustainable economy and society for all.’ In their publications, the negative experiences of peat workers and communities in the Irish Midlands are referenced, where job losses have not been ‘matched by the required supports from government or state agencies, in a manner consistent with Just Transition’ (Just Transition Alliance, 2022). They warn that without a comprehensive plan and strategy for a genuine Just Transition there will be an ‘erosion of worker confidence and public support’ for the transition to carbon neutrality (Just Transition Alliance, 2022). The alliance also calls on the government to establish a National Just Transition Commission based on social dialogue and comprised of representatives of government, trade unions, employers, affected communities and civil society. Overall, the Irish trade unions campaign for a Just transition and focus on both, *the economy and the wider society* with key areas of focus as follows: (a) Campaigning for structures and governance mechanisms (e.g. Just Transition Commission) which would enable well-informed, evidence-based, participatory dialogue, decision-making and governance (b) The protection of workers generally (and those in fossil fuel industries in particular); and (c) Building an economy that supports liveable lives for all (Doughnut Economics).

In **Spain**, trade unions have defended a unitary position: transition policies must be accompanied by social protection, industrial support and regional development policies. Furthermore, they expect these policies to be designed with a clear gender dimension. Likewise, they understand that a just transition is essential to prevent an increase in social rejection of measures to combat climate change among the most vulnerable population groups and the most affected territories. More specifically, in construction, the trade union UGT highlights concerns about job losses, working conditions, social inequalities and environmental degradation during the transition. Addressing these issues requires comprehensive strategies and cooperation among stakeholders. The importance of fair working

conditions and the protection of workers' rights in the midst of industrial change is emphasised. In addition, ensuring that all workers have access to employment and training opportunities is critical to promoting social inclusion and equality. In transportation, CSIT's involvement in promoting eco-friendly transportation underscores the importance of collaborative efforts in navigating these challenges while prioritizing fairness and inclusivity.

6.4. NGOs and other Civil Society Organisations

The debates on the twin transition among the FITTER case studies NGOs is rather limited, with some exceptions. In Germany, one of the most influential movements on digitalisation and sustainability – Bits and Baume, provides a set of recommendations with the following main points: (a) demand for global justice and regional autonomy, with indigenous and local groups to be integrated into the global digital politics and economy, with digitalisation in agriculture to serve the needs of local farmers, and a demand for responsibility in global digital trade (b) the need for the technology to serve intersectional-feminist and anti-discrimination principles, especially with technology that shall not reproduce existing anti-discriminatory power relations and (c) a recommendation for protecting digital users/digital citizens by offering free safety and authentication systems in the frame of a democratic e-governance. Another organization, The DBU, highlighted a field of tension in the discourse about the twin transition: digitalisation can lead to a more (time-)efficient use of resources, but it is questionable if this efficiency shall serve a neoliberal/capitalistic market system or if it benefits the whole society and is just. This includes the so-called *Rebound effect* – although resources can be used less and more efficiently, this saving of resources is invested in other resources, hence increasing growth and overconsumption. Besides always underlining social factors, this study only mentions groups in marginalised situations once in terms of creating a more diverse working atmosphere, and without providing any further specifics on that (DBU, 2022).

In relation to digitalisation, an **Italian** organization, The forum Disuguaglianze, focuses on digital transition and its adverse effects on knowledge and data management, pointing out that the profound transformations of contemporary society are related to data mining, storage, and processing capabilities (big data, supercomputing, artificial intelligence). It implies a crucial challenge in terms of social justice and democracy. On the one hand, the digital transition can be an opportunity for widespread improvement in people's material and social conditions; on the other, it can lead to unprecedented monopolistic concentration of knowledge and control of data, with a serious worsening of inequality.

In **Ireland**, Pavee Point, an advocacy organisation for Traveler and Roma people in Ireland, stress the importance of Travellers' inclusion in social dialogue processes on climate change; the Irish Traveller Movement highlights that the location and design of Traveller sites should be established through an environmentally just process; and the Disability Federation of Ireland stress the need for accessibility and universal design for all transportation modes. The National Youth Council of Ireland has committed to 7 climate justice principles. They explicitly reference the historical roots of inequalities, stressing the need for an understanding of the power dynamics that have created extreme inequalities *between and within countries*, as this is key to a just transition. The Irish Human Rights and Equality Commission has developed a comprehensive policy statement on a Just Transition with 15 key recommendations. These highlight the need for:

- Structures and Governance Mechanisms (e.g. Just Transition Commission; systematic data collection, monitoring, review, evaluation and reform of climate actions particularly regarding the impacts on structurally vulnerable groups; accountability mechanisms for

business and enterprise in line with international standards; promoting the Public Sector Duty)

- Systemic design practice and evidence-based actions, including targeted measures to support affected communities and structurally vulnerable groups, adopting intersectional and intergenerational perspectives.
- Participatory, and place-based approaches to policy design
- Due regard for the situation of climate refugees and proactive asylum planning
- Parallel rollouts of the Digital Ireland Framework and Climate Action Plan to ensure meaningful participation by all structurally vulnerable groups, including prioritisation of targeted offline non-digital mechanisms for consultation, including at community level (IHREC, 2023)

In relation to gender+ inequalities, the National Women's Council, together Community Work Ireland conduct a project 'Feminist Communities for Climate Justice'. This project recently carried out a baseline review of Irish environmental and climate policy, providing an overview of the current relationship between climate policy actions and 6 areas - the Just Transition; Care; Energy Poverty and Housing; Food, Land Use, Agriculture and Biodiversity; Transport; and Health. According to the project, there is evidence that across each of the areas the needs of structurally disadvantaged groups are omitted or inadequately accounted for in Irish climate policy and actions. It should be noted that, although the project aims to take an intersectional approach, in effect their approach largely foregrounds the needs of women, with an acknowledgement that women are not a homogenous group and can suffer multiple disadvantages along other grounds of discrimination. Despite this shortcoming in relation to intersectional analysis, the project is particularly informative with regard to its focus on the care economy. It stresses that care is multidimensional and includes care as a mindset (lived ethnics of care), care as an activity (concerned with the wellbeing of humans, non-humans, nature, and physical environment) and care as work (including – including paid care work in the care economy e.g. social care, childcare; and unpaid care work for families, communities etc). They note that care work is also tied to repair work (e.g. everyday repair of objects; repair work after climate change related disasters). The report is highly critical of narrow understandings of the economy, whereby carbon-intensive male-dominated sectors are identified as key areas for support and development, while completely omitting low-carbon, yet highly precaritised and feminised sectors of the economy, as targets for support. The authors stress the need to understand care jobs as 'green jobs' (Dukelow et al. 2024).

6.5. Additional issues highlighted by the case studies in relation to public debate on twin transition.

In some of the FITTER country reviews, additional issues in the public debate were highlighted. These were specifically related to potential vulnerabilities and social justice. In **Germany**, one of the most prominent discussions regarding social inequalities revolves around gender equality (binary concept) in regard to the labour market and the transition in skill requirements (gender pay gap and lead gap), but also in regard to women's political and digital inclusion, see e.g. BMWK strategy 'women in the economy' (BMWK, 2024). Furthermore, BUND and DMB published a study analysing and developing concepts to ensure that climate protection in rented housing makes progress and is socially equitable. They argue that necessary investments must be better distributed between three players: the public sector, landlords and tenants ('one-third model') (Mellwig, 2024).

In **Ireland** various themes have featured in the public debate in relation to inequalities in green and digital transition, with the former gaining more traction in the mainstream media. The most recent one is about the newly introduced 'bottle return scheme', which is an interesting example of how a green policy did not consider certain inequality grounds such as age or disability. As has been pointed out in some media outlets, self-service machines used for the scheme are not suitable for older citizens or wheelchair users, among others. There have also been voices pointing out to the inadequacy of the housing retrofit scheme which includes grants, yet significant investment is required by individual citizens (upfront payments), thus making the scheme unaffordable for many mid to low-income households. As pointed out by some, retrofit schemes do not take social inequalities into account and are most likely to result in a situation where households with lower incomes will face higher energy bills (Hearne, 2022), with some also pointing to the gendered nature of such impact, with women more likely to be negatively affected (Murphy, 2023b). The introduction of 'smart meters' for home energy use has also been criticised for its inadequacy, in the context of a very low uptake (only 11% of households which installed smart meters avail of smart tariffs). It has been argued that customers need more support and help and that this should be addressed immediately as change in energy use is crucial to addressing climate change (Byrne, 2024). These themes are crucial in the light of the debate of green policies which shift the responsibility to individual and individual behavioural change. The importance of data centres in Ireland, has also recently featured in the public discussion on twin transition, namely the need for expansion of such centres due to the digitalisation and the energy required to run these. Finally, access to broadband, particularly in rural areas, remains an important part of the political debate. One of the aspects are inequalities in relation to rural schools where the broadband can be patchy (McGuire, 2021), but also in relation to regional re-development often halted by the lack of reliable broadband in rural regions.

Finally, in **Italy** the national ecologist organisation Legambiente is currently promoting a public debate for the next European elections across the country. The focus is on the future of the green transition, technologies, socio-economic cohesion, and economic development. The campaign will promote a public debate in all regions with candidates on the pillars of the green strategy proposed by Legambiente that includes climatic justice. In the area of social innovation several initiatives often founded by the European institutions promote debates among stakeholders on sustainability of the twin transition. An example is the Social Innovation Campus organised every year by the Fondazione Triulza, where representatives from different areas (academics, students, workers, unions, firms, innovators, etc) met to discuss about the future of the twin transition. The feminist organization Stati Generali delle Donne, promotes public debates on the role women can play in the twin transition with their choices as citizens, entrepreneurs, consumers, and workers. Women promote the modernisation of production processes with low environmental impact and high impact solutions of social evolution. Women's sections of unions and professional organisations also promote public debate on the role of women in the twin transition. For instance, women from the transport unions (FILT CGIL) or the women's section of the national association of architects organise meetings to discuss the expected impacts of the twin transition on the labour market and women's employment in their respective sectors.

7. Just transition: FITTER approach

7.1. A Just Transition

7.1.1. Context

The FITTER project aims to contribute to existing research on the origins, dynamics and determinants of inequalities and enable anticipatory governance to support a fair and inclusive twin transition in Europe. A key objective is to enable timely assessments of twin transition policies under different scenarios in order to assess how policy measures may address or further exacerbate existing inequalities.

The climate crisis requires a complete transformation of energy sources, economic models, and land stewardship. Yet the scale of the transformative actions required are not yet part of transition policies (McCauley, 2023). Although the Paris Agreement set out a goal of limiting global temperatures to 1.5C rise above preindustrial levels, most climate scientists believe that that is unachievable. Current climate policies mean the world is on track for about a 2.7C rise ([Carrington, 2024a](#)). Global temperatures have risen by 1.2C above preindustrial levels and this has already caused extreme weather events, with people losing their lives or livelihoods due to 'more deadly and more frequent heatwaves, floods, wildfires and droughts brought by the climate crisis' ([Carrington, 2022](#)). A recent study estimated that 'US\$ 143 billion per year of the costs of extreme events is attributable to climatic change' (Newman and Noy, 2022). Despite this reality, there remains a real lack of political will to undertake transformative action. In a recent survey of IPCC experts, respondents were asked why the global response to the climate crisis was so slow and inadequate. Almost three-quarters of respondents cited a lack of political will; 60% also blamed vested corporate interests such as the fossil fuel industry; 27% cited lack of money as a concern; 6% cited lack green technology; and 4% cited lack of scientific understanding ([Carrington, 2024b](#))

The United Nations has established the parameters of a Just Transition starting with the International Labor Organization (ILO) Just Transition Guidelines. There are seven guiding principles. These can be summarised as the requirement that a Just Transition be

based on **social dialogue and consensus building**; realise **rights at work**; **promote equitable outcomes**, with a specific focus on being **gender transformative**; adopt a **systems approach to policy design**; create **decent jobs**; take a **place based approach** to policy design; and **foster international cooperation**.

The ILO Just Transition Guidelines are closely aligned to the achievement of the Sustainable Development Goals (SDGs); and the imperative of a Just Transition is enshrined within the Paris Agreement on Climate Change.

The Solidarity and Just Transition Silesia Declaration (2018) followed the Paris Agreement (2015). This outlined provisos to protect vulnerable workers and recognised the interlinked nature of climate change, inequity, poverty and the needs of countries in the Global South (referred to as 'developing countries'). The accompanying report on 'Implementing Just Transition after COP 24' (2018) stresses the need for social dialogue and consensus building, highlighting that:

- Stakeholders should define the scope and nature of change during the process of coalition building and policy design, policy coordination and integration, considering whether such change is transitional or transformative. Transitional change continues with the current economic model, whereas transformative change is more radical, moving

towards a broader conception of communities and more collaborative energy production and ownership.

- In order to develop adaptive capacity, change must be participatory and achieved through social dialogue. Transitional and transformational change has the potential to present more inclusive, robust, solutions, but only if stakeholders are engaged in determining what is right in each context.

The report is focused on workers in affected industries (e.g. energy) but does include indirect workers (e.g. supply chain employees), communities of fossil fuel industry (e.g. teachers), communities in other countries, and fenceline communities, in the list of required stakeholders for social dialogue and policy design processes.

The agreed ILO Just Transition Guidelines are focused on mitigation and adaptation measures to climate change and climate breakdown. They are thus forward looking and avoid significant analysis of the history of climate breakdown causation. Furthermore, they have been criticised for adopting ‘the fallacy’ of there being a win-win approach to the Just Transition. Such an approach avoids addressing the scale and urgency of the planetary emergency we face, favouring a green growth paradigm, driven by technology, which keeps us on a collision course with planetary limits (Barry 2012).

While international agreements thus far define the just transition as ensuring a fair and equitable approach to the transition away from fossil fuels for all workers, it remains within a ‘dominant economic perspective between employers and employees, driven by capitalist market understandings of inequality’ (McCauley, 2023: 245). This narrow understanding is evidenced by its operationalisation - providing support mechanisms for fossil fuel areas (which are themselves often ineffective) rather than investment in areas of future renewable technologies; and, in the EU, revising the criteria so that the just transition fund would be available for natural gas projects, which evidences the continued dominance of fossil fuel interests in the just transition agenda (McCauley 2023). Instead, an ‘urgent maturation process of the just transition concept at an international institutional level is required to avoid just transition funds being used to further embed carbon- intensive activities’ (McCauley 2023: 247).

7.1.2. Conceptual widening of the Just Transition Approach – A Socio-Ecological Contract

There are four main dimensions to inequalities within the climate-environment-social nexus.

1. Inequalities related to causation/carbon footprint
2. Inequalities related to the effects of climate change – e.g. impact of extreme weather events by geography
3. Inequalities related to adaptation capacities – e.g. protection against extreme weather events, housing, food supply
4. Unequal exposure to environmental hazards and toxic substances (Galgóczi and Akgüç, 2021)

Depending on the focus of politics and the structure of funding mechanisms, climate policies will have differential distributional effects (e.g. regressive effects of carbon taxes on low income households, low carbon products’ high upfront costs disadvantaging low income households etc) and differential employment effects (e.g. with an absence of focus on the care economy, gender inequalities will remain and possibly deepen) (see Galgóczi and Akgüç 2021).

As discussed in section 2, with regard to the climate crisis, cause and effect are both spatially and temporally distant from each other and the link between individual and collective responsibility is difficult to fully comprehend. Climate change is not perceived as an existential threat by those most responsible for GHG emissions and this is one reason for the lack of coordinated transformative effort at multiple levels (national, regional, global). Furthermore, ‘social movements, non-governmental organisations and trade unions have tended to focus on one particular form of inequality, while neglecting others and the interlinkages between them’ (Galgóczi, 2023).

Galgóczi (2023: 66) defines the just transition as concerned with

how social justice with fair burden-sharing can be applied in the context of controlling climate change, while taking climate, environmental and energy justice into account with all their dimensions.

In theory, this should be possible given the theoretical and empirical maturity of justice frameworks (Section 2). However, in practice, just transition policies are fragmented, one-dimensional, and not transformative.

Facilitating labour-market transitions is a key element of the just transition but the ‘interlinkages between the welfare state and labour-market policies are not clarified’ and the ‘responsibility of the (welfare) state in dealing with the labour-market effects of the green transformation is not defined’ (Galgóczi, 2023: 66). There are distinct views of the role of the welfare state in relation to the economy:

1. Decommodification and destratification (Esping-Andersen, 1990) – the role of the welfare state is to guarantee an acceptable standard of living for all, regardless of participation in the labour market, and redistribute resources and opportunities equally (Mandelli, 2022: 336)
2. Complementarity of the welfare state and economic growth – welfare provisions are primarily designed to incentivise participation in the labour market and the financial resources for public social expenditure are linked to economic growth (Galgóczi, 2023: 68; Mandelli, 2022)
3. Social investment – social policies should not be understood as a cost, but rather understood as an investment, supporting human capital formation and employment.

However, the environment has largely been absent as a consideration until very recently, despite significant and growing evidence base that the dominant understanding of the complementarity of the welfare state and continued economic growth are unsustainable from an ecological perspective (Meadowcroft, 2005; Gough 2016; Raworth 2017; Hickel and Kallis, 2020).

The social investment perspective holds more potential, but it would need to be updated if it is to integrate a rethinking of the welfare state as an institution that could address ‘ecological change and enable the development of new forms of working’ (Galgóczi, 2023: 68). For example, Dukelow (2022: 503) argues that revaluing the role of activation in job creation, linking it to arguments for a job guarantee within post-growth scholarship, opens up the possibility of envisaging ‘ways in which it might contribute to progressing models of sustainable welfare without growth.’

As discussed (Section 2) there are increasing demands for a move toward a **socio-ecological contract**. As summarised by Galgóczi (2023), writing from a trade union perspective, key demands include the need to address:

- Wage stagnation and income polarisation
- Informal and precarious work contracts

- Job losses due the twin transition
- Unaffordable housing, healthcare and education
- Unaffordable low carbon technologies
- Energy and transport poverty
- Gender and Race gaps
- Growing regional disparities
- Vulnerability to environmental hazards and adverse climate events (floods, heatwaves etc)

To address these, Galgóczi (2023, 72-73) argues that policies are needed to:

- Help the creation of climate-friendly quality jobs;
- Provide a just-transition policy framework that facilitates labour-market transitions, green skills development, fends off detrimental distribution effects of climate policies for vulnerable groups (correcting degressive effects) and makes low-carbon technologies affordable and accessible to all. Such a policy framework should also include regional and industrial policies
- Guarantee equal rights for all workers, regardless of their employment arrangements;
- Provide universal social protection, with the perspective of universal basic services;
- Eliminate all discrimination due to race, age, gender or other factors;
- Achieve fair taxation for both individuals and corporations at national and international level.

The United Nations Research Institute for Social Development (UNRISD) takes an approach that is more explicitly informed by feminist and decolonial theories of justice. It stresses the need to ensure human rights for all (including women and migrants); larger freedom for all including security and protection; and the transformation of economies and societies to halt climate change and environmental destruction.

The UNRISD propose an **eco-social contract** that ensures:

1. **Human rights for all** – A new eco-social contract must surpass the post-war welfare state settlements by ensuring human rights for all, including those excluded from previous social contracts or relegated to a secondary role, such as women; informal workers; ethnic, racial and religious minorities; migrants; and LGBTQIA+ persons. This requires a human rights-based approach that goes beyond formal-employment-dependent social benefits.
2. **A progressive fiscal contract** - A new eco-social contract must go hand in hand with a new fiscal contract that raises sufficient resources for climate action and SDG implementation, and fairly distributes the financing burden.
3. **Transforming economies and societies** - A new eco-social contract must be based on the common understanding that we need to transform economies and societies to halt climate change and environmental destruction and promote social inclusion and equality.
4. **A contract with nature** - A new eco-social contract must recognise that humans are part of a global ecosystem. It must protect essential ecological processes, life support systems and the diversity of life forms, and pursue harmony with nature.
5. **Addressing historical injustices** - A new eco-social contract must be decolonised, informed by indigenous knowledge, social values and capacities from the global South. It must remedy historical injustices, and combat the climate crisis fairly through just transitions.

6. **A contract for gender justice** - A new eco-social contract must recognise that previous social contracts have been built upon an unequal sexual contract. It must go hand in hand with a contract for gender justice in which activities of production and reproduction are equally shared by women and men and different genders, and where sexual orientations and expressions of gender identity are granted equal respect and rights.
7. **New forms of solidarity** - A new eco-social contract requires new bottom-up approaches to transformative change for development, bringing together social movements and progressive alliances between science, policy makers and activists. It must overcome the mindset of 'us against them', fostering instead a spirit of 'all united against' global challenges such as climate change, inequalities and social fractures.

Across the proposed socio-ecological social contracts, there is a concern for upholding worker's rights; the provision, improvement, and maintenance of life sustaining infrastructures; fair burden sharing; and the elimination of discrimination. That said, the 'how' of achieving them or even moving in the direction of institutionalising a socio-ecological contract is less clear.

7.2. Approaches to Policy Design, Implementation and Evaluation

The 'how' of the work of institutionalising a socio-ecological contract, or at least moving in that direction, necessitates an understanding of approaches to policy design, implementation, and evaluation. In this section, we consider the parameters of four approaches - eco-social policy design, systemic design practice, intersectional policy design criteria, and the capabilities approach to forecasting and evaluating policy impact.

7.2.1 Eco-Social Policy Design

Reconciling environmental, social, and economic policy goals is challenging, presenting us with what is termed the 'eco-social-growth trilemma' (Mandelli, 2022).

The 'business as usual' (BAU), economic growth-focused approach to policymaking tends to exacerbate inequalities and is increasingly unsustainable (IPCC, 2023). This is understood as 'double injustice'. On the one hand, **leaving inequalities unaddressed exacerbates climate change**. Wealthier corporations, nations and individuals remain less accountable (or not at all accountable) for high GHG emissions, unequal societies have low levels of cohesion and socio-ecological resilience, and at a global level our collective capacity for action remains underdeveloped. On the other hand, **climate change further exacerbates existing inequalities** because structurally vulnerable groups are most exposed to ecological risks, and least equipped to deal with climate change induced harms. Indeed this 'double injustice' can be seen as a 'triple injustice' when we 'consider that climate policies can further contribute to existing inequalities, due to their socially regressive nature' (Mandelli, 2022: 340). Eco-social policies seek to solve the triple injustice by not using the silo 'growth-first' logic and by 'overcoming the vicious circle between inequality and climate change, as well as other socio-ecological challenges' (Mandelli, 2022: 341).

Gough (2016) argues that a 'business as usual' (BAU) approach in OECD countries would involve the 'layering' of environmental policies on existing social policies leaving the underlying reproduction, production, and consumption systems in place. A BAU approach would further disadvantage existing structurally vulnerable cohorts including low-income households and people without the skills to transition to green jobs. To the degree that there is rectification measures, they would take the form of compensating the 'losers' of the transition within the borders of the nation state. This approach 'entails a market liberal view of both climate mitigation and social policy' (Gough, 2016: 41), leaving fair burden sharing and the principles of climate and energy justice largely unaddressed.

An alternative scenario involves stringent emissions reduction programmes integrated with radical social policies. Modelling of this (Demailly et al. 2013 cited in Gough 2016: 42) shows ‘more immediate losses to growth rates over the next two decades, as carbon pricing drives up production costs and slows down increases in purchasing power, but subsequently leads to higher growth rates in the following decades due to the move to an energy pathway less constrained by peak oil’ (Gough, 2016: 42). It requires a radical shift in priorities and mindset, and arguably some form of a socio-ecological contract. Gough (2016: 42) argues that it would require eco-social consumption and work policies.

- **Eco-social consumption policy** - the prioritisation of collective investment and consumption over private commodities; fostering local, community-based consumption; identifying high-carbon luxury consumption; improving diets to benefit both health and the environment; and moving welfare interventions ‘upstream’ to *prevent rather than ameliorate* social problems.
- **Eco-social work policy** – reduction of paid work time, fostering alternative employment contracts, developing ‘co-production’ in service delivery, encouraging low-carbon leisure activities.

The very unequal distribution of personal consumption would need to be addressed, via socialised consumption, taxation, public transfers, and ‘pre-distribution’ measures such as minimum wages, maximum rewards, and trade-union rights. These are radical shifts that would challenge dominant interests and ideas, for example ‘consumer sovereignty’ and unquestioned economic growth. They would need to be accomplished in the face of accumulated policy legacies of existing welfare and environmental states. A necessary precondition would be *extensive consensual policymaking* involving key constituencies of interest to set the frameworks for markets (Gough 2016: 42, our emphasis)

Galgóczi (2022) argues that a problem with the ‘eco-social-growth trilemma’ is that the environment, economic and social objectives are insufficiently situated within dynamic systems (incorporating knowledge systems, beliefs, norms etc). She offers a similar analysis to Gough (2016) arguing that the eco-social-growth trilemma is more usefully conceptualised as a ‘complex relationship between ‘production and consumption system’, ‘planetary ecosystems’ and ‘social redistribution systems’ (2022: 354), framed in terms of the current reality defined by climate emergency.

Mandelli (2022) seeks to further define eco-social policy design. He highlights how the existing literature largely proposes and assesses policies on their expected social and environmental *outcomes* e.g. a healthier population, low unemployment levels; rather than their *outputs* e.g. lower levels of air pollution (which may or may not lead to a healthier population when situated in the overall societal environment). Mandelli argues that an output-based definition of eco-social policies is needed because it enables empirical studies to ‘identify and ultimately evaluate eco-social policies, even when their outcomes are deficient’ (2022: 340).

An output-based definition requires that an ‘eco-social’ policy has two defining features - first that it **explicitly** pursues both environmental and social objectives, and second that it does so in an **integrated** way. An integrated policy is based on ‘a design in which, first, multiple policy goals can be coherently pursued at the same time, and, second, policy instrument mixes are consistent, in the sense of being mutually supportive in the pursuit of policy goals’ (Rayner and Howlett, 2009: 100 cited in Mandelli, 2022: 340). Integration thus aims to counter the ‘growth-first’ logic that remains dominant in the governance of the social, environmental and economic spheres (2022: 345).

Mandelli (2022) divides eco-social policies into two types – reactive and preventative.

1. **Reactive** eco-social policies **prioritise environmental objectives** in response to the climate emergency and then **add on a ‘social’ dimension**. Reactive eco-social policies are developed in immediate response to the climate emergency. They prioritise mitigation and adaptation e.g. paying the social costs of ecological catastrophes (heatwave, floods etc) or redistributing the costs

and benefits of the twin transition. These ‘reactive’ eco-social policies can be further classified as serving either (a) a ‘**protective**’ function and so **not contributing** to growth e.g. transition income support in case of loss of employment in carbon intensive sectors, or income support for energy-poor households; or (b) as serving an ‘**investment**’ function and so **contributing** to growth e.g. reskilling provisions for redundant workers in carbon-intensive sectors

2. **Preventative** eco-social policies **prioritise social objectives** and then **add on an environmental dimension**. Preventative eco-social policies can be used to reduce the carbon footprint of a region or nation state, and to promote behavioural change in the direction of sustainable practices. They are larger in scale and seek to embed long-term societal transformation. These ‘preventative’ eco-social policies can also be classified as serving either (a) a ‘**protective**’ function and **not contributing** to growth e.g. long-term income support to allow working time reductions, *or* (b) a ‘**preventive**’ function and also **contributing** to growth e.g. building energy-efficient social housing, investment in green job creation.

Mandelli’s output-based definition of eco-social policies is useful because it requires that policy design be both explicit, in terms of what is prioritised; and integrated i.e. the policy design and instruments need to work together to at least balance out economic growth with environmental and social objectives, or more radically, enable a transformation of the underlying reproduction, production and consumption systems.

7.2.2 Systemic design practice for participatory policymaking

A key element of systemic-design is that we learn by doing, we are open to reframing problems, accounting for wider systems, and we seek to effectively intervene so that we address the root causes of policy problems. There are 5 core domains of the Systemic Design Practice Wheel that can be considered in advance of systemic design practices (Blomkamp, 2022):

Domain 1 - Principles - Why and how does this work need to happen? There are six core principles of the 5 Ps framework which can be amended in light of project needs (Blomkamp, 2022: 17).

- 1) Purpose-driven – what do we want to change? What is the common goal?
- 2) Recognising complexity – we need to acknowledge the dynamism and interconnectedness of norms, structures, relationships, issues and solutions. We need to ‘work together to build understanding of the system(s) we are seeking to influence and to remain open to reframing problems and boundaries as new understandings emerge’ (Blomkamp, 2022: 17)
- 3) Self-determination – taking a rights based approach to dialogic design
- 4) Equalising power – policy practitioners can act as enablers of equality of voice, distributed leadership and devolved decision-making as far as possible.
- 5) Inclusive collaboration – a wide range of stakeholders need to be involved and engagement strategies must account for the ‘different kinds of knowledge they hold, including lived, professional and specialist expertise’ (Blomkamp, 2022: 18)
- 6) Adaptive learning - Systemic design ‘requires the creation of space, mindsets and mechanisms for collaborative problem-solving and mutual learning. Continuous feedback and a flexible accountability structure are needed to enable ongoing adaptation throughout the design process, from the early phases of problem framing through development to implementation’ (Blomkamp, 2022: 18).

Domain 2 - Place - Where is the policy issue situated in the wider socio-ecological environment? A ‘systems lens encourages us to consider when and where the work fits in relation to other activities, initiatives, organizations and networks, as well as within broader, interrelated human and ecological systems’ (Blomkamp, 2022: 16).

Domain 3 - People – who needs to be involved and in what capacity? Roles include stakeholders, enablers (e.g. civil servants), facilitators (e.g. researchers), insider-outsiders, creative designers. We need to understand our positionality and that of others, power dynamics, take care of all involved, and build relationships. The principles of inclusive collaboration, self-determination and equalizing power all apply (Blomkamp, 2022: 19). To note, it is often not possible to equalise power relations and it may be more productive to acknowledge the ways in which they are operative and adopt a conflict resolution approach (Barry et al, 2008)

Domain 4 - Process – The standard approach to policymaking involves problem identification through desk research, consultation with key stakeholders, policy development, decision-making in a closed space, and policy implementation. In contrast a design-led process ‘focuses on the lived experience of the people involved or affected by an issue, to understand the problem and its context. Taking a systemic approach often means going beyond the initial brief to explore the root causes of an issue, as well as consider different perspectives and the interrelatedness of problems’ (Blomkamp, 2022: 20). The goal is to effectively intervene in the system rather than addressing specific issues in isolation from wider systems. Design and systems-led approaches can ‘result in a reframing of the problem before potential options or solutions are explored’ (2022: 20). A key element of the FITTER process will involve interrogating the ways in which current structures are dependent on unpaid care work, low costed resource extraction which neglects to account for the full ecological costs, and/or low paid labour and resources from the Global South. Each of these contribute to the generation of intersectional inequalities.

Domain 5 - Practice – systemic-design is a learned practice, the specifics of which need to be adjusted to suit the context at hand. Key elements include - human-centeredness, convening stakeholders, dialogic process, iterative inquiry, and multiple design actions over time (Jones, 2014 cited in Blomkamp, 2022: 21). We will document our practice for each case study and reflect and learn from successes and challenges.

7.2.3 An Intersectional Framework for Policy Design and Implementation

There are persistent weaknesses in policy design and implementation that seeks to use an intersectional framework. Issues include: deploying a simple additive approach which neither conceptualises nor addresses inequalities and forms of discrimination which develop at the intersections e.g. the ways white and Black women are gendered differently; reproducing stereotypes associated with predetermined ‘vulnerable’ cohorts; adopting an ahistorical approach to policy design and not accounting for local contexts; failing to acknowledge both dynamics of privilege and oppression; and institutionalizing intersectionality as multiple forms of discrimination and thus encouraging competition for scarce resources (van der Vleuten, 2019: 54; Choo and Feree, 2010). Lombardo and Agustín (2012) develop 6 key quality criteria for intersectional policy design and evaluation:

1. Explicitness, visibility and inclusiveness – it is important to name the operative dynamics of privilege and oppression. If problems are not named, they will not be addressed.
2. Articulation – this relates to the ways in which dynamics of privilege and oppression are conceptualised, the degree to which they are explained such that all contributors can engage in productive discussions. To understand how disadvantages are mutually constitutive (rather than multiple) we need to understand the histories of categorisation. Where no ‘explanations or understandings of the nature of the relationship between the categories are expressed’ (2012: 489) the dynamics remains inarticulate.
3. Gendering/Degendering – Lombardo and Agustín argue that keeping gender as a category of analysis (i.e. not omitting gender in consideration of other categories) is a

‘quality criterion because it enhances the likelihood that gender equality is treated as an aim in itself, and that this goal is not lost when other inequalities enter the agenda’ (2012: 490). Arguably it is more useful to consider racialised gendering as a quality criterion given the tendency of some uses of intersectionality to seek to adopt a disembodied or ‘post-racial’ framework (Bilge, 2020: 2317). Adopting a disembodied framework effectively invisibilises processes of racialisation (including the ways whiteness operates as a dynamic of privilege).

4. Structural understanding of inequalities – to develop transformative policies and address power hierarchies it is important to understand inequalities as systemic i.e. it is norms, structures and institutions which create and reproduce socially stratified population groups.
5. Avoid the stigmatisation of specific groups/Challenge privileges – this is linked to developing a structural understanding of inequalities. The naming of groups who experience co-constitutive disadvantages risks stigmatising them as lacking in agency and/or responsible for their own adverse circumstances. To mitigate this risk it is important that practitioners maintain a focus on both oppression and privilege e.g. racism is *productive* of advantages of some people and disadvantages for others. This includes developing a good understanding of the various modalities of power. Empowerment mechanisms (power-to) which are disconnected from power-with others, power-within or conscientisation, and power-for (as the vision, values, and demands that orient our work), may well lead us to act in ways that uphold or reinforce relations of domination (Gaventa, 2021).
6. Consultation of civil society in the policy-making process – consultation with a wide range of stakeholders increases the possibility that policy design and implementation adopts a user-oriented approach and incorporates ‘a more explicit, articulated, transformative, and less-biased approach to intersectionality since the inclusion of embodied subjects expressing different concerns can promote greater self-reflexivity on one’s own biases’ (Lombardo and Agustín, 2012: 491). This does not always work in practice - time, resources, and network contacts can limit the range of stakeholders consulted and the most powerful actors can prevail in consultations (2012: 491).

The role of categorisation in demarcating who counts and who counts less (or not at all) has rendered the process of categorisation and research and policy-making based on categorisations suspect. The concern is that categorisation leads to demarcation, which leads to inclusion/exclusion, and exclusions underpin inequalities. Yet, given our histories, it is also impossible to not use categories in an analysis of the reproduction of symbolic violence and material inequalities. McCall (2005) distinguishes three methodological approaches to the analysis of categories, that seek to better understand the complexities of inequalities.

1. *Anticategorical* approach – the primary goal of this methodology is to deconstruct analytical categories. There is a focus on the complexities of subjects and structures. The goal is to clarify the ways in which fixed ‘master’ categories (e.g. gender, race sexuality) are historically and culturally particular constructs which operate as simplifying social fictions that produce inequalities in the process of producing differences (McCall, 2005: 1773). The value of this methodological approach is that it illuminates the constructed nature of categories, enabling us to have a more critical relationship to categories, and to better understand how they can be reworked to enable liveable lives for all (e.g. facilitating transgender inclusive practices and environments).
2. *Intracategorical* approach – this methodology inaugurated the study of intersectionality. Similar to an anticategorical approach, it understands analytical categories as historically and culturally

particular constructs, which work to include some people and exclude others. However, there is also a stress on the embeddedness of social categories – we make sense of ourselves and others through ‘our’ gender, race and sexuality for example. Unlike the anticategorical approach, this approach is more critical of overly broad acts of categorization, than of categorization in and of itself. It is also critical of single category analysis and politics e.g. the prioritisation of the needs of middle class, heterosexual, white women before (rather than as intersecting with) the needs of women from ethnic minority communities, disabled women and members of LGBTQI+ communities. With an intracategorical approach the goal is not to establish and uncritically maintain group boundaries using ever more refined categorical intersections. Instead, the goal is to focus on categorical intersections in order to better understand ‘multiple and conflicting experiences of subordination *and* power’ (McCall, 2005: 1780, our emphasis). An example of this is a case study of a particular group of people e.g. Black women in Berlin. The researcher is interested in understanding commonalities and also the diversity of experiences within the case study group (by class, citizenship status, sexuality etc) in order to uncover multiple experiences of subordination *and* power. The value of this approach is that the material and discursive importance of categories is acknowledged and deployed in service of better understanding how categories are ‘produced, experienced, reproduced, and resisted in everyday life’ (2005: 1783).

3. *Intercategorical approach* – with the first two approaches categories are understood as relational. It is only possible to deconstruct (anticategorical) or critically interrogate (intracategorical) categories as they are experienced within relational dynamics. While relationships are *the premise* of the first two approaches, charting the ‘changing relationships among multiple social groups defines *the goal*’ of the third, intercategory approach (2005: 1785). Whereas the *intracategorical* approach could be understood as ‘single case intensive’, the *intercategory* approach includes multiple groups and is systematically comparative (2005: 1786). The value of the intercategory approach is that it can track changing configurations of inequalities over time.

McCall uses the intercategory approach (3) to analyse existing wage inequality in US regional economies - through the differences between groups within standalone ‘master’ categories (e.g. gender – men vs women); and also between the intersections of groups across those categories - gender (men, women) race (Black, Asian, Latino and White) and education level (college educated, not college educated). Her findings demonstrate that distinct configurations of inequalities were evident in different regions and no ‘single dimension of overall inequality can adequately describe the full structure of multiple, intersecting and conflicting dimensions of inequality’ (2005: 1791). This led her to question the utility of general indicators of inequality as a measure of overall societal equality, and to better understand how ‘some forms of inequality seem to arise from the same conditions that might reduce other forms [of inequality]’ (2005: 1791). To note, McCall’s analysis is situated in the context of top down, single-issue policy design and implementation which entails the layering of often ‘one-off’ social or environmental measures on top of existing reproduction, production, and consumption systems.

7.2.4 The Capabilities Approach to Forecasting and Evaluating Policy Impact

The Capabilities Approach (CA) is a method used for evaluating human well-being and sustainability. It is increasingly used in policy design and evaluation. At the core of this approach is the argument that assessments of well-being, social justice and development ought not to ‘primarily focus on resources, or on people’s mental states, but on the effective opportunities that people have to lead the lives they have reason to value’ (Robeyns, 2006: 351).

The concepts of ‘functionings’ and ‘capability sets’ have undergone much change in the literature (see [Robeyns, 2017](#)) but for our current purposes we can make a working distinction between a person’s

functionings as their beings and doings (e.g. well-fed, IT literate) and their capability sets (the actual opportunities they have to realise their functionings). Capabilities are thus the ‘substantive freedoms people have to achieve certain functions’ (Przybylinski and Sidortsov, 2023: 125).

The capability approach is radically underspecified. To adopt a capabilities approach at least three decisions are necessary (Robeyns, 2006):

1. *Choosing to focus on capabilities or functionings.* Focusing on capabilities does not impose a version of the good life on others (anti-paternalistic) and respects that individuals are responsible for their own choices (responsibility-sensitivity principle). Focusing on functionings allows us to measure well-being outcomes for large groups of people. To note, most available datasets do not enable us to draw conclusions about people’s capability set so we may have to use functionings in line with data availability. A significantly different distribution of actual functionings for two groups of people can enable us to deduce that, overall, they did not have the same capability sets i.e. real opportunities to realise functionings.
2. *Selecting relevant capabilities to evaluate.* There are several ways of selecting relevant capabilities – e.g. limiting capabilities to those necessary to achieve a certain role e.g. citizen; endorsing a well-defined list of capabilities defined as moral entitlements of every human (Nussbaum, 2000); adopting a defined selection procedure (Alkire, 2002; Robeyns, 2003); or using statistical techniques such as factor analysis.
3. *Deciding on the weighting of different capabilities for an overall assessment.* This is a complex area involving intrapersonal and interpersonal aggregation of capabilities; assessing which combination of variables one functioning can be based on and the weightings given to each variable; weighting different functionings into a quality-of-life indicator; and/or weighting indicators interpersonally. The allocation of weightings could be arrived at through debate and justification, using a statistical method, or through a type of social choice procedure (see Robeyns, 2006: 356-358). When using existing indicators, it is necessary to clearly explain the data collection processes, variables, weightings etc upon which the indicator is based.

The way one chooses to answer any given question – selecting relevant functionings/capabilities, choosing specific variables on which to base functionings, deciding on weightings - will lead to different results. If we alter the choices we make at any point, the results will also change. This means that it is ‘crucial that each application of the capability approach explicitly justifies its specifications (selection of capabilities, weights, and so on) and tries, as much as possible, to analyze and discuss the extent to which these specifications affect the results’ (Robeyns, 2006: 371).

Robeyns (2003) suggests five criteria for drawing up a list of initial functionings in a research study:

1. **Explicit formulation:** the list should be explicit, discussed and defended. The options will likely be constrained by existing datasets but ‘the capabilities that would have been appropriate but for which no information is available’ (2003: 70) should also be discussed and reported on.
2. **Methodological justification:** clarify and justify the method for putting the list together. One method could be first, unconstrained brainstorming; second, refinement in light of existing literature and data on the issues of concern; third, engaging with other lists of functionings/capabilities; and fourth, debating the list with other people (Robeyns, 2003)
3. **Sensitivity to context:** the level of abstraction should be appropriate to the research objectives. Functionings relevant to economic or social justice discussions will be less abstract than those relevant to philosophical discussions.

4. Different levels of generality: This is distinct from but related to the third criterion. It states that 'if the specification aims at an empirical application, or wants to lead to implementable policy proposals, then the list should be drawn up in at least two stages. The first stage can involve drawing up a kind of "ideal" list, unconstrained by limitations of data or measurement design, or of socio-economic or political feasibility. The second stage would be drawing up a more pragmatic list which takes such constraints into account. Distinguishing between the ideal and the second-best is important, because constraints might change over time' (2003: 70-71). For example, including currently unavailable data on care labour on the 'ideal' list contributes to the case for collecting actual data on care labour. The multi-stage procedure guards against the unreflective production of lists which reproduce existing biases.
5. Exhaustion and non-reduction: listed functionings/capabilities should include all important elements and each element should not be reducible to other elements (Robeyns, 2003).

Wolff and De-Shalit (2007) stress that while the functionings people can achieve is important, it is equally as important that those functionings are secure i.e. they can sustain their current level of functionings. Thus, the risk and vulnerability of losing key functionings also needs to be accounted for. This is particularly relevant to forecasting scenarios in the context of the just transition wherein ecological protections are required to ensure current and future peoples can achieve basic functionings (Holland, 2008).

The capability approach is **methodologically individualistic but not ontologically so**. CA 'does not assume atomistic individuals nor that our functionings and capabilities are independent of our concern for others or of the actions of others' (Robeyns, 2003: 64) The degree to which we can achieve certain functionings is dependent on conversion factors that convert resources into a functioning. These include personal conversion factors which are internal to a person e.g. health, intelligence; social conversion factors which structure society e.g. norms, policies, and practices; and environmental conversion factors which relate to the physical and built environment e.g. pollution, amenities, transport networks, communication networks. Taking conversion factors into account enables a fuller evaluation of opportunities to achieve functionings.

The capability approach *is* **ethically or normatively individualistic** though - the unit of concern is each individual and the focus is on enabling individual agency. The advantage of this ethical individualism is that an individual's functionings are not subsumed under a wider units of analysis such as a households (Robeyns, 2003).

One disadvantage of ethical individualism is that the focus on enabling individual agency cannot account for *interdependencies*. For example, there is an assumption that women, as individuals, need to be more self-interested and refuse unfair allocations of unpaid care labour, when the focus could be on men's responsibility to take on their fair share of unpaid care, even though this would set constraints on men's freedoms. Applications of the capabilities approach could risk over-emphasising individual freedoms and under-emphasising the values of justice and care (Lewis and Giullari, 2005: 94). The priority that the capability approach gives to enabling individual agency is also criticised for being forward-looking only. A focus on addressing contemporary individual oppressions can work to efface the need for an analysis of the ways in which the ordering of the global economy produces these inequalities (Jaggar, 2002).

Another disadvantage of the focus on enabling individual agency is that it can leave out non-individualistic understandings of that which contributes to wellbeing. All understandings of agency and wellbeing objectives are culturally particular. If indicators only measure individual functionings they may miss out on collective activities and processes that many view as fundamental to wellbeing.

A final disadvantage is that many applications of the capability approach are not structured to explain *why inequalities are generated* i.e. the root causes. Instead, they provide a framework for evaluating *how inequalities are patterned* across groups of people, often using large datasets.

7.2.5 Key considerations for Policy Design and Implementation

Distinct philosophical underpinnings and methodological decisions produce distinct kinds of substantive knowledge. As discussed in Section 2, the social contract instituted a public/private divide and restricted its domain to the nation-state. Understandings of self and others, defined by gender, race, class, and sexuality, centred the needs of some people, and placed others outside the domain of justice (e.g. because they were in the private sphere, in the Global South, same-sex sexual activity was criminalised). If we assume that the subject of justice, and of policy, is an able-bodied, full-time employed, citizen with no significant caring responsibilities, then the experiences of all who fall outside that understanding of personhood will be less accounted for or ignored.

Current policy responses at EU level focus on ecological modernisation and social investment, albeit without sufficient acknowledgement of the climate emergency as an emergency, without the application of substantive justice principles, and without the substantive reconceptualisation of the welfare state that a reworking of the social contract in the direction of a socio-ecological contract would entail.

In part this is due to how policy-making currently operates as a top-down, largely elite-driven, often single issue, process. Policy development and review processes are assumed to be rationalistic wherein common social values can be identified and institutionalised, policy can be built on values and the evidence base, and policy can determine outcomes.

Transition management literature argues that sustainability ‘can be achieved through consensus on visions and through common effort under a process-oriented, interactive style of national policy-making, implementation and learning’ (Scrase and Smith, 2009: 708). Transition Management approaches advocate inclusive question-framing, the development of high-level visions, experimentation, reflexive learning and iterative sociotechnical change. A consensual or corporatist model of politics is assumed which leads to the development of largely technocratic strategies that favour incumbent infrastructures, technological fixes, and established economic interests (Scrase and Smith, 2009; Revez et al, 2021). Unfortunately, this approach is inadequate for the political struggles we face in the context of a climate emergency and the need for a just transition.

The slow, piecemeal response to climate change arguably relates to the power of some actors across multiple spaces. Actors can work to ‘realise their interests through decision making and the control of resources (visible power); backdoor machinations and institutional organisation (hidden); and the structural-discursive empowerment of the actor and the creation and use of discourse (invisible)’ (Hathaway, 2016: 118). In reality, values are contested, the evidence base is sometimes acted upon and sometimes ignored, and policy recommendations are subject to debate and often reshaped to suit powerful actors’ social and economic interests. This favouring of ‘business as usual’ approaches is also one of the reasons why the three undervalued, yet crucial support pillars, of the 20th century social contract – unpaid and low paid care work (mostly performed by women); inexpensive labour and raw materials from the Global South (many of which are former colonies); and the assumed ongoing availability of the planets resources as raw materials for the production of goods, and as a sink for production’s waste – continue to be assumed as a low cost to no cost given by many policy decision makers.

Taking an intersectional approach involves both analysing dynamics of oppression and dynamics of privilege. Policymakers cannot only be forward looking and ignore the root causes of climate

breakdown which necessitates the twin transition. Taking an intersectional approach also necessitates relational and systems thinking. The combined scale, urgency and complexity of the actions required to transition to a carbon neutral economy has led policy makers to recognise the need for adopting the processes and principles of systems thinking and prioritising human-centred design. This ‘participatory turn’ in public policy design, implementation, and review presents challenges to the more traditional reliance on authority, control and singular areas of expertise (Macken-Walsh, 2016). Attention must be given to the wider socio-ecological environment, the framing principles and objectives of policy, the processes through which policy is developed, and the needs of, and impacts on, end users. In the context of a just transition, relying on the existing evidence-base only can reinforce our assumptions about what to solve and how to solve it. Future-oriented policy considers the existing evidence base and also introduces the concept of ‘desirable futures’ to frame problems and structure solutions (Dekker, 2020; Nogueira and Schmidt, 2022; Blomkamp, 2022).

7.3. FITTER Approach

A methodology outlines the philosophy, methods and data upon which the research process and knowledge production are based. In this sub-section we outline the basic building blocks of the FITTER approach to an analysis of inequalities in the four sectors.

FITTER is a mixed methods study which takes a critical realist intersectional approach to addressing inequalities in each of the four sectors. Critical realism maintains that:

1. The real world is not knowable in an absolute sense because it is subject to human interpretation through diverse epistemological frameworks;
2. The real world establishes limits on knowledge such that not all interpretations are equally plausible. Reality is ‘complexly patterned but patterned nonetheless. We can determine the source of the complexity, we can describe it, and we can theorize it’ (McCall, 2005: 1794);
3. There is a role for the development of theoretical knowledge about directly unobservable phenomena (McCall, 2005: 1794).

We define the just twin transition as a process that addresses intersectional inequalities, with fair burden-sharing, while controlling climate change and taking the dimensions of environmental, climate, and energy justice into account (adapted from Galgóczi, 2023).

In line with our definition of a just transition we account for core tenets of environmental, energy and climate justice.

Environmental justice (local/national): Environmental justice can be defined as ‘the extent to which the physical and economic burdens of pollution and degradation, as well as environmental benefits, are equitably distributed across society, both spatially and temporally, and the degree to which individuals and communities most vulnerable to environmental risks can access and participate in relevant decision-making processes’ (O’Neill et al, 2022: 5)

Climate Justice (global): concerned with addressing inequitable outcomes of climate change for differing peoples, communities, contexts and generations. Aims to achieve greater equity in the distribution of climate related burdens and responsibilities, as well as greater parity of decision-making and human rights protection.

Energy Justice (global): concerned with developing mechanisms to enable justice (recognitive, representative, distributive and restorative) at all stages of the energy life cycle 1) Extraction; 2)

Production; 3) Operation and supply; 4) Consumption; 5) Waste Management (Heffron and McCauley, 2017)

7.3.1. Subject, Domain and Object of Justice

As outlined, different theories of justice have different subjects of justice (the unencumbered individual, women and gender minorities, the racialised subject); domains of justice (public institutions, the domestic, the divide between Global South and Global North); and objects of justice (maldistribution of goods or liberties, unequal responsibility for unpaid care-work, historical and contemporary relations of power that oppress (post)colonised peoples).

In line with an intersectional approach, we propose taking the ‘historically constituted and corporeally vulnerable rights-bearing individual’ as the subject of justice. This formulation acknowledges that we are all constituted through historical processes (e.g. colonialism, capitalism) and cultural norms (e.g. the primacy of the gender binary) that preceded us. We are corporeally vulnerable in the sense that we are all dependent on others in one way or another, we engage in giving and receiving care, and we are also dependent on fundamental goods such as clean air, clean water, and food. Awareness of ‘the rights-bearing individual’ as historically constituted and corporeally vulnerable could orient us toward a model of co-existence that enables liveable lives for all.

We propose taking the four FITTER sectors as they are embedded in global, national, local and household levels of power as the domain of justice.

We propose taking relations of production (economic-environmental-social) in each of the sectors; and mechanisms of distribution (economic-environmental-social) in each of the sectors as the object of justice.

7.3.2. Adopting a Grounded Approach to Justice

Based on this literature review we do not anticipate that a just transition is possible without a fundamental restructuring of economic models, land stewardship, energy sources, and mindsets. The transition to a carbon neutral economy is very unlikely with existing climate policies and the prevailing economic growth-focused mindset. The literature on the transition is largely focused on elite-level policy making models and the technologies that could drive sustainable transitions. Little attention is paid to the political realities of power dynamics, the difficulties of the transition, and its impact on already existing inequalities. As discussed above, in a recent survey of IPCC experts, respondents were asked why the global response to the climate crisis was so slow and inadequate. Almost three-quarters of respondents cited a lack of political will, with 60% blaming vested corporate interests (Carrington, 2024b).

Climate Emergency

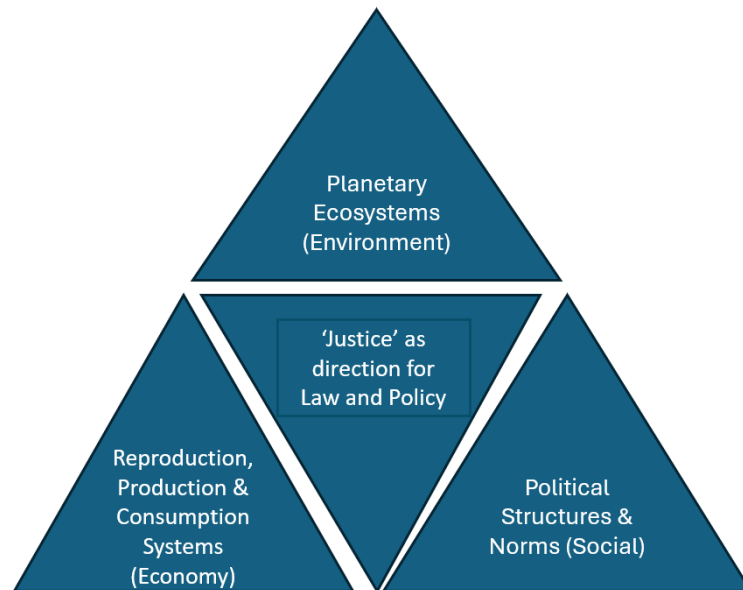


Figure 7.1: The 'Systems Trilemma' (see also Galgóczi, 2022; Heffron and McCauley, 2017)

Following and further expanding on the work of Galgóczi (2022: 354) we propose situating the 'eco-social-growth' trilemma as a 'systems trilemma' with complex relationships between 'reproduction, production and consumption systems', 'planetary ecosystems' and 'political structures and norms' framed in terms of the reality defined by the climate emergency. Following and further expanding the work of Heffron and McCauley (2017: 665), whose focus is energy justice, we propose situating 'justice' at the core of the wider triangle. Justice concerns function to provide direction for law and policy decision-making. Law and policy, guided by justice concerns, should effectively intervene to (re)structure systems so that we remain within planetary boundaries and enable liveable lives for all. Current climate policy is arguably too focused on piecemeal adjustments to the 'reproduction, production and consumption system' and neglects accounting for transformations needed with regard to our relationship to planetary ecosystems, and political structures and norms (including the possibilities for 'enabling' or 'eco-social' political structures and norms).

There are four strands of work across FITTER tasks– *assessment* of the production of inequalities; *policy design* for three scenarios; *forecasting* of policy impact under three scenarios; and the *production* of tools to assist stakeholders in addressing intersectional inequalities in the context of a just twin transition. We propose that work package leads can identify which strands of work are included within each task and follow the processes in Table 7.1.

Table 7.1: FITTER Strands of Work – Assessment, Design, Forecasting, and Production

	Steps	Qualitative	Quantitative
Assessment of the generation of inequalities	1	Problematising 'vulnerability'	
	2	Assess the degrees to which recognition justice, distributional justice and representative justice are present/absent within existing policy framework. Develop understandings of inequalities as co-constitutive (not multiple)	Follow Robeyns (2003) five criteria for drawing up a list of functionings to assess - taking all elements of the 'systems trilemma' into account (pg 69-70). Decide on weightings of functionings/indicators for overall assessment.
	3	Assess where the policy issues fit within wider systems, taking the whole production-distribution-waste mgmt life cycle into account.	Identify inequalities in functionings, patterned across groups of people (Intercategorical Approach (McCall, 2005))
Systemic Design Practice	4	Consider how to apply justice concerns in practice. Restorative justice includes fair burden sharing, prevention of harm, and repair of harm	Survey of stakeholders
	5	Assess where policy design sits on a reactive-preventative- transformative continuum	
Forecasting Policy Impact	6	N/A	Taking account of the three scenarios developed at steps 4 and 5 forecast the impact of each scenario via expected functionings
Production of Tools for Stakeholders	7	Produce tools to assist policy makers.	Produce tools to assist policy makers.

We propose adopting a grounded, bottom-up, partial approach to justice that begins with the practical problems in each sector and the injustices thereby generated. Given the limitations of the capabilities approach in assessing *how* inequalities are generated, and its focus on the individual level only (not systems and structures), we will use this as a support for the qualitative analysis. CA is used to identify patterns of inequalities between groups of people, and forecast the impact of scenarios.

Our proposed ground approach to justice is non-ideal in three senses – it begins from the empirical complexity of the real-world, it seeks improvement from the current situation, it makes no assumption of full compliance with an ideal (Valentini, 2012). In line with an intersectional approach we seek to prioritise the development of relational thinking skills by all involved in the project - developing a comprehensive understanding of the FITTER areas, identifying how current structures generate

intersectional inequalities, linking those structures and mechanisms to broader societal and global dynamics of oppression and dynamics of privilege, using systemic design practice to develop three scenarios, and developing tools that assist in policy development so that such policies address, or at least do not further exacerbate, societal inequalities across four sectors.

Assessment of the Production of Inequalities

Step 1 – Problematising Vulnerability

We propose maintaining a focus on the structures which generate intersectional inequalities. We note that redistributive justice for structurally vulnerable groups in society is prioritised within many theories of justice. However, we note that the concept of ‘vulnerability’ can be used in ways that cause further harm to those who are subject to poverty, exclusion, discrimination and/or violence.

Life sustaining infrastructures (healthcare, education, transport networks, agricultural production, clean environment) must be actively supported and maintained – they consist of reproducible and dynamic institutions and relations (Butler, 2009: 24). We are all corporeally vulnerable and a healthy environment is necessary for our survival. That said, how we each experience vulnerability depends on whether (or the degree to which) the organisation of political, environmental, economic and social relations works to our advantage. Arguably a task for politics and policy is to maintain life sustaining infrastructures in egalitarian ways such that the production of disadvantaged groups - by the configuration of the economic-environmental-social nexus - is minimal. What is in need of protection are the conditions of life which support liveable lives (Butler, 2009: 24). A more straightforward identification of structurally disadvantaged cohorts (women, people of colour, people with disabilities etc) as ‘vulnerable’ cohorts diverts attention away from the norms and conditions which produce vulnerabilities.

We note that empirical work in the research programme ‘Towards a European Theory Of juStice and fairness’ (ETHOS) found that greater understanding of the differences between sub-categories of the ‘the vulnerable’ underscored the constructed nature of ‘vulnerability’ and the injustices of treating people as reducible to a generic social category. They found that ‘categorization into specific (allegedly vulnerable) groups enhances vulnerability (and injustice as misrecognition) through stigma’ (Lepianka and Knijn, 2020: 11). They highlighted that the ‘category of ‘vulnerability’ presupposes the existence of a normatively preferred modus vivendi, governed by neoliberal ideals of self-sufficiency, responsibility and in- rather than interdependence’ (Lepianka and Knijn, 2020: 11).

Naming cohorts as vulnerable draws attention away from the systems and structures that produce advantage and disadvantage. It risks stigmatisation, ‘naturalising’ the vulnerability of these cohorts, and/or rendering them personally responsible for their adverse circumstances (Brown, 2011, 2014; Brown et al 2017).

Step 2 (Qual) - Assess the degrees to which recognition justice, distributional justice and representative justice are present/absent within existing policy frameworks

To assess the structures which generate intersectional inequalities we propose starting with three key tenets of justice – justice as recognition, justice as representation, and justice as redistribution. We build on the work of ETHOS, combining the work of Nancy Fraser (1995, 1997, 2008) and one of the frameworks which has emerged from energy justice (McCauley et al 2013; Heffron and McCauley 2014, 2017).

Nancy Fraser developed the tripartite distinction between justice as redistribution, justice as recognition and justice as representation. This approach focuses on identifying injustices generated within real-world social and institutional implementation of rights and responsibilities.

Justice as recognition

Misrecognition occurs when a person is not perceived as a legitimate participant in a given space. For example, in agriculture there is a gendered division between 'outdoor' and 'indoor' work. Traditional gender norms for men centre on being tough, physically able, stoic, tenacious, self-reliant, a good provider, and 'in charge' of outdoor work. Traditionally women were expected to be the linchpin that holds the family together, supportive of their husband and children, generous, and selfless. When women participated in outdoor work they were viewed as 'helpers' to their male partners, and participating in outdoor work led to a higher overall workload because women are not relieved of care work tasks (Brandth 2001: 112). This cultural understanding of women as 'helpers' on the farm, rather than farmers in their own right, is a deeply embedded norm which retains cultural force and creates significant obstacles to women's full participation in agriculture.

Recognitional injustices include the use of mechanisms of exclusion (ignoring someone, insulting someone, devaluing or dismissing certain perspectives without due consideration); and misrecognising people (drawing on stereotypes rather than lived experiences, failing to unlearn harmful norms).

From an energy justice perspective misrecognition has been shown to occur when local opposition to environmental policies is attributed to ignorance and/or self-interest by policy makers with insufficient evidence base for such conclusions. Taking the time to understand the core concerns of those who oppose environmental policies, developing a broad based understanding of the assumptions, values and judgements that underpin distinct positions in relation to a policy (in favour, against, undecided) and taking a conflict resolution approach to deliberative public engagement could arguably lead to the creation of a negotiated compromise that does not sacrifice social justice requirements in order to achieve environmental and economic goals (Barry et al, 2008).

Justice as redistribution

An intersectional approach to addressing distributional inequalities proposes that inequalities are structured by norms and societal structures which limit some cohorts' access to both material resources and discursive resources (Fraser, 1995; Crenshaw, 1991). Recall the three undervalued, yet crucial support pillars, of the 20th century social contract – unpaid and low paid care work (mostly performed by women); inexpensive labour and raw materials from the Global South (many of which were former colonies); and the assumed ongoing availability of the planet's resources as raw materials for the production of goods, and as a sink for production's waste. Dynamics of oppression such as sexism and racism (which set limits on the discursive resources of affected communities) underpin and legitimise unequal access to material resources (e.g. gender and race pay gaps). From an intersectional perspective, justice as redistribution does not only relate to a fair distribution of goods and liberties, but also includes a commitment to understanding the histories of discriminatory norms, their material effects, and working to transform both in the direction of more liveable lives for all.

From an environmental and energy justice perspective distributional justice further refers to the spatial distribution of environmental benefits and ills (McCauley et al, 2013).

Justice as representation

When perspectives within a polity are unjustly excluded, silenced or ignored there is a risk of 'false negative danger' - where certain public interests are ignored; 'false positive danger' - where common

interests are misrepresented and other interests are falsely presented as common interests (Pettit, 2004); and misframing – where the boundaries of the community of concern are drawn to unjustly exclude certain persons (Fraser, 2008). For example, given that we are embedded within a global economy and addressing climate change as part of a just transition is a collective problem, it would be unjust to ignore human rights abuses in countries wherein raw materials are sourced. Even though it may be unfeasible to recruit participants from affected global communities in direct consultation, policy makers can place a local energy issue in the broader energy life cycle taking global interdependencies and issues into account (Heffron and McCauley, 2017).

From an energy justice perspective, justice as representation is replaced with the concept of ‘procedural justice’. This includes the requirement that all stakeholders be included in a non-discriminatory way; that they be able to participate in decision-making; and that government and industry stakeholders commit to participation, impartiality and full information disclosure (McCauley et al 2013). This understanding of procedural justice has much in common with Fraser’s concept of justice as representation, but it is less focused on the risks of misframing. Given our focus on accounting for the three undervalued yet crucial support pillars of the 20th century social contract we will proceed with the concept of justice as representation and supplement it with an analysis based on procedural justice as necessary.

Step 2 (Quan) – Draw up a list of functionings

We propose following Robeyns (2003) five criteria for drawing up a list of initial functionings: Explicit formulation; methodological justification; sensitivity to context; appropriate level of generality; and exhaustion and non-reduction. Decide on weightings of functionings/indicators for overall assessment.

To note, we acknowledge that we initially committed to deriving a series of dimensions and indicators to measure social justice and equality in relation to the twin transition as part of this deliverable D2.1 (M04). As outlined in this deliverable this is a complex task in itself that will therefore be further progressed and reported on in relation to D3.1 (M10) and D2.2 (M13).

Step 3 (Qual) – Wider Systems Analysis

The relations of production and the mechanisms of distribution within each sector are the object of justice. An assessment of where policy issues fit within wider systems should take reproduction and consumption systems, planetary ecosystems and political structures and norms into account across the whole production-distribution-waste management life cycle (see Heffron and McCauley, 2017).

Step 3 (Quan) – Identify Patterns of Inequality between Groups of People (Intercategorical Approach (McCall, 2005))

The agreed functionings and indicators will be used to identify inequalities of functionings, patterned across large groups of people.

Systemic Design Practice

Step 4 (Qual) – Policy Design

Apply justice concerns in practice, using the concept of restorative justice. This includes fair burden sharing, prevention of harm, and repair of harm. Heffron and McCauley stress that if ‘restorative justice were applied to the energy sector it would ensure that decision-making was made in light of considering the potential harm of that decision and consequently the true cost of that decision’ (2017:

661). At present costs to the planetary ecosystems and within the care economy (reproduction system) are significantly underestimated.

Following and further expanding the work of Blomkamp (2022) we propose defining the 5 domains of the FITTER System Design Practice in terms of:

Domain 1 - Principles

- 1) Purpose-driven – we seek to enable a just twin transition which addresses intersectional inequalities, with fair burden-sharing, while controlling climate change and taking the dimensions of environmental, climate, and energy justice into account
- 2) Recognising complexity – we commit to understanding justice concerns as situated within the ‘systems trilemma’ (Figure 7.1). We commit to understanding inequalities as co-constitutive (not multiple), making dynamics of oppression and privilege explicit and articulated; gendering policies; challenging privilege (Lombardo and Agustín, 2012).
- 3) Rights-based/Self-determination – we commit to upholding human rights, taking the core tenets of energy, environmental and climate justice into account.
- 4) Addressing power relations – we commit to taking a conflict resolution approach to deliberative mechanisms, acknowledging operative power dynamics, and seeking to create a negotiated compromise that does not sacrifice social justice requirements in order to achieve environmental and economic goals. We also aim to challenge harmful exclusionary norms and extend the boundaries of what is ‘normal’ (addressing invisible power).
- 5) Inclusive collaboration – we commit to consulting a wide range of stakeholders in order to adopt a user-oriented approach and incorporate ‘a more explicit, articulated, transformative, and less-biased approach to intersectionality since the inclusion of embodied subjects expressing different concerns can promote greater self-reflexivity on one’s own biases’ (Lombardo and Agustín, 2012: 491).
- 6) Adaptive learning – we commit to collaborative problem-solving, openness to reframing problems, the unlearning of exclusionary norms, understanding inequalities as systemic, and ongoing adaptation throughout the design process.

To note, these principles can be further developed at team level and in consultation with key stakeholders.

Domain 2 - Place

As part of step 3 we will have situated the policy concerns within a wider systems analysis. We will build on this at the policy design stage taking the tenets of spatial justice into account, with a focus on effective interventions into systems that work to address co-constitutive inequalities.

Domain 3 – People

We will specify who needs to be involved at each stage taking account the roles required – e.g. enablers, facilitators, insider-outsiders, creative designers etc. We will work to understand our positionality and that of others and work to build collaborative relationships.

Domain 4 - Process

We will adopt a design-led process which involves going beyond the core policy issues in a specific sector to explore the root causes of issues and the interrelatedness of both problems and solutions. Key elements of the FITTER process will involve:

- interrogating the ways in which current structures are dependent on unpaid care work, low costed resource extraction (which neglects to account for the full ecological costs), and/or low

paid labour and resources from the Global South, as each of these contribute to the generation of intersectional inequalities; and

- developing three scenarios which address intersectional inequalities in the context of a just transition. Proposed measures could potentially be situated on a reactive-preventative-transformative continuum.

Domain 5 -Practice

Systemic-design is learned by doing, the specifics of which need to be adjusted to suit the context at hand. We will document our practice for each case study and reflect and learn from successes and challenges.

Step 4 (Quan) – Survey of Stakeholders

The survey of stakeholders will be part of the process domain within systemic design practice.

Step 5 - Assess scenarios in line with a reactive-preventative- transformative continuum

Building on and expanding the work of Mandelli (2022) we can assess the proposed policy outputs within each of the three scenarios along a reactive-preventative-transformative continuum. Reactive strategies seek to compensate for social injustices whilst leaving underlying systems in place (reproduction, production & consumption; political structures and norms; planetary ecosystems). Preventative to transformative strategies seek to enact larger scale, medium to long term changes in political structures and norms, in order to enable healthy ecosystems, and fair benefit- and burden-sharing within reproduction, production and consumption systems. We can also assess the proposed policy outputs in terms of the degree to which they adopt an intersectional framework (Lombardo and Agustín, 2012).

Step 6 – Forecasting Policy Impacts (Quantitative Analysis)

Taking account of the three scenarios developed at steps 4 and 5 forecast the impact of each scenario via expected functionings.

Step 7 – Production of Tools to Assist Policy Makers Address the ‘Systems Trilemma’

The tools produced will need to assist policy makers to identify and address:

- Discourses which reproduce harmful cultural norms (e.g. care work is primarily women’s work) as part of the work of transforming the structures and practices that derive from those harmful norms (relates to affected communities’ ‘power-within’ and ‘power-for’; addressing ‘invisible power’)
- The ways in which dynamics of inclusion/exclusion operate at multiple levels of power – household, local, national and global – and how that contributes to the climate emergency.
- The evidence base that demonstrates that leaving inequalities unaddressed exacerbates climate change, and in turn, climate change further exacerbates inequalities.
- The costs and benefits of policy outputs as situated along a reactive-preventative-transformative continuum

The step-by-step process outlined above is at an early stage and will be further discussed, amended and agreed on at team level.

By assessing the generation of intersectional injustices, primarily through Nancy Fraser’s tripartite conception of justice, combined with key elements of the energy justice framework aligned to her work (see McCauley et al 2013; Heffron and McCauley 2017) - and taking a systemic-design approach

to policy development in each of the four sectors - we will develop 3 future scenarios that account for the synergies and tensions generated by justice claims at the social-environmental-economic nexus.

We note at least two limitations at present. A key element of systemic-design is that we learn by doing, we are open to reframing problems, accounting for wider systems, and we seek to effectively intervene so that we address the root causes of the generation of intersectional inequalities. In their use of Fraser's tripartite conception of justice as a starting point, the ETHOS project team also undertook an iterative approach. They found that Fraser's - three dimensions (redistribution, recognition, representation), and two strategies for repair (affirmative, transformative), in order to enable the principle of participatory parity - are a useful means of assessment but incomplete. Depending on the issues at hand, the tripartite conception may need to be supplemented by additional dimensions of justice in order to identify, assess and redress particular injustices (e.g. historical injustices or epistemic injustices).

The ETHOS project team also caution that the realities on the ground may impede enabling justice principles in full and considerations of the 'importance of incremental transitions away from manifest injustices in the real world' are also required (Brink et al, 2020: 205). Such options would likely be situated on the reactive end of the continuum.

The ETHOS project team concluded that that 'empirically informed and action-guiding theories of justice need to be *case-based*: geared to helping us better understand, evaluate, and recommend responses to European injustices' (2020: 209). These case-based theories of justice can combine empirical research, policy analysis and normative reasoning in order to understand why a policy area generates injustices, and to connect strategies of redress with reasoning and normative principles such as the principle of parity of participation. The objective is for reasonable views 'to unite against identifications of what is manifestly unjust, even when they do not agree on what perfect justice would be' (2020: 210).

8. Conclusions

This is the first deliverable on the concepts, definitions, and methodological approach to embedding social justice in twin transition research. It will be updated in M13 and again in M21 and published as D2.2 and 2.3. The development of the concepts, definitions and methodological approach is iterative given the co-creative structure of FITTER EU.

As demonstrated throughout this report, the debate on twin transition is still in an emerging stage, which is reflected in both the policy development and the academic discourse. Within policy, there is a rather optimistic presumption that the digital and the green should go hand in hand, with the former accelerating the latter. However, there is some criticism of such an approach, highlighting that digitalisation is not always ‘climate friendly’ and that advancing digital technologies does may not result in a more sustainable society. Furthermore, there seems to be a gap in the debate on the actual ‘twin’ transition as the two are often discussed separately. This is particularly the case for the just transition debate which, to date, has mainly focused on the green transition. We propose that more attention should be given to the digital aspect of the transformation as the two are not equal and will result in different impacts on existing inequalities, if not managed carefully. This is especially important from the point of view of age as increased reliance on new technologies may disproportionately disadvantage older citizens.

In addition, the literature review shows that the main emphasis of the twin transition discourse, and of the debate on just transition, continues to be on employment. It has been argued that the transition will result in job growth, but there is a need for investment in upskilling and reskilling the existing workforce. Most of these points are usually made in relation to sectors which are high in relation the GHG emissions, such as transport, construction or energy – all of them traditionally male-dominated. Importantly, the gender aspect of the systemic shift is missing in this context, as the transformation (or losses) of jobs are deemed to affect men – yet women in the communities are most likely to share the burden. Digitalisation and its impact on jobs should also be considered here, as different groups participating in the labour market will be affected unequally.

Moreover, the classification of so-called ‘green’ jobs has also been problematised as too simplistic and focused on specific skills. It has been proposed by some authors to include care as a green job – particularly from the point of view of its growing importance. Within this context, it has also been argued that the just transition approach should go beyond the labour market lenses. Rather, a more holistic approach should be undertaken, with all aspect of human lives intertwined and impacted, and intersectional inequalities taken to account. Based on the literature reviewed for the purpose of this report, we propose the following definition of the just twin transition, based on one developed by Galgóczi (2023):

A just twin transition is a process that addresses intersectional inequalities, with fair burden-sharing, while controlling climate change and taking the dimensions of environmental, climate, and energy justice into account

We have proposed taking the ‘historically constituted and corporeally vulnerable rights-bearing individual’ as the subject of justice; the four FITTER sectors as they are embedded in global, national, local and household levels of power as the domain of justice; and both *relations of production* (economic-environmental-social) and *mechanisms of distribution* (economic-environmental-social) connected to each of the sectors as the object of justice.

We have distinguished three strands of work across FITTER (EU). We draw upon existing frameworks to further specify our approach to each strand:

Assessments of the generation of intersectional inequalities

- developing a team and participant understanding and fluency in the histories of discriminatory norms, operative dynamics of oppression and privilege, the co-constitutive nature of disadvantage based on dynamics of oppression, and their material effects
- adopting a grounded, bottom-up, partial approach to justice, identifying forms of recognitional injustice, distributive injustice, and representative injustice (Fraser, 2008; Heffron and McCauley 2017) whilst supplementing this with additional dimensions of justice as required (e.g. assessment of historical injustices).
- identifying that which is left out or assumed within existing policy approaches – i.e. assumptions of the continued availability of the planets resources as raw materials for the production of goods, and as a sink for production's waste; unpaid and low paid care work; inexpensive labour and raw materials from the Global South.

Policy development processes in response to the requirements of a just transition

- developing policy measures under 3 scenarios within each of the sectors and identifying the degree to which adaptation is conceptualised as transformative in line with a revised eco-social contract under each scenario.
- taking a systemic design approach to policy development. This involves combining/adapting:
 - ii. the systemic design practice wheel principles
 - iii. the quality criteria for intersectional policy design (explicitness, articulation, gendering, structural understanding of inequalities, challenging privileges, consultation with civil society;
 - iii. energy, environmental and climate justice principles (e.g. parity of participation, fair burden sharing, restorative justice);
 - iv. realistic, best practice approaches to accounting for power dynamics and the political realities of struggle and antagonism e.g. adopting a conflict resolution approach to deliberative processes; and
 - v. an output-based construction of eco-social policy and practice.

Forecasting the impacts of addressing the generation of intersectional inequalities under each of the three scenarios generated; and production of tools to assist policy makers address the systems trilemma.

One element which is currently under-specified in the work packages is taking a systemic design approach to policy development. Given that we propose combining that with the quality criteria for intersectional policy design we anticipate that it could be more realistic for each case study to focus on the energy sector plus one other of the three sectors (transport, food and agriculture, building and housing).

Generation of, access to, and use of energy underpins all elements of the social-environmental-economy nexus. The selection of the additional sector to focus on in each case study country could proceed through the criteria:

- Explanatory – to what degree will this sector serve a broader explanatory function in clarifying causal relationships among it and the energy sector, and in understanding the wider political, economic and environmental systems operative in the case study country

- Existing eco-social policy proposals – to what degree does existing research in this sector propose viable and transformative eco-social policy and practices. If there is already a significant research focus on the sector with a view to eco-social policy development and implementation it may be better to choose a sector wherein research and eco-social policy development is less developed.
- Reliance on undervalued resources – to what degree is the current structure of the sector reliant on and exploitative of undervalued resources including low paid and unpaid care labour, extraction of limited resources, utilisation of the planet as a sink for production's waste, and low cost labour in the Global South. These are key generators of inequalities and environmental harms which need to be addressed as part of a just transition.

Ideally each of the three additional sectors will be a focus of two of the six case study countries.

Next steps include:

1. Team review and discussion of the proposed methodology
2. Refinement and finalisation of the approach to each strand of work – assessment, policy development, forecasting and production of tools.
3. Decision making with regard to the scope of each case study
4. Development of the tasks and deliverables in the context of steps 1-3

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